

Data Analysis

Main Reference Textbook v4

Chapters 1–17 and Appendices A–B with Math Foundations source map

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Notation Guide

This guide collects the notation used across the regression portion of the text. It is not a replacement for the chapter explanations; it is a quick reference for keeping the object types straight.

Object Types

Notation	Object type	How to read it
x, y, b, p, n	scalars	Single numerical values or indices.
X, Y, ϵ	random variables	Population-level or model-level random quantities.
$\mathbf{x}, \mathbf{y}, \mathbf{e}$	vectors	Observed sample-level vectors, usually in $\mathbb{R}^n$.
$\mathbf{X}, \mathbf{H}, \mathbf{I}_n$	matrices	Data, projection, or identity matrices.
$\boldsymbol{\beta}, \boldsymbol{\epsilon}$	parameter/error vectors	Coefficient and error vectors in regression models.
$\hat{}$	estimate or fitted quantity	A hat marks something estimated from data or predicted by a fitted model.

Data and Statistical Learning Notation

For a single observation, $\mathbf{x} = (x_1, \dots, x_p)^T$ is the vector of p input values and Y is the response. The general supervised-learning model is

$$Y = f(\mathbf{X}) + \epsilon,$$

where f is the systematic relationship and ϵ is the unexplained error.

For a data set with n observations, i indexes observations and j indexes variables:

$$x_{ij} = \text{value of predictor } j \text{ for observation } i.$$

The observed response vector is

$$\mathbf{y} = (y_1, \dots, y_n)^T.$$

The notation $\mathbf{x}_j$ means the full observed column of predictor j . The notation $\mathbf{x}_i^T$ means the row of predictors for observation i .

Regression Notation

In the regression chapters, the design matrix $\mathbf{X}$ includes an intercept column unless stated otherwise. With p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)}, \quad \boldsymbol{\beta} \in \mathbb{R}^{p+1}.$$

The fitted values and residuals are

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}.$$

The true error vector is $\boldsymbol{\epsilon}$; the residual vector is $\mathbf{e}$. Errors are part of the data-generating model and are not observed. Residuals are computed after fitting a model.

The ordinary least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y},$$

when $\mathbf{X}$ has full column rank. The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T, \quad \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}.$$

The residual-maker matrix is $\mathbf{I}_n - \mathbf{H}$.

Later chapters also use $\mathbf{Z}$ for an *engineered* design matrix. Its columns may include transformed predictors, indicator/dummy variables for categories, and interaction columns. Once those columns are built, least squares treats them like any other design-matrix columns; the new work is choosing and interpreting the columns responsibly.

Observed Vector Versus Random Vector

Most chapters use $\mathbf{y}$ for the observed response vector from the data set in hand. Chapter 10 temporarily uses $\mathbf{Y}$ because it studies the fixed-design theoretical model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\mathbf{X}$ is treated as fixed and $\mathbf{Y}$ varies randomly through $\boldsymbol{\epsilon}$. After that theory chapter, the text returns to $\mathbf{y}$ when discussing a realized fitted model.

Fit, Testing, and Diagnostics

Notation	Meaning
RSS	Residual sum of squares, $\sum_i (y_i - \hat{y}_i)^2 = \ \mathbf{e}\ ^2$.
TSS	Total sum of squares, $\sum_i (y_i - \bar{y})^2$.
RegSS	Regression or explained sum of squares, often TSS – RSS when an intercept is included.
R^2	Proportion of response variation explained relative to the mean-only baseline.
RSE	Residual standard error, the estimated residual standard deviation.
h_{ii}	Leverage of observation i , the i th diagonal entry of $\mathbf{H}$.

Later chapters use λ for regularization strength, α for elastic-net mixing when needed, and K for the number of principal components retained. The notation $\|\mathbf{b}\|_1$ means a one-norm penalty based on absolute coefficient sizes, while $\|\mathbf{b}\|_2^2$ means a squared two-norm penalty.

The Big Three Regression Assumptions

The regression chapters use a common three-step assumption ladder:

1. **Mean-zero errors:** the model is centered correctly, e.g. $\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}$.
2. **Constant variance and uncorrelated errors:** $\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$.
3. **Gaussian errors:** $\epsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

The stronger assumptions support stronger inferential claims, but they are not all needed for every modeling task.

Linear Models

Early chapters use “linear” in the familiar geometric sense: a line, plane, or hyperplane in the displayed predictors. Chapter 11 sharpens the definition: for least squares, the key requirement is that the model be linear in the unknown coefficients. Curved functions of the original predictors, category indicators, and interaction terms can still be linear models if they can be written as a design matrix times a coefficient vector plus error.

Chapter 1

Data Analysis as Statistical Learning

1.1 Why This Course Begins with Statistical Learning

Data analysis is not just the act of computing summaries or fitting models. It is the disciplined process of using data to learn something about a system, make predictions about future observations, and communicate conclusions in a way that can be checked, reproduced, and defended. In this course, we use the phrase *statistical learning* to emphasize that our models are tools for learning from data rather than formulas that are true simply because we wrote them down.

This course serves as a bridge. It connects the probability, statistics, and mathematical tools developed in earlier operations analysis courses to the modeling and machine-learning methods that appear later. The immediate goal is to build the statistical modeling intuition needed for regression, exploratory data analysis, model validation, and diagnostics. The longer-term goal is to prepare you to use more advanced methods without treating them as black boxes.

A recurring theme is the statement, often summarized as “all models are wrong.” The point is not that models are useless. The point is that a model is a deliberate simplification. It keeps the structure we think is useful and ignores or absorbs the rest as noise, approximation error, measurement error, or unexplained variation. A good analyst understands both sides: what the model is trying to capture and what it is failing to capture.

1.2 What Is Data?

A data set is a collection of measurements, records, or observations. Each observation corresponds to one unit of analysis: a person, ship, aircraft, market, mission, experiment, time step, or other object being studied. Each variable records one attribute of that observation.

In statistical modeling, we usually distinguish between *input variables* and an *output variable*. Input variables are the measured quantities we use to explain or predict something. They are also called *predictors*, *features*, *covariates*, or *independent variables*. The output variable is the quantity we are trying to explain or predict. It is also called the *response* or *dependent variable*.

For example, suppose we want to model a service member’s physical fitness test score. Possible inputs might include age, training frequency, prior score, sleep, and injury status. The output could be the next test score. In a logistics setting, inputs might be demand history, location, distance,

and inventory status, while the output could be delivery time or stockout risk. In a classification problem, inputs might be sensor readings and the output might be whether a track is classified as friendly, hostile, or unknown.

We collect the input variables for a single observation into a vector. For an observation with p inputs, write

$$\mathbf{x} = (x_1, x_2, \dots, x_p)^T \in \mathbb{R}^p.$$

Here p is the number of input variables, and x_j is the value of the j th input for this observation. The corresponding output is denoted by Y or, for a realized observed value, by y .

Math Foundations Connection. Math Foundations treats a vector as an ordered collection of quantities that belong to the same object, not merely as an arrow in physical space. That is exactly how we use $\mathbf{x}$ here. The vector $\mathbf{x}$ is one observation's collection of input measurements. Its entries may be physical quantities, demographic attributes, encoded categories, time summaries, or engineered features. The geometric interpretation of vectors will become useful later, but the first interpretation is simpler: a feature vector is a structured record of measured information. See the Math Foundations textbook, Chapter 2.

1.3 A First Statistical Model

The central modeling idea in this chapter is that an output variable Y is related to input variables $\mathbf{X} = (X_1, X_2, \dots, X_p)^T$ through an unknown relationship plus random error:

$$Y = f(\mathbf{X}) + \epsilon. \tag{1.1}$$

The function f represents the systematic part of the relationship between the inputs and the output. The random variable ϵ represents the part of Y not explained by $f(\mathbf{X})$.

A simple example with two predictors is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon. \tag{1.2}$$

In this model, β_0 is an intercept, β_1 and β_2 are coefficients, and ϵ is the unobserved error. The inputs x_1 and x_2 are observed. The output Y is observed after the fact. The error term ϵ is not directly observed; it is the difference between the observed output and the systematic part of the model.

Equation (1.1) is intentionally general. The function f might be linear, nonlinear, simple, complicated, interpretable, or opaque. In this course, we often begin with linear models because they are mathematically transparent and remain important for inference. But the broader statistical learning framework is not limited to linear models.

We usually impose basic assumptions on the error term. A common starting point is

$$\mathbb{E}[\epsilon] = 0, \tag{1.3}$$

$$\text{Var}(\epsilon) > 0, \tag{1.4}$$

$$\epsilon \text{ is independent of } \mathbf{X}. \tag{1.5}$$

The mean-zero assumption says that the error does not systematically push the output upward or downward once the systematic relationship has been accounted for. The positive-variance assumption

says that there is genuine unexplained variability. The independence assumption says that the leftover error is not still carrying predictable information from the inputs. These assumptions are not decorations. They are part of the model’s claim about the world, and later chapters will show how violations of assumptions affect inference, prediction, and diagnostics.

Since f is unknown, we estimate it using data. We denote the estimated function by $\hat{f}$. Given a new input vector $\mathbf{x}$, the corresponding predicted output is

$$\hat{Y} = \hat{f}(\mathbf{x}). \quad (1.6)$$

The hat notation signals that the quantity is estimated or predicted rather than directly observed or known.

Math Foundations Connection. When the model is linear in the input variables, the systematic part can be written compactly as

$$f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = \beta_0 + \boldsymbol{\beta}^T \mathbf{x},$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$. The term $\boldsymbol{\beta}^T \mathbf{x}$ is an inner product. Equivalently, it is a linear combination of the entries of $\mathbf{x}$, with coefficients $\beta_1, \dots, \beta_p$. This is why linear algebra becomes the natural language of regression: the model combines measured inputs by weighting them and adding them together. See the Math Foundations textbook, Chapters 2, 3, and 5.

1.4 Inference and Prediction

There are two major reasons to estimate f : inference and prediction. They are related, but they are not the same task.

1.4.1 Inference

Inference is about understanding the relationship between the inputs and the output. In the linear model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon,$$

inference focuses on questions such as:

- What is the meaning of β_1 and β_2 ?
- Are these coefficients positive, negative, or plausibly zero?
- Which predictors are most relevant to Y ?
- How much does the output change when one input changes and the others are held fixed?
- What assumptions are required before we trust those conclusions?

Inference is about learning structure. If a commander asks why readiness changed, which drivers matter, or whether a particular intervention had an effect, then the analysis is inferential. Accuracy still matters, but the model must also be interpretable enough to support an explanation.

1.4.2 Prediction

Prediction is about estimating the output for new or future inputs. Given a new vector $\mathbf{x}_{\text{new}}$, we use the fitted model to compute

$$\hat{Y}_{\text{new}} = \hat{f}(\mathbf{x}_{\text{new}}).$$

Prediction focuses on questions such as:

- Given these inputs, what output should we expect?
- How accurate is the prediction likely to be?
- How much uncertainty remains even after we use the model?
- How well will the model perform on cases it has not yet seen?

Prediction is about learning what is likely to happen next. If an analyst is building a model to forecast demand, estimate casualty risk, classify a sensor return, or predict equipment failure, then the model may be judged primarily by its predictive performance.

1.4.3 The Same Model Can Serve Both Goals

The same mathematical model can sometimes support both inference and prediction, but the priorities differ. A highly interpretable model may be preferred for inference even if a more flexible model predicts slightly better. Conversely, in a pure prediction task, a less interpretable model may be acceptable if it performs substantially better on new data. A defensible analysis states which goal is primary and evaluates the model accordingly.

1.5 Regression and Classification

A first way to categorize statistical learning problems is by the type of output variable.

In a *regression* problem, the output is numerical or quantitative. Examples include predicting fuel consumption, time to complete a task, exam score, cost, weight, or sales. A regression model returns a number. A simple linear regression model has the form

$$\hat{y} = wx + b, \tag{1.7}$$

where w is a slope or weight and b is an intercept or bias term.

In a *classification* problem, the output is a category or class label. Examples include predicting pass/fail, disease/no disease, hostile/friendly/unknown, or low/medium/high risk. A classifier may return a class directly, but many classifiers first estimate probabilities for each class and then choose the most likely class.

For binary classification, where $Y \in \{0, 1\}$, logistic regression models the probability of class 1 as

$$\Pr(Y = 1 \mid \mathbf{x}) = \sigma(w_1x_1 + w_2x_2 + b), \tag{1.8}$$

where

$$\sigma(t) = \frac{1}{1 + e^{-t}} \tag{1.9}$$

is the logistic or sigmoid function. The expression $w_1x_1 + w_2x_2 + b$ is a score. The sigmoid maps that score to a probability between 0 and 1.

For multiclass classification, a model may compute several scores, often called *logits*. For example, an image classifier might produce a cat logit, dog logit, and ship logit. Those logits can then be transformed into class probabilities. The key idea is that classification models often separate two steps: first produce class-specific scores, then convert those scores into a decision or probability distribution.

1.6 Linear and Nonlinear Models

The previous examples introduce a second distinction: whether the model class is linear or nonlinear. This distinction is separate from the regression/classification distinction. We can have linear regression, nonlinear regression, linear classification, and nonlinear classification. For now, "linear" means the familiar geometric picture: lines, planes, and hyperplanes in the predictors as written. Chapter 11 later sharpens this by showing that a model can still be linear after transforming predictors, as long as it remains linear in the unknown coefficients.

A *linear regression model* predicts a numerical output using a linear function of the inputs:

$$\hat{y} = \beta_0 + \beta_1x_1 + \cdots + \beta_px_p.$$

With one predictor, this is a line. With two predictors, it is a plane. With more predictors, it is a hyperplane.

A *linear classification model* uses a linear score to separate classes. In two dimensions, the decision boundary is a line. Logistic regression is a standard example: the model computes a linear score and then converts that score into a probability.

But data are not always linear. Sometimes the pattern in a regression problem is curved rather than straight. For example, a one-dimensional nonlinear regression model might be written as

$$\hat{y} = w(x - a)^2 + b. \tag{1.10}$$

This model can represent a curved relationship because the input appears through the transformed term $(x - a)^2$.

Similarly, a nonlinear classifier may use a nonlinear score before applying the sigmoid function. For example,

$$\Pr(Y = 1 \mid \mathbf{x}) = \sigma(\gamma_0 + \gamma_1(x_1 - h)^2 + \gamma_2(x_2 - k)^2). \tag{1.11}$$

This model can create a curved decision boundary in the (x_1, x_2) plane. The boundary is no longer forced to be a straight line.

The important point is that *data shape* and *model shape* must be considered together. If the true relationship is nonlinear but the model is linear, the model may miss essential structure. If the model is extremely flexible, it may capture real structure but also chase random noise. Later chapters develop this tradeoff carefully.

1.6.1 Neural Networks as Repeated Linear and Nonlinear Operations

A fully connected neural network layer is another example of this linear/nonlinear pattern. A basic layer takes an input vector $\mathbf{x}$, forms a vector of linear combinations, and then applies a nonlinear

activation function:

$$\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b}). \quad (1.12)$$

Here W is a matrix of weights, $\mathbf{b}$ is a vector of intercepts or biases, and σ is applied component-by-component. Without σ , stacked layers are still only affine transformations (linear maps plus shifts). The nonlinearity is what lets neural networks represent more complicated functions.

This course does not begin with neural networks because the basic modeling questions are easier to see in linear regression. However, the same foundational ideas remain: inputs, outputs, parameters, loss, assumptions, prediction error, and the need to justify the modeling choices.

1.7 Statistical Concepts Used Throughout the Course

Before developing prediction error and model diagnostics, we need a small set of statistical concepts. This section reviews expectation, variance, covariance, correlation, conditioning, sampling behavior, and bias. These ideas will reappear throughout the course.

1.7.1 Expectation

The expectation of a random variable is its probability-weighted average. For a discrete random variable X with possible values $x_1, \dots, x_m$ and probabilities $p_1, \dots, p_m$, the expectation is

$$\mathbb{E}[X] = \sum_{i=1}^m p_i x_i. \quad (1.13)$$

The probabilities weight the possible outcomes. Outcomes that are more likely contribute more to the average. For a continuous random variable with density $p(x)$, the corresponding definition replaces the sum with an integral:

$$\mathbb{E}[X] = \int x p(x) dx. \quad (1.14)$$

Math Foundations Connection. If we collect the possible outcomes into a vector $\mathbf{x} = (x_1, \dots, x_m)^T$ and the probabilities into a vector $\mathbf{p} = (p_1, \dots, p_m)^T$, then

$$\mathbb{E}[X] = \mathbf{p}^T \mathbf{x}.$$

Thus expectation can be viewed as an inner product between a probability vector and an outcome vector. The probability vector must have nonnegative entries that sum to one. This connects the statistical idea of a weighted average to the linear algebra idea of an inner product. See the Math Foundations textbook, Chapters 2, 3, and 5.

Expectation has two especially important algebraic properties. First, for constants a and b ,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b. \quad (1.15)$$

Second, for random variables X and Y ,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]. \quad (1.16)$$

This second property does *not* require X and Y to be independent. This is why we call it linearity of expectation.

For the affine rule, the reasoning is straightforward:

$$\mathbb{E}[aX + b] = \mathbb{E}[aX] + \mathbb{E}[b] = a\mathbb{E}[X] + b,$$

since a constant has expectation equal to itself.

1.7.2 Variance

Variance measures the average squared deviation from the mean. If $\mu_X = \mathbb{E}[X]$, then

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2]. \quad (1.17)$$

A useful computational identity is

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (1.18)$$

To see why, expand the square:

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu_X)^2] \\ &= \mathbb{E}[X^2 - 2\mu_X X + \mu_X^2] \\ &= \mathbb{E}[X^2] - 2\mu_X \mathbb{E}[X] + \mu_X^2 \\ &= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 \\ &= \mathbb{E}[X^2] - \mu_X^2. \end{aligned}$$

Since $\mu_X = \mathbb{E}[X]$, this gives Equation (1.18).

Variance changes predictably when we scale and shift a random variable. For constants a and b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X). \quad (1.19)$$

The constant b does not affect variance because shifting all outcomes by the same amount does not change their spread. The scale factor a multiplies deviations by a , so squared deviations are multiplied by a^2 .

1.7.3 Covariance and Correlation

Variance measures how one random variable varies around its mean. Covariance measures how two random variables vary together. If $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$, then

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (1.20)$$

A useful computational identity is

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \quad (1.21)$$

This follows by expanding the centered product and applying linearity of expectation. Covariance is symmetric, and covariance with itself is variance:

$$\text{Cov}(X, Y) = \text{Cov}(Y, X), \quad \text{Cov}(X, X) = \text{Var}(X). \quad (1.22)$$

If X tends to be above its mean when Y is above its mean, the covariance tends to be positive. If X tends to be above its mean when Y is below its mean, the covariance tends to be negative. If their deviations do not move together in a systematic linear way, the covariance may be near zero.

Covariance is sensitive to scaling but not shifting. For constants a , b , c , and d ,

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y). \quad (1.23)$$

In particular, $\text{Cov}(aX + b, Y) = a \text{Cov}(X, Y)$. The shifts b and d disappear because covariance is based on deviations from means; multiplying a variable by a constant multiplies its deviations by that constant.

Covariance appears in the variance of a sum:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y). \quad (1.24)$$

A direct derivation comes from writing centered variables. Let $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Then

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E} \left[((X + Y) - (\mu_X + \mu_Y))^2 \right] \\ &= \mathbb{E} \left[((X - \mu_X) + (Y - \mu_Y))^2 \right] \\ &= \mathbb{E} [(X - \mu_X)^2] + \mathbb{E} [(Y - \mu_Y)^2] + 2\mathbb{E} [(X - \mu_X)(Y - \mu_Y)] \\ &= \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y). \end{aligned}$$

For many random quantities, the same idea gives

$$\begin{aligned} \text{Var} \left(\sum_{i=1}^n X_i \right) &= \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j). \end{aligned} \quad (1.25)$$

This is why covariance matters when random quantities are combined or averaged: cross-terms affect the total spread unless they vanish.

If X and Y are independent, then $\text{Cov}(X, Y) = 0$. The converse is not generally true: zero covariance does not necessarily imply independence. Covariance detects linear co-movement, while dependence can be nonlinear.

Correlation is a scaled version of covariance:

$$\rho_{XY} = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}. \quad (1.26)$$

Correlation is unitless and lies between -1 and 1 when both variances are positive. The scaling by the standard deviations makes correlation easier to compare across contexts than covariance.

Math Foundations Connection. For observed data, sample correlation can be interpreted geometrically. If we center two data vectors by subtracting their sample means, then their sample correlation is the cosine of the angle between the centered vectors. This is why correlation measures linear alignment: two centered data vectors pointing in nearly the same direction have correlation near 1 ; two pointing in opposite directions have correlation near -1 ; and two nearly orthogonal centered vectors have correlation near 0 . See the Math Foundations textbook, Chapters 2, 3, and 5.

1.7.4 Conditioning and Total Laws

Conditioning means restricting attention to a known event, group, or predictor value before computing a probability, mean, variance, or covariance. If B is an event with positive probability, then

$$\Pr(A | B) = \frac{\Pr(A \cap B)}{\Pr(B)}. \quad (1.27)$$

If G records which group an observation belongs to, then conditioning on $G = g$ means computing the relevant quantity within group g . When the groups form a partition of the sample space,

$$\Pr(A) = \sum_g \Pr(A | G = g) \Pr(G = g). \quad (1.28)$$

This is the law of total probability: compute the probability within each group, then average those group probabilities using the group frequencies. The same ingredients also give Bayes' rule:

$$\Pr(G = g | A) = \frac{\Pr(A | G = g) \Pr(G = g)}{\sum_h \Pr(A | G = h) \Pr(G = h)}. \quad (1.29)$$

Bayes' rule updates the probability of group membership after observing that event A occurred.

The conditional expectation $\mathbb{E}[Y | G]$ is the mean of Y after conditioning on the group. It is itself a random variable because its value depends on the group. The law of iterated expectation says

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | G]]. \quad (1.30)$$

In words: average within groups, then average the group means.

The law of total variance decomposes total variation into within-group and between-group pieces:

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | G)] + \text{Var}(\mathbb{E}[Y | G]). \quad (1.31)$$

The first term is the average conditional variance: how much Y varies within groups. The second term is the variance of the conditional means: how much the group means differ from one another.

The same idea extends to covariance:

$$\text{Cov}(X, Y) = \mathbb{E}[\text{Cov}(X, Y | G)] + \text{Cov}(\mathbb{E}[X | G], \mathbb{E}[Y | G]). \quad (1.32)$$

The first term is the average within-group covariance. The second term is the covariance of the group means. This is why a marginal relationship can differ from a within-group relationship: the overall association may be driven partly by differences among groups. Regression uses this same conditioning logic when it asks how the response behaves after accounting for predictors.

1.7.5 Statistics, Estimators, and Sampling Behavior

A *statistic* is any function computed from the observed data. The sample mean, sample variance, sample correlation, maximum value, and fitted regression coefficients are all statistics. An *estimator* is a statistic used to estimate an unknown population quantity. The distinction is about purpose: a statistic is a data summary, while an estimator is a data summary aimed at an unknown target.

Because an estimator depends on the sample, it is random before the data are observed. If we repeatedly collected new samples from the same population and recomputed the estimator, we

would get a distribution of possible estimator values. That distribution is the estimator’s *sampling distribution*. The *standard error* of an estimator is the standard deviation of its sampling distribution. Standard errors measure sampling instability: how much the estimate would tend to change from one sample to another.

An estimator $\hat{\theta}_n$ is *consistent* for a target parameter θ if it converges to θ as the sample size n grows. The Law of Large Numbers explains why the sample mean is a natural estimator of a population mean. If $X_1, \dots, X_n$ are independent observations with common mean μ , then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \longrightarrow \mu \quad \text{as } n \rightarrow \infty. \quad (1.33)$$

The sample mean stabilizes around the population mean as more observations are collected.

The Central Limit Theorem describes the approximate shape of the sample mean’s sampling distribution. Under common regularity conditions, if the observations have mean μ and variance σ^2 , then for large n ,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx N(0, 1). \quad (1.34)$$

Equivalently, $\bar{X}_n$ is approximately normal with mean μ and standard error $\sigma/\sqrt{n}$. This is one reason normal approximations, standard errors, confidence intervals, and hypothesis tests appear throughout classical statistical modeling. The theorem does not say every data set is normal; it says that many averages and estimators built from many small contributions have approximately normal sampling behavior.

1.7.6 Bias

An estimator is a rule for using data to estimate an unknown quantity. If the unknown parameter is θ and the estimator is $\hat{\theta}$, then the bias of the estimator is

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta. \quad (1.35)$$

Bias is systematic error. It asks whether the estimator tends to miss in a particular direction over repeated samples. Bias is not the same thing as random error. An estimator can be unbiased and still vary substantially from sample to sample. Conversely, a biased estimator may have low variability but systematically target the wrong value.

Bias can be induced by measurement choices, data collection procedures, model assumptions, or the deliberate use of a simplified model. This is one reason the phrase “all models are wrong” matters. A model can be useful while still introducing bias. The analyst’s job is not to pretend the model is perfect; it is to understand what kind of error the model introduces and whether that error is acceptable for the task.

1.8 Reproducibility and Defensibility

A data analysis must be more than numerically correct at one moment on one computer. It should also be reproducible and defensible.

1.8.1 Reproducibility

Reproducibility means that the same code applied to the same data produces the same results. A reproducible workflow has several features:

- the data source is clear;
- preprocessing steps are documented;
- transformations are recorded;
- modeling choices are traceable;
- assumptions and exclusions are stated;
- results can be rerun without guessing what happened.

A useful standard is this: your future self, or another analyst, should be able to rerun the analysis and understand the major decisions without needing a private explanation from you.

1.8.2 Defensibility

Defensibility concerns what we can explain to a customer, commander, decision maker, or technical reviewer. A defensible analysis answers questions such as:

- Why did we choose this model?
- What assumptions does the method rely on?
- What alternatives did we consider?
- What are the limitations of the data?
- How much uncertainty remains?
- What conclusion is supported, and what conclusion would overstate the evidence?

Reproducibility is about what we did. Defensibility is about what we can justify. Both are essential. A result can be reproducible but not defensible if it repeatedly applies the wrong method. A result can sound defensible but not be reproducible if the workflow cannot be checked. Good data analysis requires both.

1.9 Chapter Summary

This chapter introduced data analysis as statistical learning. A data set consists of observations and variables. In supervised learning, we distinguish input variables from an output or response variable. We model their relationship using

$$Y = f(\mathbf{X}) + \epsilon,$$

where f is the systematic relationship and ϵ is the unexplained error.

The two major modeling goals are inference and prediction. Inference asks what the model says about the relationship between inputs and output. Prediction asks how accurately the model

can estimate outputs for new inputs. Regression problems involve quantitative outputs, while classification problems involve categorical outputs. Models may be linear or nonlinear, and modern methods such as neural networks build complex functions by combining linear operations with nonlinear activations.

Finally, the chapter reviewed expectation, variance, covariance, correlation, conditioning identities, sampling behavior, and bias. These concepts provide the mathematical language needed to discuss model error, uncertainty, assumptions, and the limits of prediction. The next chapter uses this foundation to study prediction error, model flexibility, overfitting, and generalization.

Chapter 2

Prediction Error, Flexibility, and Generalization

2.1 Why Prediction Error Matters

The previous chapter introduced the supervised learning model

$$Y = f_0(\mathbf{X}) + \epsilon, \quad (2.1)$$

where $\mathbf{X} = (X_1, \dots, X_p)^T$ is the vector of input variables, Y is the response, f_0 is the true systematic relationship, and ϵ is the error term. In this chapter we ask a prediction-focused question:

Given the information in $\mathbf{X}$, how accurately can we predict Y ?

This question makes precise the phrase “all models are wrong.” A model is not judged by whether it perfectly describes reality. It is judged by how much useful signal it captures, how much error remains, and whether its performance holds up on data it has not already seen.

To avoid a common notation problem, this chapter uses f_0 for the true unknown function and $\hat{f}$ for a fitted model. The true function f_0 is fixed but unknown. The fitted function $\hat{f}$ is random because it depends on the training data used to fit it. If a different training data set were collected, the fitted model would generally be different.

2.2 Mean Squared Error

The most common measure of prediction error for a quantitative response is *mean squared error*. For a new observation with predictors $\mathbf{x}_0$ and response Y_0 , the prediction is

$$\hat{Y}_0 = \hat{f}(\mathbf{x}_0).$$

The squared prediction error is

$$\left(Y_0 - \hat{f}(\mathbf{x}_0)\right)^2.$$

The mean squared error, or MSE, is the expected squared prediction error:

$$\text{MSE}(\mathbf{x}_0) = \mathbb{E} \left[\left(Y_0 - \hat{f}(\mathbf{x}_0)\right)^2 \right]. \quad (2.2)$$

The expectation in Equation (2.2) averages over the randomness in the training data used to produce $\hat{f}$ and the randomness in the new response Y_0 .

For an observed data set, we often compute an empirical version of MSE. Given observations $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$ and predictions $\hat{y}_i = \hat{f}(\mathbf{x}_i)$, the empirical MSE is

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (2.3)$$

Large errors are penalized heavily because the errors are squared. This makes MSE mathematically convenient, but it also means that a few large mistakes can dominate the score.

The square root of MSE is called *root mean squared error*, or RMSE:

$$\text{RMSE} = \sqrt{\text{MSE}}. \quad (2.4)$$

RMSE is often easier to interpret because it is measured in the same units as the response. For example, if Y is measured in hours, RMSE is also measured in hours; if Y is measured in thousands of dollars, RMSE is measured in thousands of dollars.

Math Foundations Connection. Let $\mathbf{y} = (y_1, \dots, y_n)^T$ be the vector of observed responses and let $\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_n)^T$ be the vector of predictions. The residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}.$$

Using the Euclidean norm,

$$\|\mathbf{e}\| = \sqrt{e_1^2 + \dots + e_n^2},$$

the empirical MSE can be written as

$$\text{MSE} = \frac{1}{n} \|\mathbf{y} - \hat{\mathbf{y}}\|^2. \quad (2.5)$$

Thus MSE is a scaled squared distance between the observed response vector and the fitted response vector. This geometric view will become especially useful when we later interpret least squares regression as a projection problem. See the Math Foundations textbook, Chapters 2, 3, 5, and 12.

2.3 The Bias–Variance Decomposition

The bias–variance decomposition explains why prediction is limited even when we use a good model. It separates expected prediction error into three parts:

$$\text{MSE} = \text{bias}^2 + \text{model variance} + \text{noise}. \quad (2.6)$$

This equation is one of the central ideas in statistical learning.

2.3.1 Set-Up for the Decomposition

Fix a predictor value $\mathbf{x}_0$. Suppose the new response is generated by

$$Y_0 = f_0(\mathbf{x}_0) + \epsilon_0, \quad (2.7)$$

where

$$\begin{aligned}\mathbb{E}[\epsilon_0] &= 0, \\ \text{Var}(\epsilon_0) &= \sigma^2,\end{aligned}$$

with ϵ_0 independent of the training data used to construct $\hat{f}$. We want the expected squared prediction error

$$\mathbb{E} \left[\left(Y_0 - \hat{f}(\mathbf{x}_0) \right)^2 \right].$$

Substitute Equation (2.7):

$$\mathbb{E} \left[\left(Y_0 - \hat{f}(\mathbf{x}_0) \right)^2 \right] = \mathbb{E} \left[\left(f_0(\mathbf{x}_0) + \epsilon_0 - \hat{f}(\mathbf{x}_0) \right)^2 \right]. \quad (2.8)$$

Rearrange the expression inside the square:

$$f_0(\mathbf{x}_0) + \epsilon_0 - \hat{f}(\mathbf{x}_0) = \left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0) \right) + \epsilon_0.$$

Now expand the square:

$$\begin{aligned}\mathbb{E} \left[\left(\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0) \right) + \epsilon_0 \right)^2 \right] \\ = \mathbb{E} \left[\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0) \right)^2 \right] + 2\mathbb{E} \left[\epsilon_0 \left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0) \right) \right] + \mathbb{E}[\epsilon_0^2].\end{aligned}$$

The middle term is zero because ϵ_0 has mean zero and is independent of the fitted model. Also, since $\mathbb{E}[\epsilon_0] = 0$, we have $\mathbb{E}[\epsilon_0^2] = \text{Var}(\epsilon_0) = \sigma^2$. Therefore,

$$\mathbb{E} \left[\left(Y_0 - \hat{f}(\mathbf{x}_0) \right)^2 \right] = \mathbb{E} \left[\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0) \right)^2 \right] + \sigma^2. \quad (2.9)$$

The final term, σ^2 , is the noise variance. It is the variation in Y that remains even if f_0 were known perfectly.

2.3.2 Separating Bias and Variance

We now decompose

$$\mathbb{E} \left[\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0) \right)^2 \right].$$

Let

$$m(\mathbf{x}_0) = \mathbb{E} \left[\hat{f}(\mathbf{x}_0) \right]$$

be the average prediction we would get at $\mathbf{x}_0$ over many possible training data sets. Add and subtract $m(\mathbf{x}_0)$ inside the square:

$$\hat{f}(\mathbf{x}_0) - f_0(\mathbf{x}_0) = \left(\hat{f}(\mathbf{x}_0) - m(\mathbf{x}_0) \right) + \left(m(\mathbf{x}_0) - f_0(\mathbf{x}_0) \right).$$

Squaring and taking expectations gives

$$\begin{aligned}\mathbb{E} \left[\left(\hat{f}(\mathbf{x}_0) - f_0(\mathbf{x}_0) \right)^2 \right] &= \mathbb{E} \left[\left(\hat{f}(\mathbf{x}_0) - m(\mathbf{x}_0) \right)^2 \right] + (m(\mathbf{x}_0) - f_0(\mathbf{x}_0))^2 \\ &\quad + 2(m(\mathbf{x}_0) - f_0(\mathbf{x}_0)) \mathbb{E} \left[\hat{f}(\mathbf{x}_0) - m(\mathbf{x}_0) \right].\end{aligned}$$

The final expectation is zero by the definition of $m(\mathbf{x}_0)$, so the cross term vanishes. Hence,

$$\mathbb{E} \left[\left(\hat{f}(\mathbf{x}_0) - f_0(\mathbf{x}_0) \right)^2 \right] = \text{Var} \left(\hat{f}(\mathbf{x}_0) \right) + \text{Bias} \left(\hat{f}(\mathbf{x}_0) \right)^2, \quad (2.10)$$

where

$$\text{Var} \left(\hat{f}(\mathbf{x}_0) \right) = \mathbb{E} \left[\left(\hat{f}(\mathbf{x}_0) - \mathbb{E}[\hat{f}(\mathbf{x}_0)] \right)^2 \right], \quad (2.11)$$

$$\text{Bias} \left(\hat{f}(\mathbf{x}_0) \right) = \mathbb{E} \left[\hat{f}(\mathbf{x}_0) \right] - f_0(\mathbf{x}_0). \quad (2.12)$$

Combining Equations (2.9) and (2.10), we obtain

$$\boxed{\text{MSE}(\mathbf{x}_0) = \text{Bias} \left(\hat{f}(\mathbf{x}_0) \right)^2 + \text{Var} \left(\hat{f}(\mathbf{x}_0) \right) + \text{Var}(\epsilon_0).} \quad (2.13)$$

This is the bias–variance decomposition.

2.3.3 Interpreting the Three Terms

The *squared bias* term measures systematic error. It is large when the average fitted model misses the true function. A model that is too simple can have high bias. For example, if the true relationship is curved but we insist on using a straight line, the fitted line may miss the main shape of the data no matter how much data we collect.

The *model variance* term measures how much the fitted model changes from one training data set to another. A highly flexible model may fit one training sample very closely but produce a noticeably different function if the sample changes. This instability is model variance.

The *noise* term, $\text{Var}(\epsilon_0)$, is irreducible error. It is the part of the response that cannot be predicted from $\mathbf{X}$ using any method, because it comes from unmeasured factors, measurement error, randomness in the system, or other variation not captured by the inputs.

This distinction gives us the language of *reducible* and *irreducible* error. Bias and model variance are reducible in principle: we can try different model classes, collect better data, improve features, or tune the model. The noise term is irreducible with respect to the inputs we have chosen. The best possible prediction error approaches this irreducible error from above; it cannot go below it when the goal is to predict the actual future response Y .

2.4 Training Error and Test Error

Prediction models are trained on data. Let

$$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_{n_{\text{train}}}, y_{n_{\text{train}}})\}$$

be the training data used to fit $\hat{f}$. The *training MSE* is

$$\text{MSE}_{\text{train}} = \frac{1}{n_{\text{train}}} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{train}}} \left(y_i - \hat{f}(\mathbf{x}_i) \right)^2. \quad (2.14)$$

Training MSE measures how well the model performs on data it already used to learn.

The prediction problem, however, is not mainly about the training data. If we train a model using data from existing diabetes patients, we already know which of those patients had diabetes. The real goal is to make accurate predictions for future patients whose outcomes are not yet known. In the same way, a logistics forecast, readiness forecast, or maintenance forecast is useful only if it generalizes beyond the cases already used to build it.

Let

$$\mathcal{D}_{\text{test}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_{n_{\text{test}}}, y_{n_{\text{test}}})\}$$

be a separate set of observations not used to fit the model. The *test MSE* is

$$\text{MSE}_{\text{test}} = \frac{1}{n_{\text{test}}} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{test}}} \left(y_i - \hat{f}(\mathbf{x}_i) \right)^2. \quad (2.15)$$

Test MSE measures generalization: the model's ability to predict unseen data.

Training MSE is usually optimistic. Many fitting procedures explicitly choose model parameters to reduce training error. Therefore, a low training MSE does not guarantee a low test MSE. The model may simply have learned the idiosyncrasies of the training sample.

2.5 Flexibility

A model's *flexibility* is its capacity to approximate a wide range of functional shapes. A model is more flexible if it can represent a larger set of possible relationships $f(\mathbf{x})$.

For example, a straight-line model in one predictor can represent only lines:

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x.$$

A polynomial model with many terms can represent many more shapes:

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 + \dots + \hat{\beta}_d x^d.$$

A smoothing spline, decision tree, random forest, boosted model, support vector machine, or neural network may be more flexible still, depending on how it is tuned.

Flexibility is not automatically good or bad. Its value depends on the relationship between the true function, the amount of noise, and the amount of data available. More flexible methods can reduce bias because they can capture more complex structure. At the same time, more flexible methods tend to increase model variance because they are more sensitive to the particular training sample.

As a general rule:

$$\text{more flexibility} \implies \text{lower bias but higher variance.} \quad (2.16)$$

This is only a general rule, not a law that applies identically in every model class. It is nevertheless a useful guide for thinking about prediction error.

2.5.1 Degrees of Freedom as a Flexibility Measure

For some models, flexibility can be summarized by a quantity called *degrees of freedom*. In smoothing splines, for example, a low-degree-of-freedom fit is smoother and less flexible, while a high-degree-of-freedom fit is more wiggly and more flexible. In polynomial regression, the number of estimated parameters provides a simple proxy for flexibility. A model with two parameters is less flexible than a model with thirteen parameters.

The important pattern is this: as flexibility increases, training MSE usually decreases, because the model has more freedom to chase the training observations. Test MSE often decreases at first, but then may increase when the model becomes too flexible.

2.6 Overfitting and Overtraining

Overfitting occurs when a model follows noise or accidental patterns in the training data rather than the underlying relationship. The model looks impressive on the data used to fit it, but performs poorly on new observations. In the lecture slides, this is also described as *overtraining*: an overabundance of flexibility causes test MSE to increase compared with simpler models.

Overfitting is undesirable because prediction is about future or unseen data. A flexible model may fit the training data so well that it learns artifacts specific to that sample. These artifacts do not appear in future data, so test performance suffers.

A useful mental picture is the following. Suppose the true relationship is a smooth curve. A very simple model may underfit: it misses important curvature and has high bias. A moderately flexible model may capture the main trend and generalize well. A very flexible model may twist through individual noisy observations; it has low training error but high variance and worse test error.

Therefore, the goal is not to make training MSE as small as possible. The goal is to choose a model whose predictions generalize.

2.7 Train/Test Splits

A train/test split is a simple way to estimate generalization performance.

1. Randomly divide the available data into a training set and a test set.
2. Fit the model using only the training set.
3. Predict the responses in the test set.
4. Compute test MSE or test RMSE using the test-set predictions.
5. Compare candidate models using test performance, not training performance alone.

The key rule is that the test data must remain unseen during model fitting. If the test data are used to choose transformations, tune flexibility, select predictors, or adjust the model, then they are no longer a clean test of generalization. Later in the course we will use cross-validation to improve on a single train/test split, but the principle is the same: separate the data used to fit the model from the data used to evaluate the model.

2.7.1 A Simulation Example

The lecture simulation uses the model

$$Y = f_0(X) + \epsilon, \quad (2.17)$$

where

$$\begin{aligned} f_0(X) &= X^3, \\ X &\sim \text{Uniform}(-3, 3), \\ \epsilon &\sim \mathcal{N}(0, 9^2). \end{aligned}$$

The true function is cubic, but the observed data are noisy. A train/test split separates the simulated observations into data used to fit the model and data used to evaluate it.

Consider four candidate models with increasing numbers of parameters. The lecture example reports the following pattern:

Model	Number of Parameters	Training MSE	Test MSE
1	2	101.99	111.82
2	4	84.56	91.15
3	9	83.52	90.60
4	13	82.46	92.55

The training MSE decreases as the number of parameters increases. This is expected: more parameters give the fitted model more freedom to reduce training error. The test MSE, however, decreases at first and then increases. The 13-parameter model has the lowest training MSE, but not the lowest test MSE. In this example, the 9-parameter model gives the best test performance among the four candidates.

The lesson is not that nine parameters are always best. The lesson is that model selection is a balance between flexibility and test error. A model that is too rigid underfits; a model that is too flexible overfits. The best predictive model is usually somewhere between these extremes.

2.8 Parametric and Nonparametric Estimators

Another way to describe model classes is to distinguish *parametric* and *nonparametric* methods.

2.8.1 Parametric Methods

A method is parametric if the function f is described using a finite number of parameters. The model may be simple or complicated, linear or nonlinear, but it is governed by a finite parameter vector.

The standard linear model is parametric:

$$f(\mathbf{X}) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = \beta_0 + \boldsymbol{\beta}^T \mathbf{X}, \quad (2.18)$$

where

$$\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T.$$

Once we assume the form in Equation (2.18), learning f becomes the problem of estimating the finite list of parameters $\beta_0, \beta_1, \dots, \beta_p$.

Math Foundations Connection. The term $\beta^T \mathbf{X}$ is an inner product. It is also a linear combination of the entries of $\mathbf{X}$ with coefficients $\beta_1, \dots, \beta_p$. This is why linear models connect naturally to the vector notation from Math Foundations: the model turns a feature vector into a prediction by weighting its components and adding the weighted components together. See the Math Foundations textbook, Chapters 2, 3, and 5.

Parametric does not mean “good” or “simple.” For example, one could write a finite-parameter model such as

$$f(\mathbf{X}) = \beta_0 + \beta_{13}X_1 + \beta_2X_2 + \beta_4\frac{X_3X_4}{\beta_3} + \dots + \cos(\beta_pX_p).$$

This is still parametric because it is controlled by a finite number of parameters, but it is not necessarily sensible, interpretable, or easy to fit. Parametric modeling is powerful because it reduces an infinite function-estimation problem to a finite parameter-estimation problem. Its weakness is that the chosen form may be wrong.

2.8.2 Nonparametric Methods

A method is nonparametric if it does not impose a fixed finite-dimensional form on f in advance. Nonparametric methods attempt to learn the shape of f more directly from the data. They can be deceptively simple or highly complex.

The advantage of a nonparametric method is flexibility. If the true relationship is highly nonlinear, a nonparametric method may capture structure that a linear model misses. The disadvantage is that flexibility requires data. Without enough observations, a flexible method may not be able to distinguish signal from noise. This is another expression of the bias–variance tradeoff: nonparametric methods often reduce bias but may increase variance, especially when the sample size is small.

This course will use nonparametric methods, but the early chapters focus on linear models because linear models make the statistical logic visible. Understanding prediction error, bias, variance, residuals, and validation in simple models prepares us to use more complex models responsibly.

2.9 Model Flexibility and Interpretability

Different models occupy different positions on the flexibility–interpretability spectrum. Least squares linear regression is relatively inflexible but highly interpretable. Subset selection and lasso remain close to the linear-model family but modify which predictors are used or how coefficients are estimated. Generalized additive models and trees are more flexible. Bagging, boosting, support vector machines, and deep learning methods are often more flexible still, but their fitted relationships may be harder to interpret directly.

For prediction, increased flexibility can be valuable. Modern prediction problems often involve nonlinear patterns, interactions, and high-dimensional structure that a basic linear model cannot capture. For pure prediction tasks, linear models are often best treated as baselines rather than as the final word.

For inference, linear models remain extremely important. Their assumptions, estimators, uncertainty measures, and diagnostic tools are well-developed. When the goal is to explain relationships, defend assumptions, and communicate uncertainty to a decision maker, the interpretability of a linear model can be a major advantage.

The practical takeaway is not that one class of models is universally best. The task determines the model. If the goal is prediction, prioritize generalization error. If the goal is inference, prioritize interpretability, assumptions, and uncertainty quantification. When the goal includes both, state the tradeoff explicitly.

Later chapters return to this tradeoff when choosing predictor sets, shrinking coefficients with regularization, reducing dimension, and using neural networks as flexible prediction models.

2.10 Chapter Summary

This chapter developed the statistical meaning of prediction error. Mean squared error measures average squared prediction error. Root mean squared error places the score back in the units of the response. The bias–variance decomposition shows that expected prediction error consists of squared bias, model variance, and irreducible noise.

Bias is systematic error caused by the model missing the true relationship on average. Model variance is instability caused by sensitivity to the training data. Noise is irreducible error in the response itself. More flexible models tend to reduce bias but increase variance, which is why more flexibility does not automatically mean better prediction.

Training MSE measures performance on data used to fit the model. Test MSE measures performance on unseen data. Because prediction is about generalization, test error is the more important quantity. Train/test splits provide a first method for estimating generalization performance.

Finally, the chapter distinguished parametric and nonparametric estimators. Parametric methods assume a finite-parameter structure for f , while nonparametric methods avoid imposing a fixed finite-dimensional form. Both can be useful, but both must be evaluated by whether they support the goal of the analysis.

Chapter 3

Data Quality, Cleaning, and Trustworthy Variables

3.1 Why Data Quality Comes Before Modeling

Before we build a statistical model, compute a regression line, or compare prediction errors, we need to know whether the variables in the data set can be trusted. Exploratory data analysis begins with the habit of “listening to the data.” That means stepping back from the final question we want to answer and first asking what the data are trying to tell us about themselves.

Data cleaning is not a clerical step that happens before the real analysis. It is part of the analysis. A model can only learn from the variables we give it. If those variables are mislabeled, inconsistently recorded, stored in the wrong type, or contaminated by impossible values, then the model may produce output that looks mathematical but is not analytically defensible.

This chapter focuses on the practical question:

What must be true about our variables before we trust them enough to summarize, visualize, or model them?

A trustworthy variable is one whose meaning, type, units, allowable values, and missing-value behavior are understood well enough that another analyst could reproduce the cleaning decision and defend it to a customer, commander, or reviewer. The goal is not to make the data look perfect. The goal is to make the workflow repeatable and the decisions defensible.

3.2 Data Tables, Observations, and Variables

A data set is often organized as a rectangular table. The rows are observations, and the columns are variables. If there are n observations and p variables, we can represent the data as a matrix

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbb{R}^{n \times p} \quad (3.1)$$

when all entries are numeric. The entry x_{ij} is the value of variable j for observation i .

The i th row,

$$\mathbf{x}_i^T = (x_{i1}, x_{i2}, \dots, x_{ip}),$$

contains all measured information for observation i . The j th column,

$$\mathbf{x}_j = (x_{1j}, x_{2j}, \dots, x_{nj})^T,$$

contains all observed values of variable j .

This matrix notation is useful, but it can also hide an important fact: not every variable is genuinely numeric. Some columns may contain categories, ordered labels, dates, text, identifiers, or codes. A column made of numbers is not automatically a numerical variable in the statistical sense.

Math Foundations Connection. In linear algebra, a vector is a collection of entries that pertain to the same object or concept. In data analysis, a numeric variable can be viewed as a vector of observed values, and a full numeric data table can be viewed as a matrix whose columns are variable vectors. This perspective helps later when we discuss design matrices, regression, and projections. For cleaning, it helps us remember that operations applied to a column should make sense for the type of object that column represents. See the Math Foundations textbook, Chapters 2, 4, and 7.

3.3 Variable Types

A variable is an attribute that may change its value between observations in a data set. The type of a variable determines which summaries, plots, transformations, and models are meaningful.

Variable type	Meaning	Natural summaries
Continuous numeric	Values can be thought of as varying across a range, rather than only at isolated counts. The observed minimum and maximum are sample bounds, not part of the definition.	mean, median, spread, histogram
Discrete numeric	Values are usually counts or integer-valued measurements.	counts, mean, spread, bar plot or histogram
Ordinal	Categories have an order, but distances between categories may not be meaningful.	counts, proportions, median, ordered bar plot
Categorical	Categories are labels without a natural ordering.	counts, proportions, mode, bar plot

Examples of continuous numeric variables include age, temperature, height, time to complete a task, and distance traveled. Examples of discrete numeric variables include number of missions completed, number of defects found during inspection, and number of officers in a unit. Enlisted rank is an ordinal variable: the order matters, but the numerical distance from one rank to the next is not usually treated as constant. Service branch, blood type, platform type, and property view are categorical variables.

A common trap is treating numeric codes as quantitative measurements. A survey might record service branch as Army = 1, Marine Corps = 2, Navy = 3, Air Force = 4, Space Force = 5, and Coast Guard = 6. Those numbers are convenient labels, not measurements. The average of Army

and Navy is not Marine Corps, and the value 6 is not larger than the value 1 in a quantitative sense. Later, categorical variables can be encoded numerically for modeling, but those encodings are modeling tools rather than evidence that the original category has arithmetic meaning.

3.4 Why Data Become Messy

Data can be messy for many reasons. Some problems are obvious, such as impossible values. Others are subtle, such as inconsistent units or category codes that look numeric. Common sources of data problems include data-entry errors, non-standardized labels, missing or incomplete data, inconsistent survey responses, inconsistent units or scales, duplicate records, impossible values, incorrect data types, dates stored as text, categorical variables stored as integers, and numeric variables stored as strings.

The analyst’s job is not merely to find these issues. The analyst must decide what to do about them and document the decision well enough that someone else could follow the same logic.

3.5 The Data Cleaning Workflow

Data cleaning is the process of identifying, documenting, and correcting data-quality issues before analysis. A useful first-pass workflow is:

1. identify and handle missing values,
2. fix data types,
3. rename and standardize columns,
4. subset rows and columns as needed,
5. detect outliers,
6. conduct basic sanity checks.

This sequence is not strictly linear. Fixing a data type may reveal new missing values; detecting an outlier may reveal a unit mismatch; a sanity check may send you back to the original data source. The workflow is iterative.

A cleaning decision should satisfy two standards. First, it should be *repeatable*: the same code plus the same raw data should produce the same cleaned data. Second, it should be *defensible*: the analyst should be able to explain why the decision was reasonable for the variable, the data source, and the purpose of the analysis.

A minimal cleaning log should record the variable, the issue, the decision, and enough detail for another analyst to reproduce the choice. For example, “`weight_lb`: values less than or equal to 0 are physically impossible for the study population; recoded as missing.” The log does not need to be elaborate, but it should prevent the cleaning process from becoming a set of undocumented manual edits.

3.6 Missing Values and Imputation

Missing values occur when an observation should have a value for a variable but the recorded data do not contain that value. Missingness can be explicit, such as `NA`, `NaN`, blank cells, or `NULL`. Missingness can also be implicit, such as a value of 0 used as a placeholder for an impossible measurement.

For a data matrix, define a missingness indicator

$$m_{ij} = \begin{cases} 1, & \text{if } x_{ij} \text{ is missing,} \\ 0, & \text{if } x_{ij} \text{ is observed.} \end{cases} \quad (3.2)$$

Then the number of missing values in variable j is

$$M_j = \sum_{i=1}^n m_{ij}, \quad (3.3)$$

and the missingness rate is

$$r_j = \frac{M_j}{n}. \quad (3.4)$$

These simple counts identify which variables have missingness and how severe it is.

One of the most important missing-data checks is distinguishing *true zeros* from *missing zeros*. A true zero means the value is genuinely zero, such as `sibling.count = 0`. A missing zero means zero was used as a placeholder for a missing or invalid value, such as `weight_lb = 0` for an adult human. Missingness cannot be detected by a universal rule such as “replace all zeros.” The rule must depend on the variable definition.

Once missing values are identified, the analyst usually has three choices: drop, impute, or flag.

Action	When it may be reasonable	Main risk
Drop rows	Missingness is limited, the variables are required, and the complete rows are still representative.	Can remove useful data or bias the analysis.
Drop columns	A variable is unnecessary, too incomplete, unreliable, or poorly defined.	May discard a useful signal for a later analysis.
Impute	A missing value can be replaced using a defensible rule or model.	Can create false precision or distort distributions and relationships.
Flag missingness	Missingness itself may be informative.	Requires careful interpretation in summaries and models.

For complete-case analysis, if J is the set of variables required for a particular analysis, the retained row set is

$$I_{\text{complete}} = \{i : m_{ij} = 0 \text{ for all } j \in J\}. \quad (3.5)$$

The cleaned analysis table is the submatrix containing rows in I_{complete} and columns in J .

Imputation replaces missing values with reasonable substitutes. If x_{ij} is missing, an imputed value is often written as $\hat{x}_{ij}$, giving

$$x_{ij}^{\text{clean}} = \begin{cases} x_{ij}, & \text{if } x_{ij} \text{ is observed,} \\ \hat{x}_{ij}, & \text{if } x_{ij} \text{ is missing.} \end{cases} \quad (3.6)$$

Simple imputation uses only the variable with missingness. For a numeric variable X_j , mean imputation replaces missing values with

$$\hat{x}_{ij} = \bar{x}_{j,\text{obs}} = \frac{1}{n - M_j} \sum_{i:m_{ij}=0} x_{ij}. \quad (3.7)$$

Median, mode, constant, or explicit-category imputation may also be used when appropriate. Conditional imputation uses other variables to make better guesses, such as regression imputation, k -nearest-neighbor imputation, predictive mean matching, or other model-based methods. Multiple imputation creates several plausible completed data sets, analyzes each, and combines the results.

The key caution is that imputation is an assumption, not a fact. Avoid imputing automatically when “did not answer” is itself meaningful, when the variable is not needed, when missingness is too extensive, when imputation would introduce obvious bias, when the missing value represents a choice rather than a failed measurement, or when the data source is too unreliable to support replacement.

3.7 Fixing Data Types and Standardizing Variables

A software data type is how software stores and treats a variable. A statistical variable type is what the variable means analytically. Cleaning often requires aligning both.

Common data-type problems include numeric columns stored as strings, categorical variables stored as integers, dates stored as text, binary variables stored as text labels, identifiers stored as numbers, and numbers with commas, units, or symbols stored as text. The practical cleaning question is:

Does the software representation support the analysis we intend to perform?

Problem	Examples	Cleaning check
Numeric stored as text	1,250, 72 in, unknown, --	Parse valid numbers; decide whether nonnumeric entries are missingness, labels, or errors.
Category stored as integer	Service branch coded 1–6	Ask whether arithmetic on the codes is meaningful. If not, treat as categorical or ordinal.
Date stored as text	04/03/2026, 2026-04-03, 3 Apr 26	Parse to date format, then validate impossible dates, unexpected future dates, and time-zone issues.
Identifier stored as number	ID, hull number, survey response number	Preserve identity; do not average or model as a quantitative variable unless explicitly meaningful.

Column names are also part of the analysis. A useful naming convention should be consistent, descriptive, short enough to use in code, free of spaces and special characters, and stable across related data sets. Including units in the name can prevent later confusion: `height_in` and `height_cm` are safer than simply `height`.

Subsetting means selecting only part of a data set. Let $I \subseteq \{1, 2, \dots, n\}$ be a set of row indices and let $J \subseteq \{1, 2, \dots, p\}$ be a set of column indices. The submatrix

$$\mathbf{X}_{I,J}$$

keeps rows in I and columns in J . For example, if $x_{i,\text{age}}$ is the age for observation i , then

$$I_{\text{adult}} = \{i : x_{i,\text{age}} \geq 18\} \quad (3.8)$$

keeps adult observations. Column subsets are usually defined by the analysis purpose, such as keeping the response, the intended predictors, and identifiers needed for traceability.

Math Foundations Connection. Subsetting is a set operation. The row set I identifies which observations remain. The column set J identifies which variables remain. Thinking in sets helps avoid vague language such as “we removed bad rows.” A defensible cleaning rule should be expressible as a clear predicate, such as “remove rows where weight is missing or impossible.” See the Math Foundations textbook, Chapters 2 and 4.

3.8 Outliers and Sanity Checks

An outlier is an observation that is unusually far from the rest of the data or does not follow the overall pattern. Outliers matter because they can strongly affect summaries, visualizations, and fitted models. They should be examined before computing final summaries or fitting final models.

Common visual tools include histograms, boxplots, and scatterplots. A histogram can reveal extreme values, unusual gaps, skewness, or multiple modes. A boxplot can reveal observations outside the central bulk of the data. A scatterplot can reveal points that do not follow a bivariate pattern.

Outliers are not automatically errors. An outlier may be a data-entry error, a unit-conversion error, a duplicate or merged-record problem, a rare but valid observation, evidence of subgroups, or evidence that the model or summary is too simple. Removing a valid extreme observation just because it is inconvenient can bias the analysis.

A common numerical rule for outlier flagging uses the interquartile range. Let Q_1 be the first quartile, Q_3 be the third quartile, and

$$\text{IQR} = Q_3 - Q_1. \quad (3.9)$$

Tukey fences flag values outside

$$[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]. \quad (3.10)$$

This rule is a flag, not an automatic deletion command.

Sanity checks ask whether the data make sense before we treat them as evidence. One useful warning is:

Be wary of taking data at face value, especially when you do not know how it was generated.

At minimum, ask:

- Are units consistent, or might a variable mix inches with centimeters, miles with kilometers, or dollars with thousands of dollars?
- Are there impossible values, such as negative ages, probabilities outside $[0, 1]$, counts below zero, or physically impossible measurements?

- Does the number of rows match the expected row unit: one person, one transaction, one unit, one day, or one person-day?
- Are duplicates expected or suspicious?
- Are categorical levels spelled consistently?
- Do minimums, maximums, histograms, and summary statistics agree with subject-matter expectations?

Impossible values are not always easy to repair. Sometimes the correct value can be recovered from documentation or another field. Sometimes the value should be treated as missing. Sometimes the entire record should be excluded from a specific analysis. The decision should be documented.

3.9 A Trustworthy Variable Checklist

Before using a variable in a summary, plot, or model, ask the following questions:

1. **Definition:** What does the variable measure, and what is the row unit associated with the measurement?
2. **Type:** Is the variable continuous, discrete, ordinal, categorical, date/time, text, or an identifier? Does the software type match the statistical meaning?
3. **Units and scale:** Are units consistent? Are values raw, standardized, logged, percentages, or rates?
4. **Missingness:** What values indicate missingness? Are placeholders such as 0, 999, -1, or blanks being used? Is missingness itself meaningful?
5. **Validity:** What values are possible, impossible, or unusual but possible? Are there outliers, duplicates, or inconsistent labels?
6. **Defensibility:** Can the cleaning decision be explained in plain language, implemented in code, and documented in a cleaning log?

This checklist is intentionally practical. A variable is not trustworthy merely because it has numbers in a column. It becomes trustworthy when its meaning and cleaning decisions are clear enough to reproduce and defend.

3.10 Realities of Data Cleaning

Real data-driven studies often involve very large data sets. In an ideal world, errors would be fixed at the data-entry point. In practice, analysts often receive data after many upstream choices have already been made. We fix what we can, document what we cannot fix, and avoid pretending that the data are cleaner than they are.

Some issues are easier to spot than others. Numerical problems may be visible through summary statistics and visual EDA. Categorical problems may require careful inspection of labels and codes. Missingness may require knowledge of how the data were collected. Credibility issues may require comparing the data against external expectations or subject-matter knowledge.

Errors can also be informative. If a toy example contains several obvious errors, those errors may be a warning about the credibility of the data source. The right conclusion may not be “clean harder.” It may be “find a better data set.”

3.11 Representative Mini-Examples

Two common cleaning traps illustrate the chapter’s main ideas.

Categorical codes are not quantities. Suppose a survey records service branch using integer codes. A careless analysis might treat the column as numeric and compute an average branch code. That summary is meaningless. The defensible cleaning decision is to recode the integers to meaningful labels and treat the variable as categorical.

Zero may mean different things in different variables. A value of `sibling_count = 0` may be a valid true zero. A value of `weight_lb = 0` for an adult is likely missing or invalid. The same recorded number has different meanings because the variables have different definitions. Cleaning rules must be variable-specific.

3.12 Chapter Summary

Data cleaning is the foundation for trustworthy data analysis. Before modeling, the analyst must understand what each variable means, what type it is, how it is stored, what values are possible, and how missingness is represented.

The main ideas from this chapter are:

- Exploratory data analysis begins by listening to the data before asking targeted modeling questions.
- Variables may be continuous, discrete, ordinal, or categorical.
- Numeric storage does not guarantee numeric meaning.
- Missing values must be detected, counted, interpreted, and handled deliberately.
- True zeros must be distinguished from placeholder zeros.
- Dropping data is simple but can remove information or introduce bias.
- Imputation can preserve rows but introduces assumptions.
- Data types, column names, row subsets, and outliers must be checked systematically.
- Sanity checks protect the analyst from taking unreliable data at face value.
- Cleaning decisions should be repeatable and defensible.

The next chapter builds on this foundation by using numerical summaries and visual structure to describe cleaned data. Once variables are trustworthy enough to inspect, we can ask what their distributions, relationships, and patterns reveal.

Chapter 4

Exploratory Data Analysis: Summaries and Visual Structure

4.1 Why Summaries and Visual Structure Matter

The previous chapter focused on whether variables are trustworthy enough to analyze. Once we have checked definitions, missingness, units, data types, impossible values, and outliers, we can begin describing what the data appear to say.

Exploratory data analysis, or EDA, is the habit of listening to data before forcing it into a model. This chapter develops two complementary tools:

1. **single-number summary statistics**, which compress one feature of a distribution into one number; and
2. **visual EDA**, which uses plots to reveal distributional shape, relationships, subgroups, and unusual structure.

Good EDA combines both. Numbers make comparisons precise; plots make structure visible. The practical question for this chapter is:

How do we summarize and visualize variables in a way that is accurate, interpretable, and defensible?

4.2 A Numeric Variable as a Vector

Let $x_1, x_2, \dots, x_n$ be the observed values of a numeric variable X for n observations. We can collect these values in a vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n. \quad (4.1)$$

This notation is useful because most numerical summaries are functions of the entries of $\mathbf{x}$.

For example, the sample mean can be written as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (4.2)$$

Using the ones vector $\mathbf{1} \in \mathbb{R}^n$, the same quantity is

$$\bar{x} = \frac{1}{n} \mathbf{1}^T \mathbf{x}. \quad (4.3)$$

This linear algebra view is not required for basic EDA, but it helps us see that summaries are not magic. They are operations applied to the observed data vector.

Math Foundations Connection. A vector is an ordered collection of quantities that pertain to the same object or concept. In EDA, a variable column is a vector of observed values. Dot products, sums, centered vectors, and norms give compact ways to express means, variances, covariances, and correlations. See the Math Foundations textbook, Chapters 2, 3, and 5.

4.3 Single-Number Summary Statistics

A single-number summary statistic characterizes one aspect of a variable's distribution. It does not describe everything about the variable. It answers a specific question.

Summary family	Guiding question	Examples
Center	Where is a typical value?	mean, median, mode
Spread	How variable are the values?	range, IQR, variance, standard deviation
Shape	How is the distribution arranged?	skewness, kurtosis
Extremes	What are the edge cases?	minimum, maximum, outlier flags

Table 4.1: Single-number summaries answer different questions about a variable.

The most common EDA mistake is to compute a statistic without saying what question it answers. The same data set may require several summaries: a center for a typical value, a spread for variability, a shape summary for asymmetry or tails, and an extreme-value check for unusual observations.

4.4 Center and Spread

4.4.1 Measures of Center

Measures of center summarize where a distribution is located. The three most common are the mean, median, and mode.

The **mean** is the balance point:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (4.4)$$

It uses every value, which makes it natural for many numerical variables, but also sensitive to outliers.

The **median** is the middle value after sorting the data. If

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$$

denotes the ordered data, then the median is the middle ordered value for odd n and the average of the two middle ordered values for even n . The median is robust to outliers because changing an extreme value usually does not move the middle of the ordered list.

The **mode** is the most common value. It is especially useful for categorical variables. For continuous numerical variables, exact ties may be rare, so analysts often discuss peaks of a histogram or density estimate rather than an exact modal value.

Situation	Useful center summary	Reason
Symmetric numeric variable	Mean or median	Often similar
Skewed or outlier-prone numerical variable	Median	More robust to tail values
Categorical variable	Mode	Categories do not have arithmetic averages
Ordinal variable	Median or mode	Respects ordering without assuming equal spacing

4.4.2 Measures of Spread

Measures of spread summarize variability. Two variables can have the same center but very different spread.

The **range** is

$$\text{Range}(\mathbf{x}) = \max_i x_i - \min_i x_i. \quad (4.5)$$

It is easy to interpret but depends only on the two most extreme observations.

The **interquartile range** is

$$\text{IQR} = Q_3 - Q_1, \quad (4.6)$$

where Q_1 is the first quartile and Q_3 is the third quartile. Because it measures the spread of the middle half of the data, the IQR is more robust to outliers than the range.

The **sample variance** and **sample standard deviation** are

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad (4.7)$$

$$s_x = \sqrt{s_x^2}. \quad (4.8)$$

Variance is mathematically powerful but is measured in squared units. Standard deviation is often easier to interpret because it is measured in the original units of the variable.

Define the centered data vector

$$\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}. \quad (4.9)$$

Then

$$s_x^2 = \frac{1}{n-1} \tilde{\mathbf{x}}^T \tilde{\mathbf{x}}. \quad (4.10)$$

Thus spread can be interpreted geometrically: large spread means the centered data vector has large squared length.

4.5 Shape and Extremes

Measures of shape describe how a distribution is arranged beyond its center and spread. The two shape ideas emphasized here are **skewness** and **kurtosis**.

For a random variable X with mean μ and standard deviation σ , the population skewness is

$$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}. \quad (4.11)$$

Positive skewness indicates a long or heavy right tail; negative skewness indicates a long or heavy left tail. In a right-skewed distribution, the mean is often larger than the median because the right tail pulls the mean upward.

The population kurtosis is

$$\gamma_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}. \quad (4.12)$$

For EDA, it is safest to interpret kurtosis as tail weight rather than simply as “peakedness.” The normal distribution has kurtosis 3, so excess kurtosis is $\gamma_2 - 3$.

Robust analogs. Mean, variance, skewness, and kurtosis are moment-based summaries; they use arithmetic powers of deviations from the mean. Median, IQR, Bowley skewness, and Moors kurtosis are quantile-based analogs. Quantile-based summaries are often more robust when outliers or heavy tails are present. For this course, the main habit is more important than memorizing every robust formula: when outliers or skewness are visible, compare moment-based summaries with quantile-based summaries.

The simplest measures of extremes are the minimum and maximum. Extremes are not automatically bad: they may be rare but valid observations, data-entry errors, unit mistakes, duplicated records, or evidence of subgroups.

A common outlier flag uses Tukey fences. Values are flagged when they fall outside

$$[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]. \quad (4.13)$$

This rule is tied to the box plot. It is a screening tool, not an automatic deletion rule. A defensible action is: flag the value, inspect the context, and document the decision.

4.6 Visual EDA

Numerical summaries compress. Plots reveal. Visual EDA uses graphics to show the behavior of observations directly. A plot should match the variable types involved.

4.6.1 Univariate Visual EDA

Univariate EDA examines one variable at a time. For categorical or ordinal variables, a **bar plot** displays counts or proportions by category. It helps reveal common levels, rare levels, unexpected labels, and ordering problems.

For numerical variables, a **histogram** groups observations into bins and displays the number or proportion of observations in each bin. A histogram helps inspect symmetry, skewness, modality,

gaps, clusters, outliers, and approximate bell shape. Because histograms depend on bin width, it is good practice to inspect more than one bin choice before making a strong claim about shape.

A histogram may be unimodal, bimodal, or multimodal. Multiple peaks often suggest subgroups, different data sources, or different operating conditions. The next EDA step is usually to color, facet, or group the data by a relevant categorical variable.

4.6.2 Bivariate Visual EDA

Bivariate EDA examines the relationship between two variables. The plot depends on the type of each variable.

Variable 1	Variable 2	Recommended plot
Categorical / ordinal	Categorical / ordinal	Grouped bar plot
Categorical / ordinal	Numerical	Grouped box plot
Numerical	Categorical / ordinal	Grouped box plot
Numerical	Numerical	Scatterplot

Table 4.2: Recommended bivariate EDA plots by variable-type pairing.

Grouped bar plots compare category combinations, especially counts or proportions. Grouped box plots compare numerical distributions across categories, including medians, IQRs, skewness, and outlier flags. Scatterplots compare two numerical variables and are often the first place we notice whether a linear model is a reasonable first approximation. A visible curve, fan shape, cluster pattern, or outlier should be noted before modeling.

4.6.3 Multivariate Visual EDA

Multivariate EDA examines more than two variables. True high-dimensional visualization is difficult, so practical multivariate EDA often uses combinations of pairwise views.

A **scatterplot matrix** displays many pairwise relationships at once. If there are p numerical variables, there are

$$\frac{p(p-1)}{2} \tag{4.14}$$

unique pairwise scatterplots. The diagonal entries are often histograms or density plots for the individual variables.

A **correlation heatmap** displays many pairwise correlations compactly. It can reveal blocks of related variables, strongly positive or negative relationships, possible multicollinearity before regression, and variables that appear weakly related to all others. A heatmap is not truly multivariate in the sense of showing all variables simultaneously; it summarizes many pairwise linear relationships.

4.7 Applications of Standard Deviation

Standard deviation becomes especially useful when the distribution is approximately bell-shaped. If X is approximately normal with mean μ and standard deviation σ , then approximately

$$\mathbb{P}(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68, \quad (4.15)$$

$$\mathbb{P}(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95, \quad (4.16)$$

$$\mathbb{P}(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997. \quad (4.17)$$

This is the **empirical rule**. It is an approximation, not a universal law. It should be used only when the histogram is reasonably bell-shaped or when a normal model is otherwise justified.

When the data are not approximately normal, Chebyshev's inequality provides a more general lower bound. For a random variable X with finite mean μ and finite nonzero variance σ^2 , for any $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}. \quad (4.18)$$

Equivalently, for $k > 1$,

$$\mathbb{P}(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}. \quad (4.19)$$

Window	Empirical rule, if bell-shaped	Chebyshev lower bound
Within 1σ	about 68%	no useful positive bound
Within 2σ	about 95%	at least 75%
Within 3σ	about 99.7%	at least 88.9%

The practical takeaway is simple: if the histogram looks bell-shaped, the empirical rule is more informative; if the shape is unknown or non-normal, Chebyshev is safer but more conservative.

4.8 Covariance, Correlation, and Heatmaps

For two numeric variables X and Y , covariance measures whether they tend to move together. The population covariance is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (4.20)$$

The sample covariance is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}). \quad (4.21)$$

Covariance depends on units, which makes it hard to compare across variable pairs. Correlation rescales covariance into the interval $[-1, 1]$:

$$r_{xy} = \frac{s_{xy}}{s_x s_y}, \quad (4.22)$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}. \quad (4.23)$$

Let

$$\begin{aligned}\tilde{\mathbf{x}} &= \mathbf{x} - \bar{x}\mathbf{1}, \\ \tilde{\mathbf{y}} &= \mathbf{y} - \bar{y}\mathbf{1}.\end{aligned}$$

Then the sample correlation can be written as

$$r_{xy} = \frac{\tilde{\mathbf{x}}^T \tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\| \|\tilde{\mathbf{y}}\|}. \quad (4.24)$$

This is the cosine of the angle between the centered data vectors. If they point in nearly the same direction, correlation is positive and close to 1. If they point in nearly opposite directions, correlation is negative and close to -1. If they are nearly orthogonal, correlation is near 0. See the Math Foundations textbook, Ch05 Sec. 5.8.

Important caution. Correlation measures linear association. A correlation near 0 does not prove that two variables are unrelated. The relationship may be curved, segmented, or driven by subgroups. Always pair correlation with a scatterplot when possible.

If $\mathbf{X} \in \mathbb{R}^{n \times p}$ contains p numerical variables, then the sample correlation matrix is

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & r_{pp} \end{bmatrix}, \quad (4.25)$$

where r_{jk} is the sample correlation between variables j and k . For nonconstant variables, the diagonal entries satisfy $r_{jj} = 1$; constant-variable correlations are undefined. The matrix is symmetric because $r_{jk} = r_{kj}$.

4.9 A Practical EDA Workflow

A reusable EDA workflow is:

1. Confirm the variable type and check missingness.
2. Compute center, spread, shape, and extreme summaries for key numerical variables.
3. Make a univariate plot and compare the plot to the summaries.
4. For pairs of variables, choose the plot based on the two variable types.
5. For many numerical variables, inspect a scatterplot matrix or correlation heatmap.
6. Record unusual patterns, possible explanations, and follow-up questions.

The goal is not to produce every possible plot. The goal is to learn enough about the data to make the next modeling or analytic step more defensible.

Coding companion reference. Package-specific Python/R commands for EDA summaries, grouped plots, and heatmaps have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the conceptual EDA workflow and interpretation guidance.

4.10 Compact EDA Example

Suppose `pft_score` is a numerical variable. A first EDA pass might compute the mean, median, standard deviation, IQR, minimum, and maximum, then create a histogram. If the mean and median are close and the histogram is roughly symmetric, the empirical rule may be a reasonable approximation. If the histogram is skewed or multimodal, the median and IQR may better represent the typical value and spread.

If `score` is numerical and `gender` is categorical, a grouped box plot compares distributions across groups. Do not stop at comparing medians. Inspect IQRs, outlier flags, skewness within each group, group sample sizes, and overlap between groups. A grouped box plot can reveal a pattern, but it does not explain why the pattern exists.

Finally, suppose a medical data set contains variables such as `triceps`, `insulin`, and `bmi`. If zeros are impossible or implausible for those measurements, then a correlation heatmap before cleaning may be misleading because placeholder zeros create artificial relationships. Recode impossible zeros as missing, document the decision, and create summaries and plots after cleaning.

4.11 Reading EDA Output Defensibly

EDA should produce observations that are specific, cautious, and tied to evidence.

Weak statement	Better EDA statement
The data are normal.	The histogram is roughly bell-shaped and symmetric, so the empirical rule may be reasonable.
There are outliers.	Tukey fences flag three high values; check units and entry errors before removal.
These variables are related.	The scatterplot shows a positive, roughly linear association; the sample correlation is positive.
Group A is better than Group B.	Group A has a higher median and smaller IQR in this sample; this plot alone does not establish causation.
The heatmap proves multicollinearity.	The heatmap suggests strong pairwise correlations; formal modeling diagnostics can check multicollinearity later.

A defensible EDA statement distinguishes what was observed from what was inferred.

4.12 Reusable EDA Checklist

Use this compact checklist when exploring a cleaned data set:

- Identify each variable type: continuous, discrete, ordinal, categorical, date, text, or identifier.
- Check missingness, minimum, maximum, impossible values, and suspicious zeros.
- Choose center and spread summaries that match the variable and distribution.
- Inspect shape: symmetry, skewness, modality, tails, gaps, clusters, and outlier flags.

- For pairs of variables, choose the plot by variable types and look for linear, curved, grouped, or outlier-driven patterns.
- For many numerical variables, use scatterplot matrices or heatmaps to decide which relationships deserve closer inspection.
- Document the evidence, the follow-up question, and the limitation of each claim.

4.13 Chapter Summary

The main ideas from this chapter are:

- Single-number summaries describe center, spread, shape, and extremes.
- The mean is a balance point but is sensitive to outliers; the median and IQR are more robust.
- Skewness measures asymmetry; kurtosis emphasizes tail weight.
- Tukey fences flag possible outliers but do not justify automatic deletion.
- Histograms show univariate numerical structure, including skewness, modality, and approximate bell shape.
- The empirical rule is useful for approximately Gaussian distributions; Chebyshev's inequality gives a conservative bound when normality is not justified.
- Bivariate plots should be selected based on the two variable types.
- Scatterplot matrices and correlation heatmaps summarize many pairwise relationships.
- Correlation is a scaled covariance and can be interpreted geometrically as the cosine between centered data vectors.
- EDA claims should be specific, cautious, and tied to the actual summary or plot that supports them.

The next chapter begins simple linear regression. The EDA habits from this chapter will matter immediately: before fitting a line, we should understand the response distribution, inspect the predictor, check the scatterplot, and decide whether a linear relationship is a reasonable first approximation.

Chapter 5

Simple Linear Regression: Model, Assumptions, and Least Squares Estimation

5.1 The Big Picture

The previous chapters built the workflow needed before formal modeling: define trustworthy variables, clean the data, summarize distributions, and visualize relationships. This chapter begins the formal regression sequence.

The central question is:

Given one input variable and one quantitative response variable, what is the equation of the best line through the data?

Simple linear regression, or SLR, is the first model we use to answer that question. It predicts a quantitative response Y using a single predictor X . The word *simple* means one predictor. The word *linear* means that the conditional mean of Y is modeled as a straight-line function of X .

SLR is important for three reasons.

1. It is the cleanest place to see the difference between a true model and an estimated model.
2. It gives explicit formulas for the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$.
3. It introduces residuals, normal equations, and model assumptions that appear throughout the rest of regression.

5.2 Review: From Statistical Learning to Linear Regression

In the statistical learning framework, we assume that a response variable Y is related to predictors through

$$Y = f(\mathbf{X}) + \epsilon, \tag{5.1}$$

where f is the systematic part of the relationship and ϵ is random error or noise.

If $\mathbb{E}[\epsilon \mid \mathbf{X} = \mathbf{x}] = 0$, then

$$\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \mathbb{E}[f(\mathbf{X}) + \epsilon \mid \mathbf{X} = \mathbf{x}] \quad (5.2)$$

$$= f(\mathbf{x}) + \mathbb{E}[\epsilon \mid \mathbf{X} = \mathbf{x}] \quad (5.3)$$

$$= f(\mathbf{x}). \quad (5.4)$$

Thus, under the mean-zero error condition, $f(\mathbf{x})$ is the conditional mean of Y given the input values. It is the best mean-squared-error predictor of Y given the information in $\mathbf{X}$.

In simple linear regression, there is only one predictor. We replace the general unknown function $f(X)$ with a parametric straight-line model:

$$f(X) = \beta_0 + \beta_1 X. \quad (5.5)$$

The modeling assumption becomes

$$Y = \beta_0 + \beta_1 X + \epsilon. \quad (5.6)$$

The unknown parameters are:

- β_0 , the intercept;
- β_1 , the slope.

Once we estimate these parameters from data, our fitted regression function is

$$\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X. \quad (5.7)$$

In data analysis language, we say that we *regress* Y *onto* X .

5.3 Data Notation

Suppose we observe n paired data points:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n). \quad (5.8)$$

Here:

- x_i is the observed predictor value for observation i ;
- y_i is the observed response value for observation i .

It is often useful to collect the observations in vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}. \quad (5.9)$$

The fitted response vector is

$$\hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}, \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \quad (5.10)$$

The residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}, \quad e_i = y_i - \hat{y}_i. \quad (5.11)$$

Math Foundations Connection. A variable column is a vector in $\mathbb{R}^n$; see the Math Foundations textbook, Chapters 2, 3, 5, and 12. The dot product and Euclidean norm give compact ways to write sums of products and sums of squared deviations. This is why least squares can be written both as a calculus problem and as a geometry problem.

5.4 Predicted Values, Residuals, MSE, and RSS

For any proposed line

$$\hat{y} = b_0 + b_1x, \quad (5.12)$$

where b_0 and b_1 are candidate values, the residual for observation i is

$$e_i(b_0, b_1) = y_i - (b_0 + b_1x_i). \quad (5.13)$$

The empirical mean squared error is

$$\text{MSE}(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1x_i)]^2. \quad (5.14)$$

The residual sum of squares is

$$\text{RSS}(b_0, b_1) = \sum_{i=1}^n [y_i - (b_0 + b_1x_i)]^2. \quad (5.15)$$

Because $\text{MSE}(b_0, b_1) = \text{RSS}(b_0, b_1)/n$, minimizing MSE and minimizing RSS give the same fitted line.

The least squares estimators are the values of b_0 and b_1 that minimize the empirical mean squared error:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} \frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1x_i)]^2. \quad (5.16)$$

5.5 The Big Three Regression Assumptions

The source lectures present these as three increasingly strong assumptions. We keep that same three-step ladder throughout the regression chapters.

The model

$$Y = \beta_0 + \beta_1X + \epsilon \quad (5.17)$$

can be paired with three assumption levels. These levels matter because they determine what claims we are allowed to make.

5.5.1 Assumption 1: Mean-zero errors

The first assumption is

$$\mathbb{E}[\epsilon \mid X = x] = 0 \quad \text{for every } x. \quad (5.18)$$

Then

$$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x. \quad (5.19)$$

In simple linear regression, this means that the line is the conditional mean of Y given $X = x$. In this setting, the line is a statement about average behavior, not about every individual observation.

5.5.2 Assumption 2: Constant variance and uncorrelated errors

The second assumption is stated after conditioning on the observed predictor values. In other words, we treat the predictor values as fixed and study the remaining randomness in the errors:

$$\mathbb{E}[\epsilon_i \mid X_i = x_i] = 0, \quad (5.20)$$

$$\text{Var}(\epsilon_i \mid X_i = x_i) = \sigma^2 \quad \text{for all } i, \quad (5.21)$$

$$\text{Cov}(\epsilon_i, \epsilon_j \mid X_1 = x_1, \dots, X_n = x_n) = 0 \quad \text{for } i \neq j. \quad (5.22)$$

The second condition is constant error variance, also called homoskedasticity. The third condition says that, after the predictor values are fixed, errors are uncorrelated across observations.

These assumptions support the classical standard error formulas and hypothesis tests that appear later.

5.5.3 Assumption 3: Gaussian errors

The third assumption is the Gaussian version of the same fixed-predictor idea:

$$\epsilon \mid X_1 = x_1, \dots, X_n = x_n \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n). \quad (5.23)$$

Equivalently, conditional on the observed predictor values, the errors are independent normal random variables with mean zero and common variance σ^2 . This Gaussian error model supports exact finite-sample inference using t and F distributions. This assumption is powerful, but it should not be made casually. Residual diagnostics later in the course help us check whether it is plausible.

Assumption level	What it says	What it supports
Mean-zero errors	$\mathbb{E}[\epsilon \mid X = x] = 0$	Line estimates the conditional mean
Constant variance and uncorrelated errors	Conditional mean zero, constant variance, uncorrelated errors after fixing predictor values	Standard errors and approximate inference
Gaussian errors	Conditional independent normal errors with common variance	Exact classical inference under the model

Table 5.1: The three-step regression assumption ladder for simple linear regression.

5.6 Deriving the Least Squares Estimators

We now derive the formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ by minimizing the empirical MSE.

Define the objective function

$$Q(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2. \quad (5.24)$$

Let

$$r_i(b_0, b_1) = y_i - (b_0 + b_1 x_i). \quad (5.25)$$

Then

$$Q(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n r_i(b_0, b_1)^2. \quad (5.26)$$

5.6.1 Partial Derivative with Respect to b_0

Using the chain rule,

$$\frac{\partial Q}{\partial b_0} = \frac{1}{n} \sum_{i=1}^n 2r_i(b_0, b_1) \frac{\partial r_i}{\partial b_0} \quad (5.27)$$

$$= \frac{1}{n} \sum_{i=1}^n 2r_i(b_0, b_1)(-1) \quad (5.28)$$

$$= -\frac{2}{n} \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]. \quad (5.29)$$

Setting this derivative equal to zero gives

$$\frac{1}{n} \sum_{i=1}^n [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)] = 0. \quad (5.30)$$

5.6.2 Partial Derivative with Respect to b_1

Similarly,

$$\frac{\partial Q}{\partial b_1} = \frac{1}{n} \sum_{i=1}^n 2r_i(b_0, b_1) \frac{\partial r_i}{\partial b_1} \quad (5.31)$$

$$= \frac{1}{n} \sum_{i=1}^n 2r_i(b_0, b_1)(-x_i) \quad (5.32)$$

$$= -\frac{2}{n} \sum_{i=1}^n x_i [y_i - (b_0 + b_1 x_i)]. \quad (5.33)$$

Setting this derivative equal to zero gives

$$\frac{1}{n} \sum_{i=1}^n x_i [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)] = 0. \quad (5.34)$$

5.6.3 The Normal Equations

Equations (5.30) and (5.34) are called the normal equations. They can be written in a compact residual form:

$$\sum_{i=1}^n e_i = 0, \quad (5.35)$$

$$\sum_{i=1}^n x_i e_i = 0. \quad (5.36)$$

The first equation says that least squares residuals sum to zero. The second says that least squares residuals have zero dot product with the predictor vector.

Using sample averages, define

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad (5.37)$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i, \quad (5.38)$$

$$\overline{x^2} = \frac{1}{n} \sum_{i=1}^n x_i^2, \quad (5.39)$$

$$\overline{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i. \quad (5.40)$$

Then the normal equations become

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0, \quad (5.41)$$

$$\overline{xy} - \hat{\beta}_0 \bar{x} - \hat{\beta}_1 \overline{x^2} = 0. \quad (5.42)$$

5.6.4 Solving for $\hat{\beta}_1$

From (5.41),

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (5.43)$$

Substitute this into (5.42):

$$\overline{xy} - (\bar{y} - \hat{\beta}_1 \bar{x}) \bar{x} - \hat{\beta}_1 \overline{x^2} = 0 \quad (5.44)$$

$$\overline{xy} - \bar{x} \bar{y} + \hat{\beta}_1 \bar{x}^2 - \hat{\beta}_1 \overline{x^2} = 0. \quad (5.45)$$

Rearranging gives

$$\hat{\beta}_1 (\overline{x^2} - \bar{x}^2) = \overline{xy} - \bar{x} \bar{y}. \quad (5.46)$$

Therefore,

$$\boxed{\hat{\beta}_1 = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2}.} \quad (5.47)$$

Then

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.} \quad (5.48)$$

5.7 Equivalent Centered Form

The slope formula in (5.47) can be written in a more interpretable centered form. First,

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) \quad (5.49)$$

$$= \overline{xy} - \bar{x}\bar{y} - \bar{x}\bar{y} + \bar{x}\bar{y} \quad (5.50)$$

$$= \overline{xy} - \bar{x}\bar{y}. \quad (5.51)$$

Similarly,

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \overline{x^2} - \bar{x}^2. \quad (5.52)$$

Thus,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}. \quad (5.53)$$

This is often the most useful formula for interpreting the slope.

5.8 Slope as Covariance Divided by Variance

The unbiased sample covariance of x and y is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad (5.54)$$

while the unbiased sample variance of x is

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (5.55)$$

Because the same denominator $n-1$ appears in both quantities, it cancels in the ratio. Therefore,

$$\boxed{\hat{\beta}_1 = \frac{s_{xy}}{s_x^2}}. \quad (5.56)$$

Equivalently, if we use biased denominator- n versions,

$$\hat{\sigma}_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad (5.57)$$

$$\hat{\sigma}_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad (5.58)$$

then

$$\hat{\beta}_1 = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x^2}. \quad (5.59)$$

For the slope, the biased-versus-unbiased denominator choice does not change the answer because the denominator cancels.

5.8.1 Interpretation of the Ratio

The numerator measures how x and y move together. The denominator measures how much x itself varies. Thus,

The least squares slope is the co-movement of x and y , scaled by the variation in x .

If x and y tend to move together, then $s_{xy} > 0$ and the slope is positive. If larger x values tend to pair with smaller y values, then $s_{xy} < 0$ and the slope is negative. If x and y have little linear co-movement, then s_{xy} is near zero and the slope is near zero.

5.9 When the Slope Is Not Identified

The formula

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (5.60)$$

requires

$$\sum_{i=1}^n (x_i - \bar{x})^2 > 0. \quad (5.61)$$

This condition means that the observed x values are not all identical.

If every observation has the same predictor value, then there is no information about how Y changes when X changes. The intercept may still describe the mean response at that one X value, but the slope cannot be estimated from the data. This is the simple-regression version of identifiability.

5.10 Why the Critical Point Is a Minimum

The calculus derivation found a critical point. We also need to know that this critical point minimizes the objective. Let

$$Q(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2. \quad (5.62)$$

This is a sum of squared functions of b_0 and b_1 . Therefore, it is a convex quadratic function.

The Hessian matrix is

$$\nabla^2 Q(b_0, b_1) = \frac{2}{n} \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix}. \quad (5.63)$$

This matrix is positive definite when the x_i values are not all identical. Therefore, the critical point is the unique global minimizer.

5.11 Matrix and Geometry View

Math Foundations Connection. The matrix notation below uses the design matrix, transposes, matrix-vector multiplication, dot products, norms, and orthogonality. These supporting concepts are covered in the Math Foundations textbook, Chapters 5 and 7.

Although this chapter derives the estimators using calculus, the same result has a clean linear algebra form. Define the design matrix

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}. \quad (5.64)$$

Then the fitted vector is

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad (5.65)$$

and the least squares objective can be written as

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2. \quad (5.66)$$

Using the dot product definition of the Euclidean norm,

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \quad (5.67)$$

Taking derivatives in matrix form gives

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \quad (5.68)$$

Equivalently,

$$\boxed{\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}.} \quad (5.69)$$

These are the normal equations in matrix form.

The vector equation

$$\mathbf{X}^T\mathbf{e} = \mathbf{0} \quad (5.70)$$

says that the residual vector is orthogonal to both columns of $\mathbf{X}$: the ones column and the predictor column. This is the geometric meaning of least squares. The fitted vector is the point in the line-generated model space that is closest to the observed response vector.

Bridge to Later Chapters. In the next regression geometry chapters, we will generalize this same idea: least squares projects $\mathbf{y}$ onto the subspace spanned by the columns of the design matrix.

5.12 Interpreting $\hat{\beta}_0$ and $\hat{\beta}_1$

The fitted simple linear regression model is

$$\hat{\mathbb{E}}[Y \mid X = x] = \hat{\beta}_0 + \hat{\beta}_1 x. \quad (5.71)$$

5.12.1 Intercept

The intercept $\hat{\beta}_0$ estimates the expected value of Y when $X = 0$:

$$\hat{\beta}_0 \approx \mathbb{E}[Y \mid X = 0]. \quad (5.72)$$

This interpretation is meaningful only when $X = 0$ is within the scope of the data and the model.

For example, if X is years of experience, then $X = 0$ may be meaningful. If X is adult height measured in inches, then $X = 0$ is not meaningful in the observed population. In that case, the intercept is still mathematically necessary for the line, but it should not be given a literal operational interpretation.

5.12.2 Slope

The slope $\hat{\beta}_1$ estimates the change in the conditional mean of Y for a one-unit increase in X :

$$\hat{\beta}_1 \approx \mathbb{E}[Y \mid X = x + 1] - \mathbb{E}[Y \mid X = x]. \quad (5.73)$$

When the linear model is appropriate, this change is constant across x .

The units of the slope are

$$\frac{\text{units of } Y}{\text{units of } X}. \quad (5.74)$$

For example, if Y is test score and X is hours studied, then the slope is measured in test-score points per hour studied.

5.13 Association Is Not Causation

The slope is a statement about association unless the study design and assumptions justify a causal interpretation.

Suppose X is the height of mercury in a thermometer and Y is ambient temperature. A fitted regression may show a strong positive slope. But if we artificially raise the mercury level by one unit, we should not expect the ambient temperature to rise. The mercury height measures temperature; it does not cause temperature.

Before interpreting a slope causally, ask:

1. Was X randomly assigned or experimentally controlled?
2. Could another variable cause both X and Y ?
3. Is there a plausible mechanism from X to Y ?
4. Is the interpretation inside the range of observed X values?
5. Would changing X actually change Y , or only change how we measure Y ?

In this course, unless a causal design has been established, interpret regression slopes as associational.

5.14 Variance and Covariance Notation

The notation difference between variance and covariance is worth understanding.

For a single variable X , variance is a second central moment:

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2]. \quad (5.75)$$

It has squared units. If X is measured in meters, then $\text{Var}(X)$ is measured in square meters.

For two variables X and Y , covariance is a cross-moment:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (5.76)$$

It has product units. If X is measured in meters and Y is measured in seconds, then covariance is measured in meter-seconds.

This is why the sample variance is written with a square,

$$s_x^2, \quad (5.77)$$

while sample covariance is written with two subscripts,

$$s_{xy}. \quad (5.78)$$

The square in s_x^2 signals the squared unit structure of a variance. The subscripts in s_{xy} signal co-movement between two variables.

5.15 Biased and Unbiased Variance Estimators

Two common sample variance estimators are

$$\hat{\sigma}_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad (5.79)$$

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (5.80)$$

The first uses denominator n and is biased for the population variance in the usual independent-sample setting. The second uses denominator $n-1$ and is unbiased.

The unbiased estimator corrects the expected value, but it can have higher variance. This is a concrete example of the bias-variance tradeoff. Sometimes we intentionally accept bias to reduce variance or improve prediction performance.

Goal	Common choice	Reason
Classical inference	Unbiased estimators such as s_x^2	Supports standard inferential formulas
Maximum likelihood under normality	Denominator- n estimator	Matches the likelihood optimizer
Prediction-focused modeling	Sometimes biased estimators	May reduce MSE
Regularized regression	Intentionally biased estimates	Bias can reduce variance

Table 5.2: Biased estimators are not automatically bad; the choice depends on the objective.

For simple linear regression slope estimation, the denominator choice cancels if covariance and variance are computed consistently:

$$\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x^2}. \quad (5.81)$$

5.16 Worked Example: Computing the Least Squares Line by Hand

Consider three observations:

x_i	y_i
1	2
2	3
3	5

First compute the sample means:

$$\bar{x} = \frac{1+2+3}{3} = 2, \quad \bar{y} = \frac{2+3+5}{3} = \frac{10}{3}. \quad (5.82)$$

Next compute the centered numerator:

$$\sum_{i=1}^3 (x_i - \bar{x})(y_i - \bar{y}) = (1-2)\left(2 - \frac{10}{3}\right) + (2-2)\left(3 - \frac{10}{3}\right) + (3-2)\left(5 - \frac{10}{3}\right) \quad (5.83)$$

$$= (-1)\left(-\frac{4}{3}\right) + 0\left(-\frac{1}{3}\right) + (1)\left(\frac{5}{3}\right) \quad (5.84)$$

$$= 3. \quad (5.85)$$

Then compute the centered denominator:

$$\sum_{i=1}^3 (x_i - \bar{x})^2 = (-1)^2 + 0^2 + 1^2 = 2. \quad (5.86)$$

Thus,

$$\hat{\beta}_1 = \frac{3}{2} = 1.5. \quad (5.87)$$

The intercept is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{10}{3} - 1.5(2) = \frac{1}{3}. \quad (5.88)$$

The least squares line is

$$\hat{y} = \frac{1}{3} + 1.5x. \quad (5.89)$$

The fitted values and residuals are:

i	x_i	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$
1	1	2	11/6	1/6
2	2	3	10/3	-1/3
3	3	5	29/6	1/6

Check the normal equations:

$$\sum_{i=1}^3 e_i = \frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0, \quad (5.90)$$

$$\sum_{i=1}^3 x_i e_i = 1\left(\frac{1}{6}\right) + 2\left(-\frac{1}{3}\right) + 3\left(\frac{1}{6}\right) = 0. \quad (5.91)$$

The residuals are orthogonal to both the intercept column and the predictor column.

5.17 A Practical Interpretation Checklist

When interpreting a fitted SLR model,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x, \quad (5.92)$$

work through the following checklist.

1. **Response:** What is Y , and what are its units?
2. **Predictor:** What is X , and what are its units?
3. **Intercept:** Is $X = 0$ meaningful and inside the data range?
4. **Slope:** What does “one unit of X ” mean operationally?
5. **Association:** Does the slope describe association or causation?
6. **Range:** Are predictions being made inside the observed X range?
7. **Residuals:** Are residuals small enough for the model to be useful?
8. **Assumptions:** Which assumption set is needed for the claim being made?

Coding companion reference. Package-specific Python/R commands for fitting simple linear regression and producing quick residual plots have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the derivation, interpretation checklist, and common mistakes.

5.18 Common Mistakes

1. **Dropping the parentheses in the residual.** The residual is $y_i - (b_0 + b_1 x_i)$, not $y_i - b_0 + b_1 x_i$.
2. **Treating the intercept as always meaningful.** The intercept is only meaningful when $X = 0$ is meaningful in context.
3. **Calling the slope causal without design support.** Regression slopes are usually associational.
4. **Ignoring units.** The slope has units of response per predictor unit.
5. **Extrapolating outside the data range.** A fitted line may behave badly outside the observed X values.
6. **Confusing residuals with true errors.** Residuals estimate errors, but they are not the unobserved errors themselves.
7. **Forgetting the denominator condition.** If all x_i values are the same, the slope cannot be estimated.

5.19 Chapter Summary

Simple linear regression estimates the conditional mean of a quantitative response using one predictor. The model is

$$Y = \beta_0 + \beta_1 X + \epsilon. \quad (5.93)$$

The least squares estimators minimize the empirical mean squared error, equivalently the residual sum of squares. Calculus gives the normal equations, and solving them yields

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2}, \quad (5.94)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (5.95)$$

The slope is the sample covariance of x and y scaled by the sample variance of x . The intercept anchors the line through the sample means. The residuals sum to zero and are orthogonal to the predictor column.

The main interpretive discipline is this: least squares gives the best line under a squared-error criterion, but the usefulness of that line depends on modeling assumptions, residual behavior, variable meaning, and whether the analyst is making an associational or causal claim.

5.20 Practice and Workflow

Focused Study Checklist

Use this as a final check after working a simple linear regression problem.

1. Can you write the model $Y = \beta_0 + \beta_1 X + \epsilon$ and state which rung of the Big Three assumptions the task requires?
2. Can you compute $\hat{\beta}_1$ and $\hat{\beta}_0$ from the centered-sum formulas?
3. Can you write the fitted line and compute residuals $e_i = y_i - \hat{y}_i$?
4. Can you interpret the slope with units and avoid over-interpreting the intercept?
5. Can you use $\sum e_i = 0$ and $\sum x_i e_i = 0$ as algebra checks rather than as evidence that the model is correct?
6. Can you separate association from causation in your conclusion?

Chapter 6

Simple Linear Regression: Fit Quality, R^2 , and Hypothesis Testing

6.1 The Big Picture

The previous chapter built the simple linear regression model and derived the least squares line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x. \quad (6.1)$$

The next question is not how to find the line. The next question is:

Once we have a fitted line, how good is it, and is there statistical evidence that the predictor is useful?

This chapter introduces two related but different ideas.

1. **Fit quality:** How much variation in the response does the fitted line explain? This is measured by R^2 .
2. **Statistical evidence:** Is the fitted relationship distinguishable from a zero-slope model after accounting for sample size and noise? This is tested with the F -test or, equivalently in simple linear regression, the slope t -test.

The difference matters. A model can have a high R^2 and still be based on too few observations to trust. A model can have a statistically significant slope but still explain little practical variation. Good analysis reports both fit quality and inferential evidence.

6.2 Review: The Mean-Only Baseline Model

Before using a predictor, we can always estimate a quantitative response Y by its sample mean:

$$\hat{y}_i = \bar{y}, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i. \quad (6.2)$$

This is the *mean-only model*. It ignores all predictor values.

The prediction error for observation i under the mean-only model is

$$y_i - \bar{y}. \quad (6.3)$$

The mean squared error of the mean-only model is

$$\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2. \quad (6.4)$$

This is the denominator- n version of sample variance. Thus, for one-dimensional data, evaluating the mean-only estimate by MSE is essentially evaluating variance.

Math Foundations Connection. A response column $(y_1, \dots, y_n)^T$ is a vector in $\mathbb{R}^n$, and the mean-only fitted vector can be written as $\bar{y}\mathbf{1}$. The sums of squared deviations below are squared Euclidean norms; see the Math Foundations textbook, Chapters 2, 3, and 5.

6.3 Adding a Predictor

When we observe paired data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n), \quad (6.5)$$

we can compare two strategies:

1. predict all responses using $\bar{y}$;
2. predict responses using the fitted regression line

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \quad (6.6)$$

If X is useful, then the fitted line should have smaller squared prediction error than the mean-only model. If X is useless, then the fitted line will look like a horizontal line with slope near zero, and it will not improve much over the mean-only model.

This is the core intuition behind R^2 and the F -test:

Compare the error from the mean-only model to the error from the fitted regression model.

6.4 Three Sums of Squares

Let

$$e_i = y_i - \hat{y}_i \quad (6.7)$$

be the residual from the fitted regression line.

6.4.1 Total Sum of Squares

The *total sum of squares* measures how much the response varies around its sample mean:

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2. \quad (6.8)$$

This is the squared error from the mean-only model.

6.4.2 Residual Sum of Squares

The *residual sum of squares* measures how much variation remains after fitting the regression line:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2. \quad (6.9)$$

This is the squared error from the fitted regression model.

6.4.3 Regression Sum of Squares

The *regression sum of squares*, sometimes called explained sum of squares, measures how much variation the model explains relative to the mean-only baseline:

$$\text{RegSS} = \text{TSS} - \text{RSS}. \quad (6.10)$$

When the regression model includes an intercept, least squares gives the decomposition

$$\boxed{\text{TSS} = \text{RegSS} + \text{RSS}.} \quad (6.11)$$

Math Foundations Connection. The decomposition in (6.11) is a geometry statement. It relies on the residual vector being orthogonal to the fitted-value direction after centering. Dot products, norms, and orthogonality are covered in the Math Foundations textbook, Chapters 2, 3, and 5.

6.5 R^2 : Proportion of Variation Explained

The coefficient of determination is

$$\boxed{R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}.} \quad (6.12)$$

The numerator $\text{TSS} - \text{RSS}$ is the reduction in squared error achieved by using the predictor. The denominator TSS is the squared error we had before using the predictor.

Thus,

R^2 is the proportion of the response variation explained by the regression model.

If $R^2 = 0$, then the fitted model does no better than the mean-only model. If $R^2 = 1$, then the residuals are all zero and the fitted line passes exactly through every observed point.

For simple linear regression with an intercept,

$$0 \leq R^2 \leq 1. \quad (6.13)$$

This interval property depends on comparing an intercept-containing regression model to the mean-only baseline. Some nonstandard models, such as no-intercept regressions, can produce different behavior.

6.6 Interpreting R^2

Suppose two models are used to predict pediatric weight.

- Model A uses wrist circumference and has $R^2 = 0.25$.
- Model B uses height and has $R^2 = 0.50$.

Then Model B explains twice as much of the observed weight variation as Model A, at least in the data used to fit the models.

The phrase *explains variation* should be read carefully. It does not automatically imply causation. It means that the fitted line reduces squared prediction error relative to the mean-only model.

6.7 R^2 and Correlation in Simple Linear Regression

For simple linear regression with an intercept, R^2 is equal to the squared sample correlation between x and y :

$$\boxed{R^2 = r_{xy}^2}. \quad (6.14)$$

To see why, define centered values

$$x_i^c = x_i - \bar{x}, \quad y_i^c = y_i - \bar{y}. \quad (6.15)$$

Let

$$S_{xx} = \sum_{i=1}^n (x_i^c)^2, \quad S_{yy} = \sum_{i=1}^n (y_i^c)^2, \quad S_{xy} = \sum_{i=1}^n x_i^c y_i^c. \quad (6.16)$$

The least squares slope is

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}. \quad (6.17)$$

The centered fitted values satisfy

$$\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x}). \quad (6.18)$$

Therefore, the regression sum of squares is

$$\text{RegSS} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \quad (6.19)$$

$$= \sum_{i=1}^n \hat{\beta}_1^2 (x_i - \bar{x})^2 \quad (6.20)$$

$$= \hat{\beta}_1^2 S_{xx} \quad (6.21)$$

$$= \left(\frac{S_{xy}}{S_{xx}} \right)^2 S_{xx} \quad (6.22)$$

$$= \frac{S_{xy}^2}{S_{xx}}. \quad (6.23)$$

Thus,

$$R^2 = \frac{\text{RegSS}}{\text{TSS}} \quad (6.24)$$

$$= \frac{S_{xy}^2 / S_{xx}}{S_{yy}} \quad (6.25)$$

$$= \frac{S_{xy}^2}{S_{xx} S_{yy}} \quad (6.26)$$

$$= \left(\frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \right)^2 \quad (6.27)$$

$$= r_{xy}^2. \quad (6.28)$$

This equality is useful in SLR. It does not directly generalize to multiple linear regression as a single ordinary correlation between one predictor and the response.

6.8 Limitations of R^2

R^2 is useful, but it does not tell us everything.

6.8.1 A Perfect Fit Can Be Misleading

With two data points and a line with two parameters, we can fit the data perfectly as long as the two x values are different. Then

$$\text{RSS} = 0 \quad \Rightarrow \quad R^2 = 1. \quad (6.29)$$

This does not mean the model will predict future data well. It means only that a line can pass through two points.

6.8.2 R^2 Does Not Measure Statistical Evidence

R^2 measures in-sample fit. It does not directly answer:

- Is the slope distinguishable from zero?
- How much sampling uncertainty is in $\hat{\beta}_1$?
- How noisy are the observations?
- How many observations were used?

These questions require hypothesis tests, confidence intervals, and residual diagnostics.

6.8.3 R^2 Does Not Check Model Assumptions

A high R^2 does not prove that the linear model assumptions are valid. Residual plots may still reveal curvature, nonconstant variance, outliers, or dependence among observations.

6.9 From Fit Quality to Hypothesis Testing

Fit quality asks:

How much did the line reduce squared error compared to the mean-only model?

Hypothesis testing asks:

Is that reduction large enough, relative to residual noise and sample size, to count as statistical evidence that the slope is not zero?

For SLR, the inferential question is usually

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0. \quad (6.30)$$

The null hypothesis says that X does not improve the conditional mean of Y through a linear slope. The alternative says that the slope is nonzero.

Math Foundations Connection. Hypotheses are mathematical statements, and hypothesis testing depends on understanding logical alternatives and negation. Basic logical statements, negation, and truth-table reasoning are covered in the Math Foundations textbook, Chapter 1.

6.10 The F -Statistic

The F -statistic compares the mean-only model to the fitted regression model.

For a general nested-model comparison,

$$F = \frac{(\text{SS}_{\text{null}} - \text{SS}_{\text{full}}) / (p_{\text{full}} - p_{\text{null}})}{\text{SS}_{\text{full}} / (n - p_{\text{full}})}, \quad (6.31)$$

where:

- SS_{null} is the squared error from the simpler model;

- SS_{full} is the squared error from the larger model;
- p_{null} is the number of parameters in the simpler model;
- p_{full} is the number of parameters in the larger model.

For simple linear regression, the null model is the mean-only model, so

$$p_{\text{null}} = 1, \quad SS_{\text{null}} = \text{TSS}. \quad (6.32)$$

The full model is the fitted line, so

$$p_{\text{full}} = 2, \quad SS_{\text{full}} = \text{RSS}. \quad (6.33)$$

Therefore,

$$F = \frac{(\text{TSS} - \text{RSS})/(2 - 1)}{\text{RSS}/(n - 2)} \quad (6.34)$$

$$= \boxed{\frac{\text{TSS} - \text{RSS}}{\text{RSS}/(n - 2)}}. \quad (6.35)$$

The numerator is the improvement over the mean-only model. The denominator is the residual variation per leftover degree of freedom.

Math Foundations Connection. The phrase *leftover degrees of freedom* is a dimension idea. The residual variation lives in directions left over after fitting the intercept and slope. Linear independence, bases, and the fact that a linearly independent set of n -vectors can have at most n elements are introduced in the Math Foundations textbook, Chapters 2, 5, and 6.

6.11 The F -Statistic in Terms of R^2

Because

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}, \quad (6.36)$$

we can rewrite the SLR F -statistic as

$$F = \frac{\text{TSS} - \text{RSS}}{\text{RSS}/(n - 2)} \quad (6.37)$$

$$= \frac{R^2 \text{TSS}}{(1 - R^2) \text{TSS}/(n - 2)} \quad (6.38)$$

$$= \boxed{\frac{R^2}{1 - R^2}(n - 2)}. \quad (6.39)$$

This formula explains why sample size matters. If two simple linear regressions have the same R^2 , the one with more observations will have a larger F -statistic. More data can make the same proportion of explained variation more statistically detectable.

6.12 The F -Distribution and the p-Value

Under Assumption 3 from Chapter 5—the Gaussian error model—and the null hypothesis $H_0 : \beta_1 = 0$,

$$F \sim F_{1,n-2}. \quad (6.40)$$

The first degree of freedom is the numerator degree of freedom:

$$p_{\text{full}} - p_{\text{null}} = 2 - 1 = 1. \quad (6.41)$$

The second degree of freedom is the residual degree of freedom:

$$n - p_{\text{full}} = n - 2. \quad (6.42)$$

After calculating the observed statistic F_{obs} , the p-value is

$$p\text{-value} = \Pr(F_{1,n-2} \geq F_{\text{obs}} \mid H_0). \quad (6.43)$$

A small p-value means that, if the true slope were zero and the model assumptions held, an improvement this large would be unlikely by random sampling variation alone.

6.13 Why the Assumptions Matter

The F -test relies on the same three-step regression assumption ladder introduced in Chapter 5.

1. Assumption 1 (mean-zero errors) supports the interpretation of the line as the conditional mean.
2. Assumption 2 (constant variance and uncorrelated errors) supports the standard error and variance calculations.
3. Assumption 3 (Gaussian errors) supports exact finite-sample t and F distributions.

If these assumptions are badly violated, the p-value may not have the advertised interpretation. This is why residual analysis is not optional.

6.14 The Slope t -Test

Most software output also reports a t -test for the slope. The slope test is

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0. \quad (6.44)$$

The residual variance estimator is

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n - 2}. \quad (6.45)$$

The standard error of the slope is

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}. \quad (6.46)$$

The t -statistic is

$$t_{\text{obs}} = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)}. \quad (6.47)$$

Under the null hypothesis and Assumption 3 (Gaussian errors),

$$t_{\text{obs}} \sim t_{n-2}. \quad (6.48)$$

6.15 Why the SLR F -Test and Slope t -Test Agree

In simple linear regression with an intercept,

$$\boxed{F = t^2}. \quad (6.49)$$

To see why, recall that

$$\text{RegSS} = \hat{\beta}_1^2 S_{xx}. \quad (6.50)$$

Also,

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\hat{\sigma}^2}{S_{xx}}, \quad \hat{\sigma}^2 = \frac{\text{RSS}}{n-2}. \quad (6.51)$$

Thus,

$$t^2 = \frac{\hat{\beta}_1^2}{\text{SE}(\hat{\beta}_1)^2} \quad (6.52)$$

$$= \frac{\hat{\beta}_1^2}{\hat{\sigma}^2 / S_{xx}} \quad (6.53)$$

$$= \frac{\hat{\beta}_1^2 S_{xx}}{\hat{\sigma}^2} \quad (6.54)$$

$$= \frac{\text{RegSS}}{\text{RSS} / (n-2)} \quad (6.55)$$

$$= F. \quad (6.56)$$

Therefore, in SLR, the overall F -test for the model and the two-sided t -test for the slope produce the same conclusion. In multiple linear regression, the overall F -test and individual coefficient t -tests answer different questions.

6.16 Worked Example

Consider five observations:

x_i	y_i
1	2
2	3
3	5
4	4
5	6

The least squares line is

$$\hat{y} = 1.3 + 0.9x. \quad (6.57)$$

The fitted values and residuals are:

i	x_i	y_i	$\hat{y}_i$	e_i
1	1	2	2.2	-0.2
2	2	3	3.1	-0.1
3	3	5	4.0	1.0
4	4	4	4.9	-0.9
5	5	6	5.8	0.2

The sample mean is

$$\bar{y} = 4. \quad (6.58)$$

The total sum of squares is

$$\text{TSS} = (2 - 4)^2 + (3 - 4)^2 + (5 - 4)^2 + (4 - 4)^2 + (6 - 4)^2 = 10. \quad (6.59)$$

The residual sum of squares is

$$\text{RSS} = (-0.2)^2 + (-0.1)^2 + (1.0)^2 + (-0.9)^2 + (0.2)^2 \quad (6.60)$$

$$= 1.9. \quad (6.61)$$

Therefore,

$$R^2 = 1 - \frac{1.9}{10} = 0.81. \quad (6.62)$$

The model explains 81% of the observed variation in y .

The F -statistic is

$$F = \frac{\text{TSS} - \text{RSS}}{\text{RSS} / (n - 2)} \quad (6.63)$$

$$= \frac{10 - 1.9}{1.9/3} \quad (6.64)$$

$$\approx 12.79. \quad (6.65)$$

Under $H_0 : \beta_1 = 0$, this is compared to an $F_{1,3}$ distribution. Equivalently, the slope t -statistic is

$$t = \sqrt{F} \approx 3.58, \quad (6.66)$$

with the sign determined by the sign of $\hat{\beta}_1$. Since the slope is positive, $t \approx 3.58$.

6.17 How to Read Software Output

Regression software usually reports both fit-quality and hypothesis-testing information.

6.17.1 Fit Quality Fields

Common fields include:

- **R-squared:** the R^2 value;
- **Residual standard error** or **scale:** an estimate of residual noise;
- **F-statistic:** the model-versus-null comparison;
- **Prob(F-statistic):** the p-value for the F -test.

6.17.2 Coefficient Table Fields

For the predictor row, common fields include:

- **coef:** $\hat{\beta}_1$;
- **std err:** $\text{SE}(\hat{\beta}_1)$;
- **t:** the slope t -statistic;
- **two-sided p-value:** the p-value for $H_0 : \beta_1 = 0$;
- **confidence interval:** an interval estimate for the slope.

In SLR, the p-value in the predictor row and the p-value for the model F -statistic correspond to the same slope question.

6.18 Decision Template for SLR Fit and Significance

When reporting a simple linear regression, use the following template.

1. **State the fitted model:**

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x. \quad (6.67)$$

2. **Interpret the slope with units.**
3. **Report R^2 :** say what proportion of response variation is explained by the predictor.
4. **State the hypothesis test:**

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0. \quad (6.68)$$

5. **Report the test statistic and p-value.**
6. **Translate the conclusion into context.**
7. **Check residuals before overclaiming.**
8. **Avoid causal language unless the design supports causation.**

Coding companion reference. Package-specific Python/R commands for computing R^2 , F -statistics, p-values, and related output have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the conceptual guidance for reading software output.

6.19 Common Mistakes

1. **Treating high R^2 as proof of causation.** R^2 is about explained variation, not causal mechanism.
2. **Treating high R^2 as proof of validity.** A model can have high R^2 and still violate assumptions.
3. **Ignoring sample size.** The same R^2 can produce different statistical evidence depending on n .
4. **Confusing residual variance with response variance.** TSS measures response variation before the model; RSS measures variation left after the model.
5. **Forgetting degrees of freedom.** In SLR, residual variance uses denominator $n - 2$ because two parameters are estimated.
6. **Thinking $R^2 = 1$ always means a useful model.** Two points can give a perfect line, but not a reliable model.
7. **Reading p-values without context.** A small p-value does not tell you effect size, practical importance, or model adequacy.

6.20 Chapter Summary

The mean-only model predicts every response using $\bar{y}$. Its squared error is the total sum of squares,

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2. \quad (6.69)$$

The fitted regression model predicts each response with

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \quad (6.70)$$

Its squared error is the residual sum of squares,

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (6.71)$$

The coefficient of determination is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}. \quad (6.72)$$

It measures the proportion of observed response variation explained by the fitted line.

The F -test compares the fitted line to the mean-only model:

$$F = \frac{\text{TSS} - \text{RSS}}{\text{RSS}/(n-2)}. \quad (6.73)$$

Under $H_0 : \beta_1 = 0$ and Assumption 3 (Gaussian errors),

$$F \sim F_{1, n-2}. \quad (6.74)$$

In simple linear regression with an intercept, the slope t -test and model F -test are equivalent:

$$F = t^2. \tag{6.75}$$

The central distinction is this:

R^2 measures how much the model fits the observed data; the F -test and t -test measure whether the observed improvement is statistically distinguishable from a zero-slope model under the same regression assumption ladder.

6.21 Practice and Workflow

Focused Study Checklist

Use this as a final check after analyzing fit quality and statistical evidence.

1. Can you identify the mean-only baseline and compute TSS?
2. Can you compute residuals, RSS, and $R^2 = 1 - \text{RSS} / \text{TSS}$?
3. Can you explain R^2 as fit quality, not as proof that the model is useful or causal?
4. Can you state the slope test hypotheses $H_0 : \beta_1 = 0$ and $H_A : \beta_1 \neq 0$?
5. Can you connect the model F -test to the slope t -test in SLR through $F = t^2$?
6. Can you conclude in context while separating fit, statistical evidence, and residual diagnostics?

Chapter 7

Linear Regression as Projection: Design Matrix and Hat Matrix

7.1 The Big Picture

The last two chapters developed simple linear regression using calculus, sums of squares, and hypothesis tests. This chapter rebuilds the same least squares estimator using linear algebra. The payoff is substantial:

Least squares regression is an orthogonal projection of the response vector onto the subspace spanned by the columns of the design matrix.

This viewpoint explains why the normal equations appear, why residuals are orthogonal to fitted values, why the fitted values can be written with the hat matrix, and why the same formula works for multiple linear regression.

The main objects in this chapter are:

$$\mathbf{y} \in \mathbb{R}^n \quad \text{the response vector,} \quad (7.1)$$

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)} \quad \text{the design matrix,} \quad (7.2)$$

$$\hat{\boldsymbol{\beta}} \in \mathbb{R}^{p+1} \quad \text{the least squares coefficient vector,} \quad (7.3)$$

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} \quad \text{the fitted-value vector,} \quad (7.4)$$

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} \quad \text{the residual vector,} \quad (7.5)$$

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \quad \text{the hat matrix.} \quad (7.6)$$

The key formula is

$$\boxed{\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}.} \quad (7.7)$$

The matrix $\mathbf{H}$ is called the hat matrix because it puts the hat on $\mathbf{y}$: it turns observed responses into fitted responses.

Notation bridge. From this chapter onward, bold lowercase vectors such as $\mathbf{y}$ denote observed sample vectors from the data in hand, while bold uppercase symbols such as $\mathbf{X}$ denote matrices.

Chapter 10 temporarily writes $\mathbf{Y}$ when the full response vector is being treated as random conditional on a fixed design matrix. The front-matter notation guide gives the book-wide convention.

Math Foundations Connection. This chapter uses vector notation in $\mathbb{R}^n$, the ones vector $\mathbf{1}$, linear combinations, dot products, norms, orthogonality, matrix notation, transpose notation, and matrix-vector multiplication. These are covered in the Math Foundations textbook, Chapters 2, 3, 5, and 7.

7.2 Vectors Behind a Regression Scatterplot

A simple linear regression scatterplot displays points

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n). \quad (7.8)$$

In matrix language, we separate the observed predictor values and response values into vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}. \quad (7.9)$$

The scatterplot is drawn in the usual (x, y) -plane, but the regression calculation takes place in $\mathbb{R}^n$. The response vector $\mathbf{y}$ is a point or vector in n -dimensional observation space. The fitted vector $\hat{\mathbf{y}}$ is another vector in that same space.

For simple linear regression, every fitted vector has the form

$$\hat{\mathbf{y}} = \hat{\beta}_0 \mathbf{1} + \hat{\beta}_1 \mathbf{x}. \quad (7.10)$$

Thus, the fitted values live in the set of all linear combinations of $\mathbf{1}$ and $\mathbf{x}$. This set is the plane

$$\text{Col}(\mathbf{X}) = \text{span}\{\mathbf{1}, \mathbf{x}\} \quad (7.11)$$

inside $\mathbb{R}^n$, provided $\mathbf{1}$ and $\mathbf{x}$ are linearly independent.

Important interpretation. A regression line in the scatterplot corresponds to a vector $\hat{\mathbf{y}}$ on a plane in $\mathbb{R}^n$. We are not projecting points in the two-dimensional scatterplot. We are projecting the full response vector $\mathbf{y}$ onto the subspace of fitted-value vectors.

7.3 The Linear Regression Problem as an Overdetermined System

Suppose we have three data points. A perfect simple linear model would satisfy

$$y_1 = \beta_0 + \beta_1 x_1, \quad (7.12)$$

$$y_2 = \beta_0 + \beta_1 x_2, \quad (7.13)$$

$$y_3 = \beta_0 + \beta_1 x_3. \quad (7.14)$$

There are three equations and only two unknowns. With more than two points, the system is usually overdetermined. In vector form, the exact system would be

$$\mathbf{y} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}. \quad (7.15)$$

Usually $\mathbf{y} \notin \text{span}\{\mathbf{1}, \mathbf{x}\}$. Therefore, no exact solution exists.

Least squares solves the nearest possible problem:

Find the vector in $\text{span}\{\mathbf{1}, \mathbf{x}\}$ that is closest to $\mathbf{y}$ in squared Euclidean distance.

That nearest vector is the orthogonal projection of $\mathbf{y}$ onto the fitted-value plane.

7.4 Projection Onto One Vector

Before projecting onto the two-dimensional regression plane, recall projection onto a single nonzero vector $\mathbf{a}$. The projection of $\mathbf{y}$ onto the line spanned by $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}}(\mathbf{y}) = \frac{\mathbf{a}^T \mathbf{y}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}. \quad (7.16)$$

The residual from this projection is

$$\mathbf{r} = \mathbf{y} - \text{proj}_{\mathbf{a}}(\mathbf{y}). \quad (7.17)$$

It is orthogonal to $\mathbf{a}$:

$$\mathbf{a}^T \mathbf{r} = \mathbf{a}^T \left(\mathbf{y} - \frac{\mathbf{a}^T \mathbf{y}}{\mathbf{a}^T \mathbf{a}} \mathbf{a} \right) \quad (7.18)$$

$$= \mathbf{a}^T \mathbf{y} - \frac{\mathbf{a}^T \mathbf{y}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}^T \mathbf{a} \quad (7.19)$$

$$= 0. \quad (7.20)$$

Least squares regression is the same idea, but the target is not a line. The target is the column space of a design matrix.

Math Foundations Connection. The projection formula uses dot products, squared lengths, and orthogonality. Dot products and Euclidean norms are covered in the Math Foundations textbook, Chapters 5 and 12.

7.5 The Design Matrix for Simple Linear Regression

For simple linear regression, define the design matrix

$$\mathbf{X} = [\mathbf{1} \quad \mathbf{x}] = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \in \mathbb{R}^{n \times 2}. \quad (7.21)$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}. \quad (7.22)$$

Then the simple linear regression model can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (7.23)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (7.24)$$

The columns of $\mathbf{X}$ are directions in $\mathbb{R}^n$. In SLR, those directions are $\mathbf{1}$ and $\mathbf{x}$. The fitted vector $\hat{\mathbf{y}}$ is a linear combination of those columns.

7.6 The Normal Equations from Orthogonality

The least squares residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (7.25)$$

The orthogonal projection condition says that the residual vector must be orthogonal to every column of $\mathbf{X}$. For SLR, this means

$$\mathbf{1}^T \mathbf{e} = 0, \quad (7.26)$$

$$\mathbf{x}^T \mathbf{e} = 0. \quad (7.27)$$

Stacking these two equations gives

$$\mathbf{X}^T \mathbf{e} = \mathbf{0}. \quad (7.28)$$

Substitute $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$:

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}, \quad (7.29)$$

$$\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{0}, \quad (7.30)$$

$$\boxed{\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}.} \quad (7.31)$$

These are the normal equations.

If $\mathbf{X}^T \mathbf{X}$ is invertible, then

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.} \quad (7.32)$$

The invertibility condition is equivalent to the columns of $\mathbf{X}$ being linearly independent. For SLR with an intercept, this requires the observed x_i 's not all being equal.

Math Foundations Connection. The step from $\mathbf{X}^T \mathbf{e} = \mathbf{0}$ to the normal equations uses matrix transpose notation and matrix-vector multiplication. Matrix-vector multiplication can be read both as row inner products and as column linear combinations; see the Math Foundations textbook, Chapters 3, 5, 7, and 12.

7.7 Recovering the Familiar SLR Formulas

The matrix formula in (7.32) is not a different estimator. It is the same least squares estimator from the calculus derivation.

For SLR,

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} \mathbf{1}^T \mathbf{1} & \mathbf{1}^T \mathbf{x} \\ \mathbf{x}^T \mathbf{1} & \mathbf{x}^T \mathbf{x} \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix}, \quad (7.33)$$

and

$$\mathbf{X}^T \mathbf{y} = \begin{bmatrix} \mathbf{1}^T \mathbf{y} \\ \mathbf{x}^T \mathbf{y} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{bmatrix}. \quad (7.34)$$

Solving the two normal equations gives

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (7.35)$$

Thus, the matrix formula and the scalar SLR formula are two views of the same least squares solution.

7.8 The Hat Matrix

Start with the fitted values:

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}}. \quad (7.36)$$

Now substitute the least squares estimator from (7.32):

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (7.37)$$

Define

$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T}. \quad (7.38)$$

Then

$$\boxed{\hat{\mathbf{y}} = \mathbf{H} \mathbf{y}}. \quad (7.39)$$

The hat matrix is an $n \times n$ matrix. It maps observed response vectors in $\mathbb{R}^n$ to fitted-value vectors in the column space of $\mathbf{X}$:

$$\mathbf{H} : \mathbb{R}^n \rightarrow \text{Col}(\mathbf{X}). \quad (7.40)$$

7.9 Properties of the Hat Matrix

Assume $\mathbf{X}$ has full column rank, so $\mathbf{X}^T \mathbf{X}$ is invertible.

7.9.1 Symmetry

The hat matrix is symmetric:

$$\boxed{\mathbf{H}^T = \mathbf{H}}. \quad (7.41)$$

Proof:

$$\mathbf{H}^T = [\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T]^T \quad (7.42)$$

$$= (\mathbf{X}^T)^T [(\mathbf{X}^T \mathbf{X})^{-1}]^T \mathbf{X}^T \quad (7.43)$$

$$= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (7.44)$$

$$= \mathbf{H}. \quad (7.45)$$

The third line uses the fact that $\mathbf{X}^T \mathbf{X}$ is symmetric, so its inverse is also symmetric.

7.9.2 Idempotence

The hat matrix is idempotent:

$$\boxed{\mathbf{H}^2 = \mathbf{H}.} \quad (7.46)$$

Proof:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (7.47)$$

$$= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{X}) (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (7.48)$$

$$= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (7.49)$$

$$= \mathbf{H}. \quad (7.50)$$

Idempotence means that once a vector has already been projected into the fitted-value subspace, projecting it again does not change it:

$$\mathbf{H}\hat{\mathbf{y}} = \mathbf{H}(\mathbf{H}\mathbf{y}) = \mathbf{H}^2\mathbf{y} = \mathbf{H}\mathbf{y} = \hat{\mathbf{y}}. \quad (7.51)$$

7.9.3 Orthogonal Projection

A matrix that is both symmetric and idempotent is an orthogonal projection matrix. Therefore, $\mathbf{H}$ is the orthogonal projection onto $\text{Col}(\mathbf{X})$.

This is the geometric core of least squares:

$$\boxed{\hat{\mathbf{y}} = \mathbf{H}\mathbf{y} \text{ is the closest vector to } \mathbf{y} \text{ in } \text{Col}(\mathbf{X}).} \quad (7.52)$$

7.9.4 Influence

Because

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad (7.53)$$

we have componentwise

$$\hat{y}_i = \sum_{j=1}^n H_{ij} y_j. \quad (7.54)$$

Therefore,

$$\boxed{\frac{\partial \hat{y}_i}{\partial y_j} = H_{ij}.} \quad (7.55)$$

The entry H_{ij} measures how much the observed response y_j influences the fitted response $\hat{y}_i$. The diagonal entries H_{ii} are called leverages. A large leverage value means observation i 's predictor row has unusual position relative to the design matrix.

7.10 The Residual Maker Matrix

The hat matrix makes fitted values. The complementary matrix makes residuals.

Define

$$\mathbf{M} = \mathbf{I} - \mathbf{H}. \quad (7.56)$$

Then

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} \quad (7.57)$$

$$= \mathbf{y} - \mathbf{H}\mathbf{y} \quad (7.58)$$

$$= (\mathbf{I} - \mathbf{H})\mathbf{y} \quad (7.59)$$

$$= \mathbf{M}\mathbf{y}. \quad (7.60)$$

The matrix $\mathbf{M}$ is also symmetric and idempotent:

$$\mathbf{M}^T = \mathbf{M}, \quad \mathbf{M}^2 = \mathbf{M}. \quad (7.61)$$

It projects $\mathbf{y}$ onto the residual subspace, which is the orthogonal complement of $\text{Col}(\mathbf{X})$.

Because $\mathbf{X}^T \mathbf{e} = \mathbf{0}$, the residual vector is orthogonal to every column of the design matrix. Since $\hat{\mathbf{y}} \in \text{Col}(\mathbf{X})$, the residual vector is also orthogonal to the fitted-value vector:

$$\hat{\mathbf{y}}^T \mathbf{e} = 0. \quad (7.62)$$

This gives the geometric decomposition

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}, \quad \hat{\mathbf{y}} \perp \mathbf{e}. \quad (7.63)$$

7.11 Dimensions and Degrees of Freedom

If $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ has full column rank, then

$$\dim \text{Col}(\mathbf{X}) = p + 1. \quad (7.64)$$

The fitted values live in this $(p + 1)$ -dimensional subspace. The residuals live in the orthogonal complement, whose dimension is

$$n - (p + 1) = n - p - 1. \quad (7.65)$$

This is the linear algebra reason that the residual variance estimator in multiple linear regression divides by $n - p - 1$.

For SLR, $p = 1$, so the fitted-value subspace has dimension 2, and the residual subspace has dimension

$$n - 2. \quad (7.66)$$

Math Foundations Connection. The dimension claim depends on linear independence and bases. A set of linearly independent n -vectors can have at most n elements, and a basis provides unique linear-combination coordinates. See the Math Foundations textbook, Chapters 2 and 6.

7.12 Multiple Linear Regression as the Same Projection

The design matrix generalizes immediately from simple linear regression to multiple linear regression. With predictors $\mathbf{x}_1, \dots, \mathbf{x}_p$, define

$$\mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \cdots \quad \mathbf{x}_p] = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}. \quad (7.67)$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}. \quad (7.68)$$

The model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (7.69)$$

The least squares solution is still

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \quad (7.70)$$

and the fitted values are still

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (7.71)$$

The only change is the subspace. In SLR, $\text{Col}(\mathbf{X})$ is usually a two-dimensional plane in $\mathbb{R}^n$. In MLR, $\text{Col}(\mathbf{X})$ is usually a $(p+1)$ -dimensional subspace in $\mathbb{R}^n$.

7.13 Design Matrix as a Modeling Choice

The design matrix tells the model what directions it is allowed to use to approximate $\mathbf{y}$. Choosing a model means choosing columns for $\mathbf{X}$.

Examples:

$$\text{SLR:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}], \quad (7.72)$$

$$\text{Polynomial model:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x} \quad \mathbf{x}^2 \quad \mathbf{x}^3], \quad (7.73)$$

$$\text{MLR:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \cdots \quad \mathbf{x}_p], \quad (7.74)$$

$$\text{Interaction model:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_1 \odot \mathbf{x}_2]. \quad (7.75)$$

Here $\odot$ means elementwise multiplication, so the interaction column has entries $x_{i1}x_{i2}$.

The same least squares machinery works for all of these models as long as the model is linear in the coefficients. This is why regression can fit curved relationships using transformed columns while still being a linear model in parameters.

7.14 Worked Example: Projection in SLR

Consider three observations:

x_i	y_i
-1	1
0	0
1	2

Then

$$\mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}. \quad (7.76)$$

Compute

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{X}^T \mathbf{y} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}. \quad (7.77)$$

Therefore,

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}. \quad (7.78)$$

The fitted line is

$$\hat{y} = 1 + \frac{1}{2}x. \quad (7.79)$$

The fitted-value vector is

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \begin{bmatrix} 1/2 \\ 1 \\ 3/2 \end{bmatrix}. \quad (7.80)$$

The residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 1/2 \\ -1 \\ 1/2 \end{bmatrix}. \quad (7.81)$$

Check the orthogonality conditions:

$$\mathbf{1}^T \mathbf{e} = \frac{1}{2} - 1 + \frac{1}{2} = 0, \quad (7.82)$$

$$\mathbf{x}^T \mathbf{e} = (-1) \left(\frac{1}{2} \right) + 0(-1) + 1 \left(\frac{1}{2} \right) = 0. \quad (7.83)$$

Thus, the residual vector is orthogonal to both columns of $\mathbf{X}$. This confirms that $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\text{span}\{\mathbf{1}, \mathbf{x}\}$.

For this example, the hat matrix is

$$\mathbf{H} = \begin{bmatrix} 5/6 & 1/3 & -1/6 \\ 1/3 & 1/3 & 1/3 \\ -1/6 & 1/3 & 5/6 \end{bmatrix}. \quad (7.84)$$

Then

$$\mathbf{H}\mathbf{y} = \begin{bmatrix} 1/2 \\ 1 \\ 3/2 \end{bmatrix} = \hat{\mathbf{y}}. \quad (7.85)$$

Coding companion reference. Package-specific Python/R commands for building design matrices, hat matrices, fitted values, residuals, and leverage values have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the projection geometry and matrix interpretation.

7.15 Common Mistakes

1. **Projecting the scatterplot instead of the response vector.** The projection takes place in $\mathbb{R}^n$, not in the two-dimensional scatterplot.
2. **Forgetting the intercept column.** With an intercept, the first column of $\mathbf{X}$ is $\mathbf{1}$. Removing the intercept changes the projection subspace.
3. **Assuming $(\mathbf{X}^T \mathbf{X})^{-1}$ always exists.** It exists only when the columns of $\mathbf{X}$ are linearly independent.
4. **Confusing random errors and residuals.** The model errors $\boldsymbol{\epsilon}$ are unobserved. The residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ are computed after fitting the model.
5. **Thinking the hat matrix estimates coefficients.** The hat matrix maps $\mathbf{y}$ to $\hat{\mathbf{y}}$. The coefficient estimator is $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$.
6. **Ignoring design-matrix columns.** Transformed predictors, interaction terms, and categorical encodings are all columns of $\mathbf{X}$.
7. **Treating high leverage as a residual problem only.** Leverage comes from $\mathbf{X}$, not from y . It measures unusual predictor geometry.

7.16 Chapter Summary

Simple linear regression can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}]. \quad (7.86)$$

The fitted values are linear combinations of the columns of $\mathbf{X}$:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (7.87)$$

Least squares chooses the fitted vector in $\text{Col}(\mathbf{X})$ closest to $\mathbf{y}$. The residual vector is orthogonal to every column of $\mathbf{X}$, so

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \quad (7.88)$$

This gives the normal equations

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}. \quad (7.89)$$

If $\mathbf{X}^T \mathbf{X}$ is invertible, then

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (7.90)$$

The fitted values can be written as

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (7.91)$$

The hat matrix is symmetric and idempotent, so it is an orthogonal projection matrix. The diagonal entries of $\mathbf{H}$ are leverage values, and the full matrix entries describe how observed responses influence fitted responses.

The same projection formula generalizes immediately to multiple linear regression by adding more columns to the design matrix. In MLR, when $\mathbf{X}$ has full column rank, the fitted values are the projection of $\mathbf{y}$ onto the $(p + 1)$ -dimensional column space of $\mathbf{X}$, and the residuals live in the $(n - p - 1)$ -dimensional orthogonal complement. If $\text{rank}(\mathbf{X}) = r$, replace $p + 1$ by r and $n - p - 1$ by $n - r$. Chapter 8 uses these dimensions to explain degrees of freedom and identifiability.

7.17 Practice and Workflow

Focused Study Checklist

Use this as a final check after translating a regression problem into projection language.

1. Can you build the design matrix $\mathbf{X}$ with the intercept column and state its dimensions?
2. Can you explain why least squares projects $\mathbf{y}$ onto $\text{Col}(\mathbf{X})$?
3. Can you derive or recognize the normal equations $\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}$?
4. Can you write $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}$?
5. Can you explain what symmetry and idempotence of $\mathbf{H}$ mean for projection?
6. Can you state the geometric meaning of $\mathbf{e} \perp \text{Col}(\mathbf{X})$?

Chapter 8

Degrees of Freedom, Subspaces, and Identifiability

8.1 The Big Picture

The previous chapter showed that linear regression is an orthogonal projection. The response vector $\mathbf{y} \in \mathbb{R}^n$ is projected onto the column space of the design matrix $\mathbf{X}$. The fitted values live in one subspace, and the residuals live in the orthogonal complement.

This chapter uses that geometry to explain three ideas that appear repeatedly in regression:

1. **Degrees of freedom:** how many independent dimensions remain after we estimate model parameters.
2. **Subspaces:** where fitted values, residuals, coefficient information, and non-identifiable directions live.
3. **Identifiability:** when regression coefficients are uniquely determined by the data, and when many coefficient vectors produce the same fitted values.

The central message is:

Degrees of freedom are leftover dimensions. Identifiability fails when the design matrix has a nonzero null space.

For a full-rank multiple linear regression model with p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)}, \quad \text{rank}(\mathbf{X}) = p + 1, \quad (8.1)$$

and the residual degrees of freedom are

$$\boxed{n - p - 1}. \quad (8.2)$$

This is why the denominator of the residual variance estimator and the denominator degrees of freedom in the overall regression F -test use $n - p - 1$.

Math Foundations Connection. This chapter uses vector notation, the ones vector, dot products, orthogonality, matrix dimensions, matrix transposes, linear independence, bases, matrix-vector multiplication, left-inverse intuition, and null-space/subspace ideas. These are covered in the Math Foundations textbook, Chapters 2, 5, 6, 7, 10, and 11.

8.2 A First Definition: Degrees of Freedom as Independent Movement

In data analysis, degrees of freedom are often introduced as a formula to memorize. For example:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2. \quad (8.3)$$

The natural question is: why $n-1$?

The geometric answer is that n observations define a vector in $\mathbb{R}^n$. Before estimating anything, the vector $\mathbf{y}$ has n independent coordinates:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n. \quad (8.4)$$

Once we estimate the mean $\bar{y}$, the residual deviations must satisfy one linear constraint:

$$\sum_{i=1}^n (y_i - \bar{y}) = 0. \quad (8.5)$$

One independent constraint removes one dimension. Therefore, the residual variation around the sample mean has $n-1$ degrees of freedom.

Reusable intuition. Each independently estimated parameter usually consumes one dimension. The leftover dimensions are available for residual variation.

8.3 Decomposing a Data Vector Around the Sample Mean

The simplest projection problem is the sample mean. Let

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^n. \quad (8.6)$$

The sample mean vector is

$$\bar{y}\mathbf{1}. \quad (8.7)$$

The residual vector around the sample mean is

$$\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1}. \quad (8.8)$$

Therefore,

$$\boxed{\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{y} - \bar{y}\mathbf{1})}. \quad (8.9)$$

This is a decomposition of the data vector into two orthogonal pieces:

1. $\bar{y}\mathbf{1}$, the projection onto the one-dimensional subspace $\text{span}\{\mathbf{1}\}$, and
2. $\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1}$, the residual vector in the orthogonal complement.

To verify orthogonality, compute

$$\mathbf{1}^T \mathbf{r} = \mathbf{1}^T (\mathbf{y} - \bar{y}\mathbf{1}) \quad (8.10)$$

$$= \sum_{i=1}^n y_i - \bar{y} \sum_{i=1}^n 1 \quad (8.11)$$

$$= n\bar{y} - n\bar{y} \quad (8.12)$$

$$= 0. \quad (8.13)$$

Thus,

$$\mathbf{r} \in \text{span}\{\mathbf{1}\}^\perp. \quad (8.14)$$

Since $\text{span}\{\mathbf{1}\}$ has dimension 1, its orthogonal complement in $\mathbb{R}^n$ has dimension

$$n - 1. \quad (8.15)$$

This is the geometric reason for the $n - 1$ denominator in sample variance.

8.4 Worked Example: Mean Decomposition with Three Observations

Suppose

$$\mathbf{y} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}. \quad (8.16)$$

The sample mean is

$$\bar{y} = \frac{2 + 5 + 8}{3} = 5. \quad (8.17)$$

Then

$$\bar{y}\mathbf{1} = 5 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}, \quad (8.18)$$

and

$$\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix}. \quad (8.19)$$

Check the residual constraint:

$$\mathbf{1}^T \mathbf{r} = (-3) + 0 + 3 = 0. \quad (8.20)$$

The residual vector lives on the plane

$$\{\mathbf{v} \in \mathbb{R}^3 : \mathbf{1}^T \mathbf{v} = 0\}. \quad (8.21)$$

That plane has dimension 2, which equals $n - 1 = 3 - 1$.

8.5 The Mean Projection Matrix

The sample mean is itself a projection. Define

$$\mathbf{H}_0 = \frac{1}{n} \mathbf{1} \mathbf{1}^T. \quad (8.22)$$

Then

$$\mathbf{H}_0 \mathbf{y} = \frac{1}{n} \mathbf{1} \mathbf{1}^T \mathbf{y} = \frac{1}{n} \mathbf{1} \sum_{i=1}^n y_i = \bar{y} \mathbf{1}. \quad (8.23)$$

The residual-maker matrix for the mean-only model is

$$\mathbf{M}_0 = \mathbf{I} - \mathbf{H}_0. \quad (8.24)$$

Then

$$\mathbf{M}_0 \mathbf{y} = \mathbf{y} - \bar{y} \mathbf{1}. \quad (8.25)$$

The matrix $\mathbf{H}_0$ is symmetric and idempotent, so it is an orthogonal projection matrix. Its rank and trace are both 1:

$$\text{rank}(\mathbf{H}_0) = \text{tr}(\mathbf{H}_0) = 1. \quad (8.26)$$

Likewise,

$$\text{rank}(\mathbf{M}_0) = \text{tr}(\mathbf{M}_0) = n - 1. \quad (8.27)$$

This trace perspective becomes useful in regression because the hat matrix for linear regression is also a symmetric idempotent projection matrix.

8.6 Regression Decomposes $\mathbb{R}^n$ Into Two Orthogonal Pieces

For multiple linear regression with an intercept and p predictors, write

$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \mathbf{X} \in \mathbb{R}^{n \times (p+1)}. \quad (8.28)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{H} \mathbf{y}, \quad \mathbf{H} = \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T, \quad (8.29)$$

when $\mathbf{X}$ has full column rank.

The residuals are

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H}) \mathbf{y} = \mathbf{M} \mathbf{y}. \quad (8.30)$$

Regression decomposes the observation space as

$$\boxed{\mathbb{R}^n = \text{Col}(\mathbf{X}) \oplus \text{Null}(\mathbf{X}^T)}. \quad (8.31)$$

Here $\oplus$ means a direct sum: every vector in $\mathbb{R}^n$ can be written uniquely as a piece in $\text{Col}(\mathbf{X})$ plus a piece in $\text{Null}(\mathbf{X}^T)$. These two pieces are orthogonal.

In regression terms:

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X}), \quad (8.32)$$

$$\mathbf{e} \in \text{Null}(\mathbf{X}^T), \quad (8.33)$$

$$\hat{\mathbf{y}}^T \mathbf{e} = 0. \quad (8.34)$$

The fitted values live in the column space. The residuals live in the left null space.

8.7 Residual Degrees of Freedom

Assume $\mathbf{X}$ has full column rank. Then

$$\text{rank}(\mathbf{X}) = p + 1. \quad (8.35)$$

Therefore,

$$\dim \text{Col}(\mathbf{X}) = p + 1. \quad (8.36)$$

The column space consumes $p + 1$ dimensions of the original n -dimensional observation space. The leftover dimensions are the residual degrees of freedom:

$$\boxed{\text{df}_{\text{resid}} = n - (p + 1) = n - p - 1.} \quad (8.37)$$

This explains the residual variance estimator:

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n - p - 1} = \frac{\mathbf{e}^T \mathbf{e}}{n - p - 1}. \quad (8.38)$$

It also explains the denominator degrees of freedom in the overall F -test for MLR.

Special cases.

Model	Fitted subspace dimension	Residual df
Mean-only model	1	$n - 1$
SLR with intercept	2	$n - 2$
MLR with intercept and p predictors	$p + 1$	$n - p - 1$
Full rank model with design rank r	r	$n - r$

8.8 A Subtle Point: Model Degrees of Freedom Versus Fitted-Subspace Dimension

There are two closely related uses of degrees of freedom in regression:

1. **Fitted-subspace dimension:** with an intercept and p predictors, $\text{Col}(\mathbf{X})$ has dimension $p + 1$.
2. **Model improvement degrees of freedom:** when comparing the full model to the intercept-only model, the predictors add p dimensions beyond the mean direction.

That is why an ANOVA-style regression decomposition often uses

$$\text{Total df} = n - 1, \quad \text{Regression df} = p, \quad \text{Residual df} = n - p - 1. \quad (8.39)$$

The $n - 1$ total degrees of freedom already subtract the mean direction. The regression degrees of freedom p count the additional fitted directions beyond the intercept-only model.

8.9 Trace of the Hat Matrix

For a full-rank linear regression model, the hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (8.40)$$

It is symmetric and idempotent. For symmetric idempotent matrices, the trace equals the rank. Therefore,

$$\boxed{\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H}) = \text{rank}(\mathbf{X})}. \quad (8.41)$$

If $\mathbf{X}$ has full column rank, this becomes

$$\text{tr}(\mathbf{H}) = p + 1. \quad (8.42)$$

The residual-maker matrix is $\mathbf{M} = \mathbf{I} - \mathbf{H}$, so

$$\text{tr}(\mathbf{M}) = \text{tr}(\mathbf{I}) - \text{tr}(\mathbf{H}) \quad (8.43)$$

$$= n - (p + 1) \quad (8.44)$$

$$= n - p - 1. \quad (8.45)$$

This is another way to see residual degrees of freedom.

Connection to leverage. The diagonal entries of $\mathbf{H}$, denoted h_{ii} , are leverages. Since the trace is the sum of diagonal entries,

$$\sum_{i=1}^n h_{ii} = p + 1. \quad (8.46)$$

The average leverage is therefore

$$\bar{h} = \frac{p + 1}{n}. \quad (8.47)$$

Large leverage means an observation is using more than its average share of the fitted-subspace degrees of freedom.

8.10 The Big Four Subspaces of the Design Matrix

Let

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)} \quad (8.48)$$

with rank r . Linear algebra associates four fundamental subspaces with $\mathbf{X}$. In regression, these subspaces have direct meanings.

Subspace	Where it lives	Regression meaning	Dimension
$\text{Col}(\mathbf{X})$	$\mathbb{R}^n$	Fitted values $\hat{\mathbf{y}}$	r
$\text{Null}(\mathbf{X}^T)$	$\mathbb{R}^n$	Residual vectors $\mathbf{e}$	$n - r$
$\text{Row}(\mathbf{X}) = \text{Col}(\mathbf{X}^T)$	$\mathbb{R}^{p+1}$	Identifiable coefficient directions	r
$\text{Null}(\mathbf{X})$	$\mathbb{R}^{p+1}$	Non-identifiable coefficient directions	$(p + 1) - r$

The first two subspaces live in observation space $\mathbb{R}^n$. The last two live in coefficient space $\mathbb{R}^{p+1}$. This distinction prevents a common mistake: residuals and fitted values are vectors with n entries, while coefficient directions have $p + 1$ entries.

8.11 Column Space and Left Null Space

The fitted values satisfy

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (8.49)$$

Therefore, $\hat{\mathbf{y}}$ is a linear combination of the columns of $\mathbf{X}$:

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X}). \quad (8.50)$$

The residual vector satisfies the normal equations:

$$\mathbf{X}^T \mathbf{e} = \mathbf{0}. \quad (8.51)$$

Therefore,

$$\mathbf{e} \in \text{Null}(\mathbf{X}^T). \quad (8.52)$$

Since $\text{Null}(\mathbf{X}^T)$ is the orthogonal complement of $\text{Col}(\mathbf{X})$, every residual vector is orthogonal to every fitted-value direction.

This is the geometric core of regression diagnostics:

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}, \quad \hat{\mathbf{y}} \perp \mathbf{e}. \quad (8.53)$$

8.12 Row Space and Null Space

The row space and null space of $\mathbf{X}$ live in coefficient space. They answer a different question:

Which coefficient directions change fitted values, and which coefficient directions are invisible to the data?

A vector $\mathbf{v} \in \mathbb{R}^{p+1}$ is in the null space of $\mathbf{X}$ if

$$\mathbf{X}\mathbf{v} = \mathbf{0}. \quad (8.54)$$

If $\mathbf{v} \neq \mathbf{0}$, then changing the coefficient vector in the direction $\mathbf{v}$ does not change fitted values:

$$\mathbf{X}(\boldsymbol{\beta} + c\mathbf{v}) = \mathbf{X}\boldsymbol{\beta} + c\mathbf{X}\mathbf{v} = \mathbf{X}\boldsymbol{\beta} \quad (8.55)$$

for any scalar c . Therefore, $\mathbf{v}$ is a non-identifiable direction.

The row space contains coefficient directions that the data can see. The null space contains coefficient directions that the data cannot see.

8.13 Identifiability

A parameter is identifiable when different parameter values imply different observable behavior. In linear regression, the coefficient vector $\boldsymbol{\beta}$ is identifiable when the mapping

$$\boldsymbol{\beta} \mapsto \mathbf{X}\boldsymbol{\beta} \quad (8.56)$$

is one-to-one.

This happens exactly when

$$\text{Null}(\mathbf{X}) = \{\mathbf{0}\}. \quad (8.57)$$

Equivalently, the columns of $\mathbf{X}$ must be linearly independent.

For $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$, the following statements are equivalent:

1. $\boldsymbol{\beta}$ is identifiable.
2. $\text{Null}(\mathbf{X}) = \{\mathbf{0}\}$.
3. The columns of $\mathbf{X}$ are linearly independent.
4. $\text{rank}(\mathbf{X}) = p + 1$.
5. $\mathbf{X}^T \mathbf{X}$ is invertible.
6. The least squares coefficient vector

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (8.58)$$

is unique.

Why $\mathbf{X}^T \mathbf{X}$ matters. For any $\mathbf{v}$,

$$\mathbf{v}^T \mathbf{X}^T \mathbf{X} \mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2. \quad (8.59)$$

If $\mathbf{X}\mathbf{v} = \mathbf{0}$ for some nonzero $\mathbf{v}$, then $\mathbf{X}^T \mathbf{X}$ cannot be invertible. Conversely, if $\mathbf{X}^T \mathbf{X}$ is singular, then there is a nonzero $\mathbf{v}$ such that $\mathbf{X}\mathbf{v} = \mathbf{0}$. Thus, $\mathbf{X}^T \mathbf{X}$ is invertible exactly when $\mathbf{X}$ has no nonzero null directions.

8.14 What Fails When the Model Is Not Identified?

When $\text{Null}(\mathbf{X}) \neq \{\mathbf{0}\}$, there are infinitely many coefficient vectors that produce the same fitted values. If $\mathbf{v} \in \text{Null}(\mathbf{X})$, then

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\hat{\boldsymbol{\beta}} + c\mathbf{v}) \quad (8.60)$$

for every scalar c . The data cannot distinguish $\hat{\boldsymbol{\beta}}$ from $\hat{\boldsymbol{\beta}} + c\mathbf{v}$.

Two important consequences follow:

1. The coefficient vector is not unique.
2. The fitted values may still be unique, because the projection of $\mathbf{y}$ onto $\text{Col}(\mathbf{X})$ is unique.

Thus, non-identifiability is primarily a coefficient problem. The model may still make fitted predictions, but the individual coefficients cannot be interpreted uniquely.

8.15 Non-Identifiability Example: Celsius and Fahrenheit

Suppose a model includes an intercept, temperature measured in Celsius, and temperature measured in Fahrenheit:

$$y_i = \beta_0 + \beta_C C_i + \beta_F F_i + \epsilon_i. \quad (8.61)$$

The two temperature predictors are exactly related:

$$F_i = \frac{9}{5}C_i + 32. \quad (8.62)$$

The design matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & C_1 & F_1 \\ 1 & C_2 & F_2 \\ \vdots & \vdots & \vdots \\ 1 & C_n & F_n \end{bmatrix}. \quad (8.63)$$

The columns are linearly dependent because

$$\mathbf{x}_F - \frac{9}{5}\mathbf{x}_C - 32\mathbf{1} = \mathbf{0}. \quad (8.64)$$

Equivalently,

$$\mathbf{X} \begin{bmatrix} -32 \\ -9/5 \\ 1 \end{bmatrix} = \mathbf{0}. \quad (8.65)$$

Therefore,

$$\begin{bmatrix} -32 \\ -9/5 \\ 1 \end{bmatrix} \in \text{Null}(\mathbf{X}). \quad (8.66)$$

This means that coefficient vectors can move in this direction without changing predictions.

For example, the coefficient vector

$$\boldsymbol{\beta} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix} \quad (8.67)$$

gives

$$\hat{y} = 10 + 2C. \quad (8.68)$$

But adding the null-space vector gives

$$\beta^* = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -32 \\ -9/5 \\ 1 \end{bmatrix} = \begin{bmatrix} -22 \\ 1/5 \\ 1 \end{bmatrix}. \quad (8.69)$$

This gives

$$\hat{y}^* = -22 + \frac{1}{5}C + F \quad (8.70)$$

$$= -22 + \frac{1}{5}C + \left(\frac{9}{5}C + 32\right) \quad (8.71)$$

$$= 10 + 2C. \quad (8.72)$$

The predictions are identical, but the coefficient interpretations are different. Therefore, the individual Celsius and Fahrenheit coefficients are not identified.

8.16 Common Sources of Non-Identifiability

Exact non-identifiability usually comes from exact linear dependence among columns of the design matrix. Common examples include:

1. **Duplicate variables:** including the same predictor twice.
2. **Unit conversions:** including meters and feet, or Celsius and Fahrenheit, in a model with an intercept.
3. **The dummy variable trap:** including an intercept and all category indicator columns.
4. **Derived totals:** including all component variables and their exact sum.
5. **Too many predictors:** if $p + 1 > n$, then $\mathbf{X}$ cannot have full column rank.

Dummy variable trap. Suppose a categorical variable has three levels: Army, Navy, and Marine Corps. Let $\mathbf{d}_A, \mathbf{d}_N, \mathbf{d}_M$ be the three indicator columns. For every row,

$$d_{iA} + d_{iN} + d_{iM} = 1. \quad (8.73)$$

Thus,

$$\mathbf{d}_A + \mathbf{d}_N + \mathbf{d}_M = \mathbf{1}. \quad (8.74)$$

If the design matrix includes $\mathbf{1}$, $\mathbf{d}_A$, $\mathbf{d}_N$, and $\mathbf{d}_M$, then its columns are linearly dependent. The usual repair is to include the intercept and drop one category indicator as the reference group.

8.17 Near Non-Identifiability: Multicollinearity

Exact linear dependence causes exact non-identifiability. Near linear dependence causes instability. This is multicollinearity.

If one column of $\mathbf{X}$ is almost a linear combination of other columns, then $\mathbf{X}^T \mathbf{X}$ may still be invertible, but it is close to singular. The least squares coefficients exist, but small changes in the data can produce large changes in the coefficient estimates.

This matters because:

1. coefficient standard errors become large,
2. signs and magnitudes can be unstable,
3. individual coefficient interpretations become fragile, and
4. predictions may remain reasonable even while coefficients are hard to interpret.

Operational distinction.

Condition	Design-matrix geometry	Consequence
Exact dependence	Nonzero Null($\mathbf{X}$)	Coefficients not unique
Near dependence	Columns nearly dependent	Coefficients unstable
Full rank and well-conditioned	Columns independent and not nearly redundant	Coefficients more stable

8.18 Worked Example: Counting Regression Degrees of Freedom

Suppose we have $n = 40$ observations and fit an MLR model with an intercept and $p = 5$ predictors. If the design matrix has full column rank, then

$$\mathbf{X} \in \mathbb{R}^{40 \times 6}, \quad \text{rank}(\mathbf{X}) = 6. \quad (8.75)$$

The fitted values live in a 6-dimensional subspace:

$$\dim \text{Col}(\mathbf{X}) = 6. \quad (8.76)$$

The residuals live in the orthogonal complement:

$$\dim \text{Null}(\mathbf{X}^T) = 40 - 6 = 34. \quad (8.77)$$

Therefore,

$$\boxed{\text{df}_{\text{resid}} = 34.} \quad (8.78)$$

For the overall F -test comparing the full model to the intercept-only model,

$$\text{numerator df} = 5, \quad \text{denominator df} = 34. \quad (8.79)$$

The numerator has 5 degrees of freedom because the full model adds five predictor directions beyond the mean-only model.

Coding companion reference. Package-specific Python/R commands for checking rank, residual degrees of freedom, and non-identifiability have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the degrees-of-freedom and subspace interpretation.

8.19 Common Mistakes

1. **Forgetting the intercept.** With an intercept and p predictors, the design matrix has $p + 1$ columns, not p .
2. **Mixing up fitted-space dimension and regression F -test numerator df.** The fitted subspace has dimension $p + 1$, but the overall F -test numerator df is p when comparing to the intercept-only model.
3. **Thinking residuals live in predictor space.** Residuals are n -vectors. They live in $\mathbb{R}^n$, not in $\mathbb{R}^p$.
4. **Thinking coefficient null-space directions are residual directions.** $\text{Null}(\mathbf{X})$ lives in coefficient space, while $\text{Null}(\mathbf{X}^T)$ lives in observation space.
5. **Using $(\mathbf{X}^T \mathbf{X})^{-1}$ when $\mathbf{X}$ is rank deficient.** If columns are dependent, the inverse does not exist.
6. **Assuming fitted values are not unique when coefficients are not unique.** Coefficients may be non-unique while the projection $\hat{\mathbf{y}}$ is still unique.
7. **Checking only pairwise correlations for identifiability.** A column can be a combination of several others even if no single pair looks perfectly correlated.
8. **Including every dummy variable plus an intercept.** This creates exact linear dependence.
9. **Treating multicollinearity as a prediction failure only.** Multicollinearity often harms coefficient interpretation before it visibly harms prediction.

8.20 Chapter Summary

Degrees of freedom are best understood as dimensions. An n -vector of observations begins with n dimensions. Estimating the sample mean projects $\mathbf{y}$ onto $\text{span}\{\mathbf{1}\}$, leaving residual variation in an $(n - 1)$ -dimensional orthogonal complement.

Regression generalizes the same idea. The design matrix $\mathbf{X}$ defines the fitted-value subspace. If $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ has full column rank, then fitted values live in a $(p + 1)$ -dimensional column space, and residuals live in an $(n - p - 1)$ -dimensional orthogonal complement. Thus,

$$\text{df}_{\text{resid}} = n - p - 1. \quad (8.80)$$

The four fundamental subspaces of $\mathbf{X}$ organize the regression geometry:

$$\text{Col}(\mathbf{X}) \quad \text{contains fitted values,} \quad (8.81)$$

$$\text{Null}(\mathbf{X}^T) \quad \text{contains residuals,} \quad (8.82)$$

$$\text{Row}(\mathbf{X}) \quad \text{contains identifiable coefficient directions,} \quad (8.83)$$

$$\text{Null}(\mathbf{X}) \quad \text{contains non-identifiable coefficient directions.} \quad (8.84)$$

Identifiability requires $\text{Null}(\mathbf{X}) = \{\mathbf{0}\}$. If the design matrix has linearly dependent columns, then infinitely many coefficient vectors can produce the same fitted values. The Celsius/Fahrenheit example shows this directly: including an intercept, Celsius, and Fahrenheit creates exact linear dependence because Fahrenheit is an affine transformation of Celsius.

8.21 Practice and Workflow

Focused Study Checklist

Use this as a final check after a degrees-of-freedom or identifiability problem.

1. Can you identify n , p , and the dimensions of $\mathbf{X}$?
2. Can you determine $r = \text{rank}(\mathbf{X})$?
3. Can you state fitted-space degrees of freedom r and residual degrees of freedom $n - r$?
4. In the full-rank intercept model, can you simplify residual degrees of freedom to $n - p - 1$?
5. Can you explain why identifiability requires $\text{Null}(\mathbf{X}) = \{\mathbf{0}\}$?
6. Can you recognize exact dependence, such as redundant columns or the dummy-variable trap?
7. Can you distinguish exact non-identifiability from near dependence or multicollinearity?

Chapter 9

Multiple Linear Regression in Practice: Predictors, Hyperplanes, and the Advertising Example

9.1 The Big Picture

The previous chapters built the geometry of least squares. We now use that geometry in a practical multiple linear regression setting. The goal is to move from the algebra

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{9.1}$$

to the workflow an analyst actually uses when fitting, interpreting, and defending a regression model.

The central message of this chapter is:

Multiple linear regression is the same projection problem as simple linear regression, but the fitted values now live in a higher-dimensional subspace determined by multiple predictors.

The practical challenge is that the model is still mathematically simple, but it becomes harder to visualize and harder to interpret casually. With one predictor, the least squares fit is a line. With two predictors, the fit is a plane. With three or more predictors, the fit is a hyperplane. The word changes, but the algebra stays the same.

The main questions in this chapter are:

1. Is at least one predictor useful in predicting the response?
2. Do all predictors help, or is only a subset useful?
3. How well does the model fit the data?
4. Given new predictor values, what response should we predict, and how accurate is that prediction?

We will use the `Advertising.csv` data as the running example. The response is `Sales`, and the predictors are advertising budgets for `TV`, `Radio`, and `Newspaper` across 200 markets.

We keep the Chapter 7 sample-level notation: $\mathbf{y}$ is the observed response vector for the current data set, $\mathbf{X}$ is the design matrix, and $\hat{\mathbf{y}}$ is the fitted vector.

Math Foundations Connection. This chapter uses vector notation, matrix dimensions, matrix-vector multiplication, column-space ideas, linear combinations, dot products, transposes, orthogonality, and rank/full-column-rank intuition. These are covered in the Math Foundations textbook, Chapters 2, 3, 5, 7, 10, 11, and 12.

9.2 Simple Linear Regression and Multiple Linear Regression Use the Same Model Form

In simple linear regression, there is one predictor. For observation i , the model is

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i. \quad (9.2)$$

In vector form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (9.3)$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}. \quad (9.4)$$

In multiple linear regression with p predictors, observation i has predictor values

$$x_{i1}, x_{i2}, \dots, x_{ip}. \quad (9.5)$$

The model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i. \quad (9.6)$$

In vector form, it is still

$$\boxed{\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.} \quad (9.7)$$

The only change is the number of columns in $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbb{R}^{n \times (p+1)}. \quad (9.8)$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \in \mathbb{R}^{p+1}. \quad (9.9)$$

Key point. The notation $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ is not unique to multiple regression. It already describes simple linear regression. Multiple regression simply adds more predictor columns to the design matrix.

9.3 The Design Matrix View

The design matrix stores the predictors in a way that lets one matrix-vector multiplication generate all fitted values at once. For a coefficient vector $\boldsymbol{\beta}$,

$$\mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \cdots + \beta_p x_{1p} \\ \beta_0 + \beta_1 x_{21} + \cdots + \beta_p x_{2p} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \cdots + \beta_p x_{np} \end{bmatrix}. \quad (9.10)$$

Thus, the i th fitted value is the inner product of row i of $\mathbf{X}$ with $\boldsymbol{\beta}$:

$$\hat{y}_i = \mathbf{x}_{i,:}^T \hat{\boldsymbol{\beta}}. \quad (9.11)$$

There is an equally important column-space interpretation:

$$\hat{\mathbf{y}} = \hat{\beta}_0 \mathbf{1} + \hat{\beta}_1 \mathbf{x}_1 + \hat{\beta}_2 \mathbf{x}_2 + \cdots + \hat{\beta}_p \mathbf{x}_p. \quad (9.12)$$

Here $\mathbf{x}_j$ is the j th predictor column. Therefore,

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X}). \quad (9.13)$$

Regression chooses the linear combination of design-matrix columns that gets closest to $\mathbf{y}$ in squared-error distance.

Looking ahead. The columns of a design matrix do not have to be only raw numeric predictors. Later chapters use the same matrix machinery after adding encoded categorical predictors, transformed predictors, and interaction columns. The least-squares problem stays the same; what changes is the analyst's responsibility to choose columns that are interpretable, identifiable, and defensible.

9.4 What Changes Geometrically? Lines, Planes, and Hyperplanes

A simple linear regression fit looks like a line in a two-dimensional scatterplot. With two predictors, the fitted relationship is a plane:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2. \quad (9.14)$$

This plane lives in the ordinary three-dimensional plotting space with axes X_1 , X_2 , and Y .

With three predictors, the fitted relationship is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3. \quad (9.15)$$

This is a hyperplane. It is not easy to draw because the graph would need one response axis plus three predictor axes.

With p predictors, the fitted relationship is

$$\boxed{\hat{Y} = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j X_j.} \quad (9.16)$$

This is a p -dimensional hyperplane in $(p + 1)$ -dimensional predictor-response plotting space.

Do not confuse two spaces. There are two geometric pictures in play:

1. The **predictor-response plot**: a line, plane, or hyperplane in the space with predictor axes and a response axis.
2. The **least-squares projection**: an orthogonal projection of $\mathbf{y} \in \mathbb{R}^n$ onto $\text{Col}(\mathbf{X}) \subseteq \mathbb{R}^n$.

Both are useful. The first helps us visualize the model. The second explains the algebra.

9.5 Least Squares Estimation in Multiple Regression

The fitted coefficient vector minimizes the residual sum of squares:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b} \in \mathbb{R}^{p+1}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2. \quad (9.17)$$

The normal equations are

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \quad (9.18)$$

Equivalently,

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}. \quad (9.19)$$

If $\mathbf{X}$ has full column rank, then $\mathbf{X}^T \mathbf{X}$ is invertible, and

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.} \quad (9.20)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H}\mathbf{y}, \quad (9.21)$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (9.22)$$

is the hat matrix.

The residuals are

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}. \quad (9.23)$$

The residuals are orthogonal to every column of $\mathbf{X}$:

$$\boxed{\mathbf{X}^T \mathbf{e} = \mathbf{0}.} \quad (9.24)$$

This remains true whether there is one predictor or many.

9.6 The Advertising Data

The `Advertising.csv` data set contains observations from 200 markets. The selected response variable is

$$\text{Sales}, \quad (9.25)$$

measured in thousands of units sold. The predictors are advertising budgets, measured in thousands of dollars:

$$\text{TV}, \quad \text{Radio}, \quad \text{Newspaper}. \quad (9.26)$$

A natural multiple linear regression model is

$$\text{Sales}_i = \beta_0 + \beta_1 \text{TV}_i + \beta_2 \text{Radio}_i + \beta_3 \text{Newspaper}_i + \epsilon_i. \quad (9.27)$$

In vector form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (9.28)$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & \text{TV}_1 & \text{Radio}_1 & \text{Newspaper}_1 \\ 1 & \text{TV}_2 & \text{Radio}_2 & \text{Newspaper}_2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \text{TV}_{200} & \text{Radio}_{200} & \text{Newspaper}_{200} \end{bmatrix}. \quad (9.29)$$

Here

$$n = 200, \quad p = 3, \quad \mathbf{X} \in \mathbb{R}^{200 \times 4}. \quad (9.30)$$

If the columns are linearly independent, then the residual degrees of freedom are

$$n - p - 1 = 200 - 3 - 1 = 196. \quad (9.31)$$

9.7 A Scatterplot Matrix Is the First Reconnaissance Tool

With one predictor, a scatterplot can show the relationship between the predictor and response. With three predictors, we need to inspect several pairwise relationships. A scatterplot matrix displays the pairwise relationships among

$$\text{Sales}, \quad \text{TV}, \quad \text{Radio}, \quad \text{Newspaper}. \quad (9.32)$$

A scatterplot matrix helps answer two first-pass questions:

1. Which predictors appear most strongly associated with `Sales`?
2. Which predictors appear most strongly associated with each other?

The first question is about predictive signal. The second question is about collinearity, an early warning sign for the multicollinearity diagnostics developed later. A predictor can look useful by itself but become less useful once another correlated predictor is included in the model.

Important caution. Pairwise plots are not the final analysis. They are reconnaissance. Multiple regression coefficients are interpreted after holding the other included predictors fixed. Pairwise plots do not automatically show that adjusted relationship.

9.8 Question 1: Is At Least One Predictor Useful?

The first model-level question is whether at least one predictor is useful in predicting the response. For the Advertising model,

$$\text{Sales}_i = \beta_0 + \beta_1 \text{TV}_i + \beta_2 \text{Radio}_i + \beta_3 \text{Newspaper}_i + \epsilon_i, \quad (9.33)$$

the null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0. \quad (9.34)$$

The alternative is

$$H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \text{ is nonzero.} \quad (9.35)$$

The test statistic is the overall regression F -statistic:

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)}. \quad (9.36)$$

Here

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad (9.37)$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (9.38)$$

For the Advertising full model, the numerator degrees of freedom are $p = 3$, and the denominator degrees of freedom are $n - p - 1 = 196$. A large F -statistic provides evidence that at least one advertising budget is associated with **Sales**.

9.9 Question 2: Which Predictors Help?

Once the global F -test suggests that at least one predictor is useful, the next question is which predictors matter after adjusting for the others.

For each coefficient β_j , the usual individual test is

$$H_0 : \beta_j = 0 \quad \text{versus} \quad H_A : \beta_j \neq 0. \quad (9.39)$$

The corresponding t -statistic has the form

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}. \quad (9.40)$$

In the full Advertising model, the fitted coefficient estimates are commonly reported approximately as

Term	Estimate	Std. error	t-statistic	p-value
Intercept	2.939	0.312	9.42	very small
TV	0.046	0.0014	32.81	very small
Radio	0.189	0.0086	21.89	very small
Newspaper	-0.001	0.0059	-0.18	0.86

These results suggest that **TV** and **Radio** are useful after adjusting for the other included predictors. **Newspaper** does not appear useful after adjusting for **TV** and **Radio**.

Interpretation of the coefficient for TV. Holding **Radio** and **Newspaper** fixed, an additional 1 thousand dollars in **TV** advertising is associated with about 0.046 thousand additional units sold. That is about 46 additional units sold.

Interpretation of the coefficient for Radio. Holding **TV** and **Newspaper** fixed, an additional 1 thousand dollars in **Radio** advertising is associated with about 0.189 thousand additional units sold. That is about 189 additional units sold.

Interpretation of the coefficient for Newspaper. Holding **TV** and **Radio** fixed, an additional 1 thousand dollars in **Newspaper** advertising is associated with an estimated change near zero. In this model, **Newspaper** does not add meaningful predictive information after accounting for the other two media.

9.10 Why a Predictor Can Look Useful Alone but Not in the Full Model

A simple linear regression of **Sales** on **Newspaper** can show a positive association. But the full multiple regression model can assign **Newspaper** a coefficient near zero.

This is not a contradiction. It is a lesson about adjustment.

A simple regression asks:

As **Newspaper** changes, how does **Sales** tend to change, ignoring other predictors?

A multiple regression coefficient asks:

As **Newspaper** changes, how does **Sales** tend to change while **TV** and **Radio** are held fixed?

If **Newspaper** budgets are correlated with **Radio** budgets, then **Newspaper** may appear useful by itself because it is partly acting as a surrogate for **Radio**. Once **Radio** is included directly, **Newspaper** may no longer have its own contribution.

Reusable diagnostic phrase. Simple regression measures marginal association. Multiple regression measures adjusted association.

9.11 Question 3: How Well Does the Model Fit?

Two common fit summaries are the residual standard error and R^2 .

The residual standard error is

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n - p - 1}}. \quad (9.41)$$

It measures a typical residual size in the response units. For the Advertising full model, RSE is about 1.69. Since **Sales** is measured in thousands of units, this corresponds to a typical residual size of about 1,690 units.

The coefficient of determination is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}. \quad (9.42)$$

It measures the proportion of sample variation in **y** explained by the fitted model. For the Advertising full model, R^2 is about 0.897, meaning the model explains about 89.7 percent of the sample variation in **Sales**.

Important limitation. Training-set R^2 does not prove that the model will predict well on new data. It also does not prove that the model assumptions are correct. It is a fit summary, not a full validation report.

9.12 Adjusted R^2 as a Quick Complexity Penalty

Ordinary R^2 cannot decrease when a predictor is added to the model, because adding a column to **X** can only reduce or maintain the residual sum of squares. This can make ordinary R^2 too generous when comparing models with different numbers of predictors.

Adjusted R^2 introduces a degrees-of-freedom penalty:

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS} / (n - p - 1)}{\text{TSS} / (n - 1)}. \quad (9.43)$$

Unlike ordinary R^2 , adjusted R^2 can decrease when a predictor is added. It asks whether the reduction in RSS was large enough to justify the extra parameter.

Practical use. Adjusted R^2 is a quick model-comparison tool, but it should not replace out-of-sample validation, residual diagnostics, or domain reasoning.

9.13 Question 4: What Should We Predict and How Accurate Is the Prediction?

For new predictor values

$$\mathbf{x}_0 = [1 \quad x_{01} \quad x_{02} \quad \cdots \quad x_{0p}]^T, \quad (9.44)$$

the point prediction is

$$\boxed{\hat{y}_0 = \mathbf{x}_0^T \hat{\boldsymbol{\beta}}}. \quad (9.45)$$

For the Advertising model, suppose a market has

$$\text{TV} = 100, \quad \text{Radio} = 20, \quad \text{Newspaper} = 30, \quad (9.46)$$

where each value is in thousands of dollars. Using the approximate full-model estimates,

$$\widehat{\text{Sales}} \approx 2.939 + 0.046(100) + 0.189(20) - 0.001(30) \quad (9.47)$$

$$= 2.939 + 4.600 + 3.780 - 0.030 \quad (9.48)$$

$$= 11.289. \quad (9.49)$$

The predicted sales are about 11.289 thousand units, or 11,289 units.

There are two common uncertainty intervals:

1. A **confidence interval for the mean response**: uncertainty about the average sales among many markets with these same predictor values.
2. A **prediction interval for a new response**: uncertainty about the sales in one individual new market.

The prediction interval is wider because it includes both uncertainty in the estimated mean and the irreducible noise for a single future market.

9.14 Worked Example: Interpreting a Full Advertising Model

Assume the fitted model is approximately

$$\widehat{\text{Sales}} = 2.939 + 0.046\text{TV} + 0.189\text{Radio} - 0.001\text{Newspaper}. \quad (9.50)$$

All variables are in their original Advertising units: **Sales** in thousands of units and budgets in thousands of dollars.

Example 1: Increase TV while holding other media fixed

Suppose **TV** increases by 10, meaning 10 thousand dollars, while **Radio** and **Newspaper** are held fixed. Then

$$\Delta\widehat{\text{Sales}} = 0.046(10) = 0.460. \quad (9.51)$$

The predicted increase is about 0.460 thousand units, or 460 units.

Example 2: Increase Radio while holding other media fixed

Suppose **Radio** increases by 10, meaning 10 thousand dollars, while **TV** and **Newspaper** are held fixed. Then

$$\Delta\widehat{\text{Sales}} = 0.189(10) = 1.890. \quad (9.52)$$

The predicted increase is about 1.890 thousand units, or 1,890 units.

Example 3: Increase all three budgets

Suppose the budgets change by

$$\Delta\text{TV} = 10, \quad \Delta\text{Radio} = 10, \quad \Delta\text{Newspaper} = 10. \quad (9.53)$$

Then

$$\widehat{\Delta\text{Sales}} = 0.046(10) + 0.189(10) - 0.001(10) \quad (9.54)$$

$$= 0.460 + 1.890 - 0.010 \quad (9.55)$$

$$= 2.340. \quad (9.56)$$

The predicted increase is about 2.340 thousand units, or 2,340 units.

Operational interpretation. The fitted model suggests that, after adjusting for the other media, **Radio** has the largest estimated marginal effect per thousand dollars among the three predictors in this model. That does not automatically mean all future spending should go to **Radio**; the model is local to the observed data, linear in the included predictors, and does not yet account for nonlinear saturation, interactions, campaign constraints, or causal design.

9.15 Assumptions Behind the Practical Regression Workflow

For estimating a useful least squares line or hyperplane, the weakest core condition is often written as

$$\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}. \quad (9.57)$$

This says that the included predictors capture the conditional mean structure, leaving errors with mean zero after conditioning on the design matrix.

For the usual standard errors, t -tests, and F -tests, a common working assumption is

$$\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}. \quad (9.58)$$

This says the errors have constant variance and are uncorrelated across observations.

For exact finite-sample normal-theory inference, we often further assume

$$\epsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}). \quad (9.59)$$

This assumption is stronger than what is needed for least squares estimation itself, but it supports exact t - and F -distribution calculations in the classical model.

9.16 Why Graphics Become Difficult in Higher Dimensions

In simple linear regression, one scatterplot can show the predictor, response, fitted line, and residual pattern. In multiple regression, one plot cannot show everything.

With two predictors, a 3D plot can show a regression plane, but even then it can be hard to see residual patterns, leverage, and interactions. With three or more predictors, no ordinary static plot can show the full fitted hyperplane.

Therefore, the practical workflow shifts toward:

1. scatterplot matrices for pairwise reconnaissance,
2. correlation matrices for predictor relationships,
3. coefficient tables for adjusted effects,
4. residual plots for assumption checks,
5. partial residual or component-plus-residual plots for predictor-specific patterns,
6. test-set or cross-validation performance for generalization.

9.17 A Practical Modeling Workflow for the Advertising Example

A defensible multiple linear regression workflow for the Advertising data could look like this:

1. State the response and predictor units.
2. Create scatterplots of `Sales` against each predictor.
3. Create a scatterplot matrix for `Sales`, `TV`, `Radio`, and `Newspaper`.
4. Check predictor-predictor relationships, especially possible collinearity.
5. Fit the full model with all three predictors.
6. Use the overall F -test to ask whether at least one predictor is useful.
7. Use coefficient tests and domain reasoning to ask which predictors contribute after adjustment.
8. Compare simpler models such as `Sales` on `TV` and `Radio` only.
9. Report RSE and R^2 , but do not stop there.
10. Check residual plots for nonlinearity, heteroskedasticity, and unusual observations.
11. Use prediction or confidence intervals depending on whether the question is about one future market or the mean response at a predictor setting.
12. Document all modeling decisions so the analysis is reproducible and defensible.

Coding companion reference. Package-specific Python/R commands for scatterplot matrices, full and reduced multiple-regression models, prediction intervals, and residual plots have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the applied modeling workflow and interpretation guidance.

9.18 Common Mistakes

1. **Forgetting that coefficients are adjusted effects.** In MLR, $\hat{\beta}_j$ is interpreted while holding the other included predictors fixed.
2. **Treating pairwise plots as final proof.** Scatterplot matrices are useful reconnaissance, but they do not replace adjusted model interpretation.

3. **Confusing units.** In the Advertising data, `Sales` is in thousands of units and budgets are in thousands of dollars.
4. **Saying a nonsignificant coefficient proves no relationship.** It means the fitted model did not provide strong evidence for an adjusted linear effect under the model assumptions.
5. **Ignoring correlation among predictors.** Predictor-predictor relationships can change coefficient interpretation and standard errors.
6. **Using ordinary R^2 alone to choose variables.** Ordinary R^2 increases or stays the same when predictors are added.
7. **Mixing confidence intervals and prediction intervals.** Confidence intervals are for mean response; prediction intervals are for individual future responses.
8. **Drawing the hyperplane but skipping residual checks.** A fitted plane can still be the wrong shape if the true relationship is nonlinear or contains interactions.
9. **Assuming predictive association is causal.** Regression on observational data can estimate associations, but causal claims require design, assumptions, or additional evidence.

9.19 Chapter Summary

Multiple linear regression extends simple linear regression by adding predictor columns to the design matrix. The model remains

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (9.60)$$

With p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)}. \quad (9.61)$$

Least squares chooses the coefficient vector that minimizes squared residual length:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2. \quad (9.62)$$

If $\mathbf{X}$ has full column rank,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (9.63)$$

The fitted values live in $\text{Col}(\mathbf{X})$, and residuals are orthogonal to every column of $\mathbf{X}$. With one predictor, the fitted relationship is a line. With two predictors, it is a plane. With three or more predictors, it is a hyperplane.

The Advertising example illustrates the main practical questions in MLR. The overall F -test asks whether at least one predictor is useful. Individual coefficient tests ask whether a predictor contributes after adjusting for the others. RSE and R^2 summarize model fit. Prediction and confidence intervals quantify uncertainty for new predictor settings.

Most importantly, MLR coefficients are adjusted effects. A predictor can look useful in a simple regression but lose usefulness in a multiple regression after another correlated predictor is included.

9.20 Practice and Workflow

Focused Study Checklist

Use this as a final check after fitting and interpreting a multiple linear regression model.

1. Can you identify the response, predictors, and their units?
2. Can you write both the scalar model and the design-matrix form?
3. Can you interpret each coefficient as an adjusted effect with other predictors held fixed?
4. Can you use the global F -test for the question “is at least one predictor useful?”
5. Can you use coefficient tests for the question “which predictors contribute after adjustment?”
6. Can you report RSE, R^2 , and adjusted R^2 without relying on any one of them alone?
7. Can you use residual plots and EDA to check whether the fitted model is defensible?
8. Can you distinguish a confidence interval for a mean response from a prediction interval for a single future response?

Chapter 10

Truth + Error, Estimator Behavior, and Residual Geometry

10.1 The Big Picture

The previous chapter treated multiple linear regression as a practical modeling tool. We used the design matrix, the coefficient vector, and the least squares fit to answer applied questions about predictors, model fit, and prediction. This chapter steps back into theory for one reason:

We want to understand what the least squares estimates and residuals are doing, so that later diagnostics and hypothesis tests are more than button-clicking.

The model is still

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (10.1)$$

The difference is that we now treat this equation as a *truth plus error* representation. The term $\mathbf{X}\boldsymbol{\beta}$ is the systematic part of the response, and $\boldsymbol{\epsilon}$ is the random part that the model does not explain.

Notation bridge. Chapters 7, 9, 11, and 12 use $\mathbf{y}$ for the observed response vector from a realized data set. This chapter temporarily switches to $\mathbf{Y}$ because the full response vector is being treated as random once the design matrix $\mathbf{X}$ is fixed.

The central questions are:

1. If $\boldsymbol{\beta}$ is the true coefficient vector, how does $\hat{\boldsymbol{\beta}}$ behave?
2. Is $\hat{\boldsymbol{\beta}}$ centered on the true value $\boldsymbol{\beta}$?
3. How variable is $\hat{\boldsymbol{\beta}}$?
4. How do the residuals $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ relate to the true errors $\boldsymbol{\epsilon}$?
5. Why do residual plots tell us whether our modeling assumptions are plausible?

This chapter is the bridge from regression geometry to regression diagnostics. It explains why residuals are useful, but also why they are not the same thing as the true errors.

Math Foundations Connection. This chapter uses vector notation, transposes, matrix-vector multiplication, identity matrices, orthogonality, rank, column-space/residual-space geometry, least-squares normal equations, and the $\mathbf{A}^T \mathbf{A}$ Hessian. These connections are covered in the Math Foundations textbook, Chapters 2, 5, 7, 10, 11, 12, and 13.

10.2 The Regression Model with Fixed Design Matrix

We start with the multiple linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (10.2)$$

where

$$\mathbf{Y} \in \mathbb{R}^n, \quad \mathbf{X} \in \mathbb{R}^{n \times (p+1)}, \quad \boldsymbol{\beta} \in \mathbb{R}^{p+1}, \quad \boldsymbol{\epsilon} \in \mathbb{R}^n. \quad (10.3)$$

The design matrix $\mathbf{X}$ includes an intercept column of ones and p predictor columns.

In this chapter, we use a fixed-design view:

Treat $\mathbf{X}$ as fixed, and let $\mathbf{Y}$ vary randomly because $\boldsymbol{\epsilon}$ varies randomly.

That is the answer to the question, “Where does the randomness come from?” It comes from the noise vector $\boldsymbol{\epsilon}$. Once $\mathbf{X}$ is fixed, the only random object on the right side of (10.2) is $\boldsymbol{\epsilon}$, so the only random object on the left side is $\mathbf{Y}$.

Conditional view. If the predictors are themselves random in the data-generating process, the results in this chapter are usually read conditionally on the observed $\mathbf{X}$. In words: after we observe the predictor values, we analyze the randomness in the response around the regression surface.

10.3 The Big Three Regression Assumptions

Chapter 5 introduced a three-step regression assumption ladder. We now restate that same ladder in matrix form. The stronger assumptions support stronger inferential claims.

Assumption 1: Mean-zero errors

The first assumption is

$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}. \quad (10.4)$$

Equivalently, for each observation,

$$\mathbb{E}[\epsilon_i \mid \mathbf{X}] = 0. \quad (10.5)$$

This says the model is centered correctly. After accounting for the predictors in $\mathbf{X}$, the remaining error should not systematically lean positive or negative.

Under this assumption,

$$\mathbb{E}[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}. \quad (10.6)$$

Thus $\mathbf{X}\boldsymbol{\beta}$ is the conditional mean of the response.

Assumption 2: Constant variance and uncorrelated errors

The second assumption is

$$\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n. \quad (10.7)$$

This compact matrix statement contains two ideas:

1. Each error has the same variance: $\text{Var}(\epsilon_i \mid \mathbf{X}) = \sigma^2$.
2. Different errors are uncorrelated: $\text{Cov}(\epsilon_i, \epsilon_j \mid \mathbf{X}) = 0$ for $i \neq j$.

The first part is called homoskedasticity. The second part rules out error correlation. Violations of this assumption are why residual plots, time-order plots, and variance checks matter.

Assumption 3: Gaussian errors

The third assumption is

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n). \quad (10.8)$$

This says the full error vector is multivariate normal with mean zero and covariance $\sigma^2 \mathbf{I}_n$. This is stronger than the first two assumptions. It supports exact finite-sample t - and F -based inference for the usual linear model quantities.

Practical interpretation. The first assumption gives correct centering. The second assumption gives the usual variance formulas. The third assumption gives exact distributional statements. In practice, diagnostics help us decide how believable these assumptions are.

10.4 Recall: Least Squares Estimator

This section restates the Chapter 7 least-squares formula in the fixed-design notation of this chapter. The formula is not new; the new question is how it behaves when $\mathbf{Y}$ is random.

The least squares estimator minimizes the squared residual length:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b} \in \mathbb{R}^{p+1}} \|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2. \quad (10.9)$$

Assuming $\mathbf{X}$ has full column rank, the normal equations give

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{Y}, \quad (10.10)$$

and therefore

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}.} \quad (10.11)$$

The fitted response vector is

$$\hat{\mathbf{Y}} = \mathbf{X} \hat{\boldsymbol{\beta}}. \quad (10.12)$$

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}. \quad (10.13)$$

10.5 Dimension Check

The formulas in this chapter are much easier to trust if the dimensions work. Let $q = p + 1$, the number of columns in $\mathbf{X}$. Then:

Object	Meaning	Dimension
$\mathbf{Y}$	response vector	$n \times 1$
$\mathbf{X}$	design matrix	$n \times q$
$\boldsymbol{\beta}$	true coefficient vector	$q \times 1$
$\hat{\boldsymbol{\beta}}$	estimated coefficient vector	$q \times 1$
$\boldsymbol{\epsilon}$	true error vector	$n \times 1$
$\hat{\mathbf{Y}}$	fitted response vector	$n \times 1$
$\mathbf{e}$	residual vector	$n \times 1$
$\mathbf{H}$	hat matrix	$n \times n$
$\mathbf{I}_n - \mathbf{H}$	residual-maker matrix	$n \times n$

The matrix

$$\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (10.14)$$

has dimension $q \times n$. It maps the response vector $\mathbf{Y} \in \mathbb{R}^n$ into the coefficient estimate $\hat{\boldsymbol{\beta}} \in \mathbb{R}^q$:

$$\hat{\boldsymbol{\beta}} = \mathbf{A} \mathbf{Y}. \quad (10.15)$$

10.6 Truth + Error Representation for $\hat{\boldsymbol{\beta}}$

Start with the least squares estimator:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}. \quad (10.16)$$

Substitute the regression model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \quad (10.17)$$

$$= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon} \quad (10.18)$$

$$= \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon}. \quad (10.19)$$

Therefore,

$$\boxed{\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon}.} \quad (10.20)$$

This is the truth plus error representation for the coefficient estimator. It says:

The estimated coefficient vector equals the true coefficient vector plus a transformed version of the random noise vector.

Each coefficient estimate has its own associated error term. That error is not a new source of randomness; it is produced by the same $\boldsymbol{\epsilon}$ that created the random response values.

10.7 Unbiasedness of $\hat{\beta}$

Using (10.20), take conditional expectations given $\mathbf{X}$:

$$\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \mathbb{E}[\beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon \mid \mathbf{X}] \quad (10.21)$$

$$= \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbb{E}[\epsilon \mid \mathbf{X}]. \quad (10.22)$$

If $\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}$, then

$$\boxed{\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta.} \quad (10.23)$$

So $\hat{\beta}$ is unbiased under the mean-zero error assumption.

What unbiased means here. Unbiased does not mean that $\hat{\beta}$ equals β in a particular dataset. It means that if we repeatedly observed new response vectors from the same design matrix and same true model, the average of the coefficient estimates would center on the true coefficient vector.

10.8 Variance of $\hat{\beta}$

The truth plus error representation also makes the variance calculation direct. Define

$$\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (10.24)$$

Then

$$\hat{\beta} = \beta + \mathbf{A}\epsilon. \quad (10.25)$$

Since β is fixed, it has no variance. Therefore,

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \text{Var}(\mathbf{A}\epsilon \mid \mathbf{X}). \quad (10.26)$$

The matrix variance rule is

$$\text{Var}(\mathbf{A}\mathbf{Z}) = \mathbf{A} \text{Var}(\mathbf{Z}) \mathbf{A}^T. \quad (10.27)$$

Using $\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$,

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \mathbf{A} \text{Var}(\epsilon \mid \mathbf{X}) \mathbf{A}^T \quad (10.28)$$

$$= \mathbf{A}(\sigma^2 \mathbf{I}_n) \mathbf{A}^T \quad (10.29)$$

$$= \sigma^2 \mathbf{A} \mathbf{A}^T. \quad (10.30)$$

Now substitute $\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$:

$$\mathbf{A} \mathbf{A}^T = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T [(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T]^T \quad (10.31)$$

$$= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} [(\mathbf{X}^T \mathbf{X})^{-1}]^T. \quad (10.32)$$

Because $\mathbf{X}^T \mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,

$$\mathbf{A} \mathbf{A}^T = (\mathbf{X}^T \mathbf{X})^{-1}. \quad (10.33)$$

So

$$\boxed{\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}.} \quad (10.34)$$

What this means. The coefficient uncertainty depends on two things:

1. the error variance σ^2 , and
2. the geometry of the design matrix through $(\mathbf{X}^T \mathbf{X})^{-1}$.

If the errors are noisy, coefficient estimates are noisy. If the predictor columns are poorly arranged, nearly dependent, or badly scaled, then $(\mathbf{X}^T \mathbf{X})^{-1}$ can contain large entries, making coefficient estimates unstable.

10.9 Standard Errors of the Coefficients

The diagonal entries of $\text{Var}(\hat{\boldsymbol{\beta}} | \mathbf{X})$ give the variances of the individual coefficient estimates. Let

$$\mathbf{C} = (\mathbf{X}^T \mathbf{X})^{-1}. \quad (10.35)$$

Then

$$\text{Var}(\hat{\beta}_j | \mathbf{X}) = \sigma^2 C_{jj}. \quad (10.36)$$

The standard error of $\hat{\beta}_j$ is

$$\text{SE}(\hat{\beta}_j) = \sigma \sqrt{C_{jj}}. \quad (10.37)$$

Because σ is unknown, we estimate it using the residuals:

$$\hat{\sigma}^2 = \frac{\mathbf{e}^T \mathbf{e}}{n - p - 1} = \frac{\text{RSS}}{n - p - 1}. \quad (10.38)$$

Thus the estimated standard error is

$$\boxed{\widehat{\text{SE}}(\hat{\beta}_j) = \hat{\sigma} \sqrt{C_{jj}}.} \quad (10.39)$$

This is the quantity reported in regression output tables. It is the denominator of the usual t -statistic.

10.10 Multicollinearity and Coefficient Instability

Chapter 8 described exact non-identifiability: if one design column is an exact linear combination of the others, then $\mathbf{X}^T \mathbf{X}$ is singular and the coefficient vector is not uniquely determined. Multicollinearity is the nearby practical problem. The columns may not be exactly dependent, so the least-squares estimate is unique, but they can be close enough to dependent that individual coefficients become fragile.

The reason is already visible in the variance formula:

$$\text{Var}(\hat{\boldsymbol{\beta}} | \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}. \quad (10.40)$$

When columns of $\mathbf{X}$ are nearly linearly dependent, $\mathbf{X}^T \mathbf{X}$ is close to singular. Its inverse may then have large entries. In particular, large diagonal entries of $(\mathbf{X}^T \mathbf{X})^{-1}$ inflate the variances and standard errors of the corresponding coefficient estimates.

The practical consequence is a separation between fitted values and individual coefficient interpretation. The model may estimate a combined direction of two related predictors well enough for useful fitted values or prediction, while struggling to assign stable separate credit to each predictor. Thus multicollinearity is mainly an interpretation and inference warning, not automatically a prediction failure.

10.11 Distribution of $\hat{\beta}$

If we also assume Gaussian errors,

$$\epsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n), \quad (10.41)$$

then $\hat{\beta}$ is a linear transformation of a multivariate normal random vector. Linear transformations of multivariate normal vectors are multivariate normal. Therefore,

$$\boxed{\hat{\beta} \mid \mathbf{X} \sim N(\beta, \sigma^2(\mathbf{X}^T \mathbf{X})^{-1})}. \quad (10.42)$$

For an individual coefficient,

$$\hat{\beta}_j \mid \mathbf{X} \sim N(\beta_j, \sigma^2 C_{jj}). \quad (10.43)$$

After estimating σ , the usual test statistic is

$$t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{\widehat{\text{SE}}(\hat{\beta}_j)}, \quad (10.44)$$

where $\beta_{j,0}$ is the hypothesized value, usually 0. Under the Gaussian model and the null hypothesis,

$$t_j \sim t_{n-p-1}. \quad (10.45)$$

Connection to inference. This is why regression output can include coefficient tests and confidence intervals. The theory starts with the truth plus error representation, then uses the error assumptions to describe the sampling behavior of $\hat{\beta}$.

10.12 Takeaways for Coefficients

The coefficient estimator has three central properties under the standard linear model assumptions:

1. **Truth plus error:**

$$\hat{\beta} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon. \quad (10.46)$$

2. **Unbiasedness under mean-zero errors:**

$$\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta. \quad (10.47)$$

3. **Variance under homoskedastic uncorrelated errors:**

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}. \quad (10.48)$$

If the Gaussian assumption is added, then $\hat{\beta}$ has a multivariate normal distribution.

10.13 The Hat Matrix Revisited

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} \quad (10.49)$$

$$= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}. \quad (10.50)$$

Define the hat matrix

$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T.} \quad (10.51)$$

Then

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}. \quad (10.52)$$

The matrix $\mathbf{H}$ puts the hat on $\mathbf{Y}$, which is why it is called the hat matrix.

The hat matrix is an orthogonal projection matrix onto $\text{Col}(\mathbf{X})$. It has two key properties:

1. Symmetry:

$$\mathbf{H}^T = \mathbf{H}. \quad (10.53)$$

2. Idempotence:

$$\mathbf{H}^2 = \mathbf{H}. \quad (10.54)$$

Symmetry and idempotence are the algebraic signatures of an orthogonal projection.

10.14 The Residual Vector

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}. \quad (10.55)$$

Using $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$,

$$\mathbf{e} = \mathbf{Y} - \mathbf{H}\mathbf{Y} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}. \quad (10.56)$$

The matrix

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H} \quad (10.57)$$

is often called the residual-maker matrix because

$$\mathbf{e} = \mathbf{M}\mathbf{Y}. \quad (10.58)$$

Now substitute $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$:

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \quad (10.59)$$

$$= (\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (10.60)$$

Because $\mathbf{H}$ projects onto $\text{Col}(\mathbf{X})$, it leaves every vector in $\text{Col}(\mathbf{X})$ unchanged. Since $\mathbf{X}\boldsymbol{\beta} \in \text{Col}(\mathbf{X})$,

$$\mathbf{H}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}. \quad (10.61)$$

Therefore,

$$(\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}, \quad (10.62)$$

and

$$\boxed{\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.} \quad (10.63)$$

This is the truth plus error representation for residuals.

Important interpretation. Residuals are not the true errors. The true errors are ϵ . The residuals are projected errors:

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\epsilon. \quad (10.64)$$

Least squares removes from ϵ the component that lies in $\text{Col}(\mathbf{X})$. What remains is the component orthogonal to the fitted-value space.

10.15 Why $\mathbf{I}_n - \mathbf{H}$ Is Also an Orthogonal Projection

If $\mathbf{H}$ is an orthogonal projection, then it is symmetric and idempotent. Let

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}. \quad (10.65)$$

We show that $\mathbf{M}$ is also an orthogonal projection.

First, symmetry:

$$\mathbf{M}^T = (\mathbf{I}_n - \mathbf{H})^T \quad (10.66)$$

$$= \mathbf{I}_n^T - \mathbf{H}^T \quad (10.67)$$

$$= \mathbf{I}_n - \mathbf{H} \quad (10.68)$$

$$= \mathbf{M}. \quad (10.69)$$

Second, idempotence:

$$\mathbf{M}^2 = (\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H}) \quad (10.70)$$

$$= \mathbf{I}_n - 2\mathbf{H} + \mathbf{H}^2 \quad (10.71)$$

$$= \mathbf{I}_n - 2\mathbf{H} + \mathbf{H} \quad (10.72)$$

$$= \mathbf{I}_n - \mathbf{H} \quad (10.73)$$

$$= \mathbf{M}. \quad (10.74)$$

Therefore $\mathbf{I}_n - \mathbf{H}$ is also an orthogonal projection matrix.

Geometrically, $\mathbf{H}$ projects $\mathbf{Y}$ onto $\text{Col}(\mathbf{X})$, while $\mathbf{I}_n - \mathbf{H}$ projects $\mathbf{Y}$ onto the orthogonal complement of $\text{Col}(\mathbf{X})$. That orthogonal complement is where residuals live.

10.16 Expectation of the Residual Vector

Using (10.63),

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\epsilon. \quad (10.75)$$

Take conditional expectations:

$$\mathbb{E}[\mathbf{e} \mid \mathbf{X}] = (\mathbf{I}_n - \mathbf{H})\mathbb{E}[\epsilon \mid \mathbf{X}]. \quad (10.76)$$

If $\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}$, then

$$\boxed{\mathbb{E}[\mathbf{e} \mid \mathbf{X}] = \mathbf{0}.} \quad (10.77)$$

This is why we expect residual plots to be centered around zero when the mean structure is correctly specified.

10.17 Variance of the Residual Vector

Again use

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (10.78)$$

By the matrix variance rule,

$$\text{Var}(\mathbf{e} \mid \mathbf{X}) = (\mathbf{I}_n - \mathbf{H}) \text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X})(\mathbf{I}_n - \mathbf{H})^T. \quad (10.79)$$

If $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$, then

$$\text{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2 (\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H})^T \quad (10.80)$$

$$= \sigma^2 (\mathbf{I}_n - \mathbf{H})^2 \quad (10.81)$$

$$= \sigma^2 (\mathbf{I}_n - \mathbf{H}). \quad (10.82)$$

Thus

$$\boxed{\text{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2 (\mathbf{I}_n - \mathbf{H})}. \quad (10.83)$$

This formula has an important consequence. The i th residual has variance

$$\text{Var}(e_i \mid \mathbf{X}) = \sigma^2 (1 - h_{ii}), \quad (10.84)$$

where h_{ii} is the i th diagonal entry of the hat matrix. The off-diagonal entries give residual covariances:

$$\text{Cov}(e_i, e_j \mid \mathbf{X}) = -\sigma^2 h_{ij}, \quad i \neq j. \quad (10.85)$$

Key diagnostic warning. Even if the true errors are uncorrelated and have equal variance, the raw residuals generally do not have equal variance and are generally correlated. This is why leverage-adjusted residuals and studentized residuals are useful in diagnostics.

10.18 Distribution of the Residual Vector

If the Gaussian assumption holds,

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n), \quad (10.86)$$

then $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}$ is a linear transformation of a multivariate normal vector. Therefore,

$$\boxed{\mathbf{e} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 (\mathbf{I}_n - \mathbf{H}))}. \quad (10.87)$$

The covariance matrix is singular because $\mathbf{e}$ is constrained to lie in the residual subspace, which has dimension $n - p - 1$, not dimension n . The residual vector cannot point anywhere it wants in $\mathbb{R}^n$; it must be orthogonal to every column of $\mathbf{X}$.

10.19 Residual Degrees of Freedom

The residual degrees of freedom are the dimensions left over after fitting the model. Algebraically,

$$\text{df}_{\text{res}} = \text{tr}(\mathbf{I}_n - \mathbf{H}). \quad (10.88)$$

Because $\text{tr}(\mathbf{I}_n) = n$ and $\text{tr}(\mathbf{H}) = p + 1$ when $\mathbf{X}$ has full column rank,

$$\boxed{\text{df}_{\text{res}} = n - p - 1.} \quad (10.89)$$

This also explains the denominator in the residual variance estimate:

$$\hat{\sigma}^2 = \frac{\mathbf{e}^T \mathbf{e}}{n - p - 1}. \quad (10.90)$$

The numerator $\mathbf{e}^T \mathbf{e}$ is the residual sum of squares. The denominator is the number of leftover residual dimensions.

10.20 Worked Mini Example: A Three-Point Simple Regression

Consider three observations with one predictor and an intercept:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}. \quad (10.91)$$

Here $n = 3$, $p = 1$, and $p + 1 = 2$. The fitted values live in a two-dimensional plane inside $\mathbb{R}^3$, and the residuals live in the one-dimensional orthogonal complement.

Compute

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad (10.92)$$

and

$$(\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}. \quad (10.93)$$

Thus the coefficient variance matrix is

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \frac{\sigma^2}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}. \quad (10.94)$$

So

$$\text{Var}(\hat{\beta}_0) = \frac{5}{6}\sigma^2, \quad \text{Var}(\hat{\beta}_1) = \frac{1}{2}\sigma^2. \quad (10.95)$$

The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix}. \quad (10.96)$$

The residual-maker matrix is

$$\mathbf{I}_3 - \mathbf{H} = \begin{bmatrix} \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \end{bmatrix}. \quad (10.97)$$

Notice that

$$\text{tr}(\mathbf{H}) = 2, \quad \text{tr}(\mathbf{I}_3 - \mathbf{H}) = 1. \quad (10.98)$$

This matches the geometry: two fitted-value degrees of freedom and one residual degree of freedom.

10.21 What Residual Plots Are Really Checking

The residual representation

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon} \quad (10.99)$$

explains why residual analysis works. If the true errors have mean zero, constant variance, and approximate normality, then the residuals should show compatible behavior.

Common diagnostic checks include:

Diagnostic plot	What we hope to see	Possible problem if violated
Residuals vs fitted values	random cloud around zero	nonlinearity or missing structure
Residuals vs predictor	random cloud around zero	predictor transformation needed
Scale-location plot	similar vertical spread	heteroskedasticity
Q-Q plot	points close to a line	non-normal error behavior
Residuals vs time/order	no runs or waves	correlated errors
Residuals vs leverage	no extreme influential points	high leverage or outliers

A good residual plot does not prove that the model is true. It only suggests that the most visible violations are not obvious. A bad residual plot is more informative: it gives evidence that something in the modeling assumptions or model form may need attention.

10.22 Good Residual Behavior

A residual plot with no visible pattern is usually a good sign. A healthy residual-vs-fitted plot should look like a random scatter of points around the horizontal line at zero. The points do not need to be perfectly symmetric, but they should not show a systematic curve, trend, funnel, or clustering pattern.

A random-looking residual cloud suggests:

1. the conditional mean may be reasonably specified,
2. the variance may be roughly stable over the fitted range, and
3. there may not be an obvious missing nonlinear term.

This is why residuals are the natural next object after coefficients. Coefficients tell us what the fitted model says. Residuals tell us what the fitted model failed to explain.

10.23 Bad Residual Behavior

Bad residual behavior usually points to one of a few issues:

1. **Curvature:** residuals form a U-shape or other smooth pattern. This suggests the mean function is not linear in the current predictors.
2. **Funnel shape:** residual spread increases or decreases with fitted values. This suggests nonconstant variance.
3. **Runs over time:** residuals stay positive or negative for long stretches in time order. This suggests correlated errors.
4. **Extreme residuals:** a few residuals are much larger than the rest. This suggests outliers or observations not explained by the model.
5. **High leverage:** some rows of $\mathbf{X}$ are far from the rest. These observations may strongly influence the fitted model.

Residual analysis is not just a plot-making exercise. It is a theory-backed health check of the assumptions that allowed us to interpret $\hat{\beta}$, standard errors, t -statistics, and prediction intervals.

10.24 Residuals, Leverage, and Studentization

The diagonal hat values

$$h_{ii} \tag{10.100}$$

measure leverage. An observation has high leverage when its predictor pattern is unusual relative to the rest of the design matrix.

Because

$$\text{Var}(e_i \mid \mathbf{X}) = \sigma^2(1 - h_{ii}), \tag{10.101}$$

raw residuals do not all have the same variance. A leverage-adjusted residual divides by its estimated standard deviation:

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}. \tag{10.102}$$

This is often called an internally studentized residual. It gives a more comparable scale for judging which residuals are unusually large.

Why leverage matters. A point with a large residual but low leverage may be an outlier in the response direction. A point with high leverage may control the fitted hyperplane even if its residual is not large. A point with both high leverage and a large residual is especially important to inspect.

Chapter 12 turns these objects into an applied influence workflow. The diagonal values h_{ii} , leverage-adjusted residuals, and Cook's distance help identify observations that deserve investigation. High influence is not an automatic deletion order; it is a signal to check data quality, model form, and sensitivity of the conclusion.

10.25 A Clean Interpretation of Diagnostics

The following table connects each theoretical assumption to the residual behavior we expect to see.

Assumption about ϵ	Residual implication	Diagnostic clue
$\mathbb{E}[\epsilon \mathbf{X}] = 0$	residuals centered near zero	no systematic trend
$\text{Var}(\epsilon \mathbf{X}) = \sigma^2 \mathbf{I}$	stable spread, no strong correlation	no funnel or tracking
Gaussian errors	residuals roughly normal after adjustment	Q-Q plot roughly linear
Correct linear mean structure	no missing pattern in residuals	no curve or clusters
Full column rank	unique coefficients	no exact multicollinearity

10.26 Why This Matters for Modeling Decisions

The coefficient theory and residual theory combine into a modeling workflow:

1. Fit a model.
2. Interpret $\hat{\beta}$, standard errors, and model fit.
3. Check residual behavior.
4. If residuals behave badly, consider model repair.

Model repair might include transforming predictors, transforming the response, adding interactions, adding nonlinear terms, using weighted least squares, using robust standard errors, or moving to a different model class. The next chapters build toward these tools.

The key point is that residual diagnostics are not separate from the math. They come directly from

$$\hat{\beta} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon \quad (10.103)$$

and

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\epsilon. \quad (10.104)$$

Coding companion reference. Package-specific Python/R commands for extracting fitted values, residuals, leverage, studentized residuals, diagnostic plots, and coefficient covariance matrices have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the theoretical residual and estimator behavior.

10.27 Common Mistakes

Mistake 1: Treating residuals as the true errors

The residuals are not ϵ . They are

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\epsilon. \quad (10.105)$$

They are the part of the error vector left after projecting away the fitted-value component.

Mistake 2: Forgetting the fixed-design condition

The cleanest formulas are conditional on $\mathbf{X}$. If the predictors are random, we usually interpret the results after conditioning on the observed design matrix.

Mistake 3: Thinking unbiased means correct in this dataset

Unbiased means centered correctly over repeated samples. It does not mean the estimate is close in one observed dataset.

Mistake 4: Ignoring multicollinearity in the variance formula

The variance formula

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1} \quad (10.106)$$

shows why nearly dependent predictor columns can cause large standard errors.

Mistake 5: Using residual plots as proof

A clean residual plot is not proof that assumptions hold. It is evidence that obvious violations are not visible. A bad residual plot is stronger evidence because it points to a concrete failure mode.

10.28 Operational Briefing Checklist

When briefing a regression model, be ready to answer the following questions:

1. What is the response variable, and what are the predictors?
2. What is the design matrix dimension $n \times (p + 1)$?
3. Is $\mathbf{X}$ full column rank, or is there an identifiability concern?
4. What assumptions are needed for the coefficient standard errors?
5. What assumptions are needed for exact t -tests and confidence intervals?
6. What do the residual plots show?
7. Are residuals centered near zero?
8. Is residual spread roughly stable?
9. Are there patterns suggesting nonlinearity, correlated errors, or missing variables?
10. Are there high leverage points or unusually large studentized residuals?

10.29 Chapter Summary

The core equations of this chapter are:

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon}, \quad (10.107)$$

$$\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}, \quad (10.108)$$

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}, \quad (10.109)$$

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H}) \boldsymbol{\epsilon}, \quad (10.110)$$

$$\text{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2 (\mathbf{I}_n - \mathbf{H}). \quad (10.111)$$

The coefficient estimator is the truth plus a transformed error. Under mean-zero errors, it is unbiased. Under constant-variance uncorrelated errors, its variance is governed by $\sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$. Under Gaussian errors, it has a multivariate normal distribution.

The residual vector is the true error vector after projection onto the residual subspace. This explains why residuals can be used to diagnose model assumptions. Residuals should be centered around zero, have roughly stable spread, and show no systematic patterns if the model assumptions are reasonable. Chapter 12 turns these facts into a practical diagnostic and repair workflow.

10.30 Practice and Workflow

Focused Study Checklist

Use this as a final check after studying the truth-plus-error representation.

1. Can you write $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ and explain why $\mathbf{Y}$ is random under fixed design?
2. Can you state the coefficient decomposition

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon}?$$

3. Can you identify which assumption gives unbiasedness and which assumptions support the usual variance and distribution results?
4. Can you write the residual relation $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}$?
5. Can you explain why residuals are useful diagnostics but not the same as the true errors?
6. Can you connect a residual pattern to a likely assumption failure?

Chapter 11

Linear Models Beyond Straight Lines: Transformations, Categorical Predictors, and Interactions

11.1 The Big Picture

The previous chapters built multiple linear regression from two viewpoints:

1. Algebraically, the fitted values are produced by a design matrix and a coefficient vector.
2. Geometrically, the fitted response vector is the projection of $\mathbf{y}$ onto the column space of the design matrix.

This chapter makes an important expansion. Earlier chapters used "linear" mainly through the geometric picture of lines, planes, and hyperplanes in the displayed predictors. That picture was useful, but incomplete:

A linear model does not require the response to be a straight-line function of the original predictor values. It only requires the model to be linear in the unknown coefficients.

That distinction is easy to miss. The model

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon \quad (11.1)$$

is curved as a function of x , but it is still a linear model in the parameters $\beta_0, \beta_1, \beta_2$. Once we create the transformed predictor column x^2 , ordinary least squares can be used exactly as before. The same idea lets us use category indicators and interaction columns while keeping the model linear in the coefficients.

The central question is therefore not, "Is the graph a straight line?" The better question is:

Can the model be written as a design matrix times a coefficient vector, plus error?

If the answer is yes, then the linear regression machinery still applies.

We also return to the sample-level notation of Chapters 7 and 9: $\mathbf{y}$ is the observed response vector from the data set being fit.

Math Foundations Connection. This chapter uses the Math Foundations ideas of vectors, sets, linear combinations, bases, matrix-vector multiplication, submatrices, linear independence, null spaces, and full-column-rank inverse-like formulas. Transformed regression, categorical regression, and interaction modeling are all ways of building columns for the design matrix. Those columns form candidate directions in $\mathbb{R}^n$, and least squares chooses the best linear combination of those directions. See the Math Foundations textbook, Chapters 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12.

11.2 What “Linear” Means Here

A model is linear in parameters when it can be written as

$$m(x; \boldsymbol{\theta}) = \theta_0 g_0(x) + \theta_1 g_1(x) + \cdots + \theta_q g_q(x), \quad (11.2)$$

where the functions $g_0, g_1, \dots, g_q$ are known functions of the input values, and the unknowns are the coefficients $\theta_0, \theta_1, \dots, \theta_q$. Usually $g_0(x) = 1$, which gives the intercept column.

The key is that the unknown parameters appear only as coefficients multiplying known columns. They are not multiplied by each other, raised to powers, placed inside nonlinear functions, or used as exponents.

For observations $i = 1, \dots, n$, define

$$z_{ij} = g_j(x_i), \quad j = 0, 1, \dots, q. \quad (11.3)$$

Then the model becomes

$$Y_i = \theta_0 z_{i0} + \theta_1 z_{i1} + \cdots + \theta_q z_{iq} + \epsilon_i. \quad (11.4)$$

Stacking the observations gives

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad (11.5)$$

where

$$\mathbf{Z} = \begin{bmatrix} z_{10} & z_{11} & \cdots & z_{1q} \\ z_{20} & z_{21} & \cdots & z_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n0} & z_{n1} & \cdots & z_{nq} \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_q \end{bmatrix}. \quad (11.6)$$

This is the same matrix form as ordinary multiple linear regression. The only difference is that the design matrix may contain transformed versions of the original variables, category indicators, and interaction columns.

Original predictor space versus feature space

It is helpful to distinguish two spaces:

1. **Original predictor space:** the variables as measured, such as x , x_1 , x_2 , temperature, distance, or time.
2. **Feature space:** the columns actually used in the design matrix, such as x , x^2 , $\log x$, $\sin x$, $x_1 x_2$, or category indicators.

Linear regression is linear in the columns of the design matrix. Those columns can be created from nonlinear transformations of the original predictors.

11.3 The Least Squares Problem with Transformed Columns

Once the transformed design matrix $\mathbf{Z}$ is built, the least squares objective is

$$\min_{\mathbf{b} \in \mathbb{R}^{q+1}} \|\mathbf{y} - \mathbf{Z}\mathbf{b}\|^2. \quad (11.7)$$

The normal equations are

$$\mathbf{Z}^T \mathbf{Z} \hat{\boldsymbol{\theta}} = \mathbf{Z}^T \mathbf{y}. \quad (11.8)$$

If $\mathbf{Z}$ has full column rank, then

$$\boxed{\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}.} \quad (11.9)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{Z} \hat{\boldsymbol{\theta}}. \quad (11.10)$$

The hat matrix for the transformed model is

$$\mathbf{H}_Z = \mathbf{Z}(\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T, \quad (11.11)$$

and therefore

$$\hat{\mathbf{y}} = \mathbf{H}_Z \mathbf{y}, \quad \mathbf{e} = (\mathbf{I}_n - \mathbf{H}_Z) \mathbf{y}. \quad (11.12)$$

Thus the geometry from Chapter 7 and the residual theory from Chapter 10 still apply. The projection subspace has changed because the design matrix columns changed, but the projection logic is the same.

11.4 A Dimension Check

Let $q + 1$ be the number of columns in the transformed design matrix $\mathbf{Z}$. Then:

Object	Meaning	Dimension
$\mathbf{y}$	response vector	$n \times 1$
$\mathbf{Z}$	transformed design matrix	$n \times (q + 1)$
$\boldsymbol{\theta}$	coefficient vector for transformed columns	$(q + 1) \times 1$
$\hat{\boldsymbol{\theta}}$	least squares estimate	$(q + 1) \times 1$
$\hat{\mathbf{y}}$	fitted response vector	$n \times 1$
$\mathbf{e}$	residual vector	$n \times 1$
$\mathbf{H}_Z$	hat matrix for transformed model	$n \times n$

The model is valid as a least squares linear model if the columns of $\mathbf{Z}$ are well-defined and have enough independent information to identify $\boldsymbol{\theta}$. If $\mathbf{Z}^T \mathbf{Z}$ is singular, then the coefficient vector is not uniquely identified.

11.5 Mayer's Moon Motion Problem

The motivating example in the source slides is the Moon motion problem. The Moon is tidally locked to Earth, but not perfectly. Its apparent motion includes libration in longitude, latitude, and diurnal effects. Because of this wobble, observers can see more than exactly half of the Moon's near side over time.

Tobias Mayer collected measurements to describe this apparent wobbling motion, called libration. The important statistical point is that the response did not need to be modeled as a straight line in the original physical angle. Instead, Mayer's equation used transformed predictors:

$$Y = \beta + \alpha x_1 + \gamma x_2 + \epsilon, \quad (11.13)$$

where

$$Y = \text{arc/libration measurement}, \quad x_1 = \sin(\text{angle}), \quad x_2 = \cos(\text{angle}). \quad (11.14)$$

For observation i , write

$$Y_i = \beta + \alpha \sin(\theta_i) + \gamma \cos(\phi_i) + \epsilon_i. \quad (11.15)$$

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & \sin(\theta_1) & \cos(\phi_1) \\ 1 & \sin(\theta_2) & \cos(\phi_2) \\ \vdots & \vdots & \vdots \\ 1 & \sin(\theta_n) & \cos(\phi_n) \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \beta \\ \alpha \\ \gamma \end{bmatrix}. \quad (11.16)$$

Then the model is simply

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}. \quad (11.17)$$

This is a linear model because β , α , and γ enter linearly. The sine and cosine transformations are part of the known design matrix, not unknown parameters.

Why the plot may not look linear. If we plot the response against the raw angles, the shape may look curved or oscillatory. But once the columns $\sin(\theta)$ and $\cos(\phi)$ are created, the fitted response vector is a linear combination of those columns and the intercept column. In the transformed feature space, the fit is a plane.

11.6 Basis Functions

A basis function is a known function used to build a column of the design matrix. In the notation from (11.2), the functions

$$g_0(x), g_1(x), \dots, g_q(x) \quad (11.18)$$

are basis functions.

A model using basis functions has the form

$$f(x) = \theta_0 g_0(x) + \theta_1 g_1(x) + \dots + \theta_q g_q(x). \quad (11.19)$$

This is a linear combination of known functions. The coefficients determine how much of each basis direction is used.

Common basis choices include:

Basis type	Example columns	Useful for
Polynomial	$1, x, x^2, x^3$	smooth curvature
Interaction	$1, x_1, x_2, x_1x_2$	combined predictor effects
Trigonometric	$1, \sin x, \cos x$	cyclic or angular behavior
Logarithmic	$1, \log x$	diminishing returns and scale effects
Indicator/step	$I(x < c), I(x \geq c)$	piecewise constant effects
Spline	piecewise polynomial basis columns	flexible smooth curves

The design choice is not automatic. The analyst chooses basis functions based on domain knowledge, EDA, residual behavior, predictive performance, and interpretability.

11.7 Polynomial Regression

A polynomial regression model of degree d is

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \epsilon_i. \quad (11.20)$$

Although this model is nonlinear in x , it is linear in the coefficients $\beta_0, \beta_1, \dots, \beta_d$.

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ 1 & x_2 & x_2^2 & \cdots & x_2^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^d \end{bmatrix}. \quad (11.21)$$

This is a Vandermonde-style design matrix. The Math Foundations textbook uses a square Vandermonde matrix to fit an interpolating polynomial through n points; see Ch08 Sec. 8.7. In data analysis, we usually do something more modest: choose a small degree d , keep $d + 1 \ll n$, and solve a least squares problem instead of forcing the curve through every point.

Interpolation versus regression

Polynomial interpolation and polynomial regression can look similar algebraically, but their goals differ.

Task	Matrix shape	Goal
Interpolation	usually square $n \times n$	pass exactly through all points
Regression	usually tall $n \times (d + 1)$	find a stable trend with residual error

Interpolation can be useful in numerical methods, but in noisy data analysis it often overfits. Regression accepts residuals because the data include noise and because the goal is usually inference, prediction, or explanation rather than exact passage through every observed point.

Degree choice

A degree-one polynomial is the ordinary straight-line model. A degree-two polynomial allows one bend. A degree-three polynomial allows more shape. But high-degree polynomials can behave badly, especially near the edges of the observed data range.

A practical rule:

Use the lowest-degree polynomial that addresses the visible pattern and is defensible for the problem.

Residual plots and test error help decide whether the added flexibility is helping or merely fitting noise.

11.8 Categorical Predictors as Indicator Columns

A categorical predictor cannot be used as an arbitrary numerical label unless the numbers have quantitative meaning. If a curriculum is coded as OR = 1, Physics = 2, Applied Math = 3, and so on, the number 3 is not “larger” than the number 1 in the same way that a height or budget is larger. The category must first be encoded as design-matrix columns.

For a binary category, define an indicator variable

$$X_i = \begin{cases} 1, & \text{if observation } i \text{ is in category } C, \\ 0, & \text{otherwise.} \end{cases} \quad (11.22)$$

Using the source-slide example, suppose Y is IQ for an NPS student and $X_i = 1$ if the student is in the OR curriculum. The model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad (11.23)$$

has two fitted group means:

$$\hat{Y} \mid X = 0 = \hat{\beta}_0, \quad (11.24)$$

$$\hat{Y} \mid X = 1 = \hat{\beta}_0 + \hat{\beta}_1. \quad (11.25)$$

Thus $\hat{\beta}_0$ is the fitted mean for the reference category, and $\hat{\beta}_1$ is the contrast from the reference category to the coded category. In this example, $\hat{\beta}_1$ is the fitted OR adjustment from the non-OR baseline; as the slides joke, we hope it is positive.

Changing the reference category changes the intercept and the sign of the contrast, but it does not change the fitted values. The reference category is an interpretation choice, not a different least-squares method.

11.9 Multi-Level Categorical Predictors

If a categorical predictor has k levels and the model includes an intercept, use $k - 1$ indicator columns. Pick one level as the reference category and create indicators for the remaining levels.

For example, let curriculum have levels

$$\{\text{OR, Physics, Applied Math, Engineering, Other}\}. \quad (11.26)$$

If Other is the reference category, define

$$X_{i1} = 1\{\text{student } i \text{ is in OR}\}, \quad (11.27)$$

$$X_{i2} = 1\{\text{student } i \text{ is in Physics}\}, \quad (11.28)$$

$$X_{i3} = 1\{\text{student } i \text{ is in Applied Math}\}, \quad (11.29)$$

$$X_{i4} = 1\{\text{student } i \text{ is in Engineering}\}. \quad (11.30)$$

Then

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \epsilon_i. \quad (11.31)$$

The fitted reference mean is $\hat{\beta}_0$. The fitted OR mean is $\hat{\beta}_0 + \hat{\beta}_1$, the fitted Physics mean is $\hat{\beta}_0 + \hat{\beta}_2$, and so on.

Why not use all k indicators? With an intercept and all five curriculum indicators, every row satisfies

$$X_{i,\text{OR}} + X_{i,\text{Physics}} + X_{i,\text{Applied Math}} + X_{i,\text{Engineering}} + X_{i,\text{Other}} = 1. \quad (11.32)$$

That sum is exactly the intercept column. The design matrix is therefore rank deficient, so the coefficient vector is not uniquely identified. This is the dummy-variable trap. The fix is to drop one level as the reference category or to remove the intercept and use all group indicators with a different interpretation.

11.10 Mixed Numeric and Categorical Predictors

Numeric and categorical predictors can appear in the same design matrix. For example,

$$Y_i = \beta_0 + \beta_1 \text{hours}_i + \beta_2 \text{OR}_i + \epsilon_i \quad (11.33)$$

uses a numeric predictor and a binary curriculum indicator. If $\text{OR}_i = 0$, the fitted line is

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{hours}_i. \quad (11.34)$$

If $\text{OR}_i = 1$, the fitted line is

$$\hat{Y}_i = (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 \text{hours}_i. \quad (11.35)$$

The two groups share the same slope but have different intercepts. That is a modeling assumption. If the slope should differ by group, we need an interaction.

11.11 Interactions as Transformed Columns

An interaction means that the effect of one predictor depends on another predictor. It is not merely correlation between predictors. In the Advertising example, Newspaper and Radio may be correlated because marketing budgets are allocated together, but an interaction asks a different question: does the marginal effect of Radio change when Newspaper spending changes?

With two numeric predictors, the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i \quad (11.36)$$

is linear in the coefficients. The interaction column is

$$z_{i3} = x_{i1} x_{i2}, \quad (11.37)$$

and the design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{11}x_{12} \\ 1 & x_{21} & x_{22} & x_{21}x_{22} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n1}x_{n2} \end{bmatrix}. \quad (11.38)$$

The model can be rewritten as

$$Y_i = \beta_0 + (\beta_1 + \beta_3 x_{i2}) x_{i1} + \beta_2 x_{i2} + \epsilon_i. \quad (11.39)$$

Thus the slope with respect to x_1 depends on x_2 .

Numeric-by-categorical interactions

Suppose Y is health-insurance charges, x is BMI, and $S_i = 1$ if person i is a smoker. The model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 S_i + \beta_3 x_i S_i + \epsilon_i \quad (11.40)$$

allows smokers and nonsmokers to have different intercepts and different BMI slopes:

$$\hat{Y}_i \mid S_i = 0 = \hat{\beta}_0 + \hat{\beta}_1 x_i, \quad (11.41)$$

$$\hat{Y}_i \mid S_i = 1 = (\hat{\beta}_0 + \hat{\beta}_2) + (\hat{\beta}_1 + \hat{\beta}_3) x_i. \quad (11.42)$$

The interaction coefficient $\hat{\beta}_3$ is the difference between the smoker and nonsmoker BMI slopes.

Factor crossings. When two categorical predictors interact, the model allows each combination of levels to have its own adjustment relative to a reference cell. This is useful when the categories combine in a way that is not additive. The cost is more columns, more degrees of freedom, and more care in interpretation.

Hierarchy principle. If a model includes an interaction such as $x_1 x_2$, it is usually good practice to include the corresponding main effects x_1 and x_2 . This makes interpretation more stable and avoids treating the combined effect as meaningful while excluding its component directions. Exceptions require a clear domain reason.

Coding companion reference. Package-specific commands for factors, reference levels, dummy-variable inspection, and formula syntax belong in the Coding and Software Cookbook companion. The main text focuses on the design-matrix logic and coefficient interpretation.

11.12 Transforming the Response

So far, the transformations have been applied to predictors. Sometimes the response is transformed instead. Examples include

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (11.43)$$

or

$$\sqrt{Y_i} = \beta_0 + \beta_1 x_i + \epsilon_i. \quad (11.44)$$

A response transformation changes the scale on which the model is linear. For example, if

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (11.45)$$

then the model says that the conditional mean structure is simpler on a log scale than on the original scale. This can help when residual spread grows with the fitted values or when effects are multiplicative rather than additive.

Important caution. A transformed-response model is not the same as an untransformed-response model. If we fit $\log(Y)$, then predictions are naturally produced on the log scale. Transforming back with $\exp(\cdot)$ gives a prediction on the original scale, but uncertainty and bias require care because

$$\mathbb{E}[\exp(W)] \neq \exp(\mathbb{E}[W]) \quad (11.46)$$

in general.

11.13 Classifying Models: Linear or Not Linear?

The quickest check is to ask whether the model can be written as $\mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ for a design matrix $\mathbf{Z}$ that can be computed from the observed data without knowing the parameters.

Model expression	Linear model?	Why
$\beta_0 + \beta_1 x$	Yes	coefficients multiply known columns
$\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$	Yes	polynomial columns are known
$\beta_0 x^{\beta_1}$	No	parameter appears as exponent
$\beta_0 + \beta_1 x_1 + \beta_0 \beta_1 x_1^2$	No	parameters are multiplied together
$1/(\beta_0 + \beta_1 x)$	No	parameters are inside reciprocal transformation
$\beta_1 \sin x + \beta_2 \cos x + \beta_3 \sin x \cos x$	Yes	transformed columns are known
$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$	Yes	interaction column is known

The logistic regression mean function

$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} \quad (11.47)$$

is not a linear regression model for $p(x)$. It is, however, linear in the logit scale:

$$\log\left(\frac{p(x)}{1 - p(x)}\right) = \beta_0 + \beta_1 x. \quad (11.48)$$

That is why logistic regression belongs to the generalized linear model family rather than ordinary least squares linear regression.

11.14 Projection Geometry Still Works

A transformed design matrix changes the subspace, not the projection rule. In ordinary MLR, fitted values live in $\text{Col}(\mathbf{X})$. In this chapter, fitted values live in

$$\text{Col}(\mathbf{Z}). \quad (11.49)$$

Least squares chooses

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{Z}) \quad (11.50)$$

so that

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} \quad (11.51)$$

is orthogonal to every column of $\mathbf{Z}$. Therefore,

$$\mathbf{Z}^T \mathbf{e} = \mathbf{0}. \quad (11.52)$$

This is the transformed-model version of the normal equations.

Residual interpretation. If a residual plot shows curvature after fitting a straight-line model, we can add a transformed column that represents the missing shape. If the new transformed model is appropriate, the residual pattern should weaken or disappear. This does not prove the model is true, but it provides evidence that the new basis is capturing structure the old basis missed.

11.15 Truth Plus Error with Transformed Designs

All of the estimator behavior from the previous chapter carries over with $\mathbf{X}$ replaced by $\mathbf{Z}$. For the theoretical expectation and variance calculations, treat the response vector as random. If

$$\mathbf{Y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad (11.53)$$

then

$$\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{Y} \quad (11.54)$$

$$= (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T (\mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}) \quad (11.55)$$

$$= \boldsymbol{\theta} + (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \boldsymbol{\epsilon}. \quad (11.56)$$

Thus,

$$\boxed{\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \boldsymbol{\epsilon}.} \quad (11.57)$$

For the realized data set, we compute the coefficient estimate using the observed response vector:

$$\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}. \quad (11.58)$$

The truth-plus-error expression above is used to study the estimator's behavior; it is not a computational formula because $\boldsymbol{\theta}$ and $\boldsymbol{\epsilon}$ are unknown.

If $\mathbb{E}[\epsilon \mid \mathbf{Z}] = \mathbf{0}$, then

$$\mathbb{E}[\hat{\boldsymbol{\theta}} \mid \mathbf{Z}] = \boldsymbol{\theta}. \quad (11.59)$$

If $\text{Var}(\epsilon \mid \mathbf{Z}) = \sigma^2 \mathbf{I}_n$, then

$$\text{Var}(\hat{\boldsymbol{\theta}} \mid \mathbf{Z}) = \sigma^2 (\mathbf{Z}^T \mathbf{Z})^{-1}. \quad (11.60)$$

This matters because transformed columns do not exempt us from regression assumptions. They only change the systematic part of the model.

11.16 Model Comparison for Transformations

When comparing candidate transformations, avoid relying on a single number or a single test in isolation. Useful evidence includes:

1. residual plots,
2. test MSE or cross-validation error,
3. adjusted R^2 , AIC, or BIC,
4. nested-model F -tests when the models are nested,
5. coefficient interpretability,
6. domain plausibility.

Nested transformed models

Suppose a reduced model has residual sum of squares RSS_R and degrees of freedom df_R , and a full model has residual sum of squares RSS_F and degrees of freedom df_F . If the reduced model is nested inside the full model, the usual comparison statistic is

$$F = \frac{(RSS_R - RSS_F)/(df_R - df_F)}{RSS_F/df_F}. \quad (11.61)$$

For example, the straight-line model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad (11.62)$$

is nested inside the quadratic model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i. \quad (11.63)$$

The nested test asks whether the added x^2 column explains enough additional variation to justify the extra parameter.

Prediction-focused comparison

For prediction, a transformation that improves training RSS may not improve performance on new data. Use a train/test split or cross-validation when the modeling goal is predictive accuracy. This connects directly to the bias-variance ideas from earlier chapters: more basis columns increase flexibility, which can reduce bias but increase variance.

11.17 Practical Transformation Patterns

Polynomial terms

Use polynomial terms when residual plots suggest smooth curvature. Example:

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon. \quad (11.64)$$

Interpretation is about the whole curve, not just one coefficient in isolation.

Log predictor

Use $\log x$ when the effect of x appears to show diminishing returns or when x spans orders of magnitude:

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon. \quad (11.65)$$

This requires $x > 0$.

Log response

Use $\log Y$ when the response is positive and variability grows with the mean:

$$\log(Y) = \beta_0 + \beta_1 x + \epsilon. \quad (11.66)$$

This requires $Y > 0$.

Square-root response

Use $\sqrt{Y}$ for nonnegative responses, especially count-like responses where variance grows with the mean:

$$\sqrt{Y} = \beta_0 + \beta_1 x + \epsilon. \quad (11.67)$$

This requires $Y \geq 0$.

Trigonometric terms

Use sine and cosine terms for cyclic predictors such as time of day, heading, angle, season, or orbital position:

$$Y = \beta_0 + \beta_1 \sin(t) + \beta_2 \cos(t) + \epsilon. \quad (11.68)$$

For a period P , use

$$\sin\left(\frac{2\pi t}{P}\right), \quad \cos\left(\frac{2\pi t}{P}\right). \quad (11.69)$$

Interactions

Use interactions when the effect of one predictor plausibly depends on another:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon. \quad (11.70)$$

11.18 Numerical Stability and Identifiability

Transformed columns can create numerical problems. For example, if x is large, then x^2 , x^3 , and x^4 may have very different scales. This can make $\mathbf{Z}^T \mathbf{Z}$ ill-conditioned.

Centering before powers

A common fix is to center before creating powers:

$$u_i = x_i - \bar{x}, \quad (11.71)$$

and then fit

$$Y_i = \beta_0 + \beta_1 u_i + \beta_2 u_i^2 + \epsilon_i. \quad (11.72)$$

Centering often improves interpretability and numerical stability.

Orthogonal polynomial columns

Another fix is to use orthogonal polynomial basis columns. This is a computational version of the Gram-Schmidt idea from Math Foundations: transform a set of related columns into a better-conditioned set of orthogonal directions; see the Math Foundations textbook, Ch09 Secs. 9.1, 9.3, and 9.11. The fitted values can represent the same polynomial space, but the coefficient estimation is more stable.

Full column rank

The transformed design matrix must still have full column rank for the usual inverse formula to apply. Problems occur when columns are exact linear combinations of each other. Examples include:

1. including duplicate columns,
2. creating indicators that sum to the intercept column without dropping a reference category,
3. including high-degree terms with too few distinct predictor values,
4. creating transformations that are constant in the observed data.

The regression problem is not saved by calling a column “transformed.” If the resulting columns are linearly dependent, the model is still not identified.

Multicollinearity cross-reference. Even when the transformed design is full rank, engineered columns such as indicators, powers, interactions, and transformed variables can be nearly collinear. When coefficient interpretation or inference matters, check both rank/identifiability and the multicollinearity diagnostics in Chapter 12.

11.19 Interpretation After Transformations

Transformations improve flexibility, but they change interpretation.

Polynomial coefficient interpretation

In

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon, \quad (11.73)$$

the coefficient β_1 is not simply the global slope. The instantaneous slope of the fitted mean is

$$\frac{d}{dx} (\beta_0 + \beta_1 x + \beta_2 x^2) = \beta_1 + 2\beta_2 x. \quad (11.74)$$

Thus the effect of changing x depends on where you are on the curve.

Categorical coefficient interpretation

For an indicator variable, the coefficient is a contrast from the reference category. In a model with one indicator,

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad (11.75)$$

β_0 is the fitted reference mean and β_1 is the fitted adjustment for the coded group. In a model with $k - 1$ indicators, each contrast is interpreted relative to the omitted reference level.

Interaction coefficient interpretation

In

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon, \quad (11.76)$$

the slope with respect to x_1 is

$$\frac{\partial}{\partial x_1} \mathbb{E}[Y \mid x_1, x_2] = \beta_1 + \beta_3 x_2. \quad (11.77)$$

The coefficient β_3 tells us how the slope in x_1 changes as x_2 increases.

Log predictor interpretation

In

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon, \quad (11.78)$$

small percentage changes in x correspond approximately to additive changes in Y . The exact change from x to cx is

$$\Delta \hat{Y} = \beta_1 \log(c). \quad (11.79)$$

Log response interpretation

In

$$\log(Y) = \beta_0 + \beta_1 x + \epsilon, \quad (11.80)$$

a one-unit increase in x changes the fitted log-response by β_1 . On the original scale, this corresponds approximately to multiplying the response by e^{β_1} , before accounting for retransformation issues.

11.20 Worked Example: Building a Polynomial Design Matrix

Suppose we observe four data points (x_i, y_i) and want to fit a quadratic model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i. \quad (11.81)$$

Let

$$x = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 5 \end{bmatrix}. \quad (11.82)$$

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}. \quad (11.83)$$

The model is

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (11.84)$$

Here $\boldsymbol{\beta}$ is the coefficient vector for this quadratic transformed design; it plays the same role as $\boldsymbol{\theta}$ above. The least squares estimate is

$$\hat{\boldsymbol{\beta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}. \quad (11.85)$$

The fitted curve is then

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2. \quad (11.86)$$

Although the graph is a parabola, the computation is ordinary least squares.

Coding companion reference. Package-specific Python/R commands for polynomial terms, interactions, response transformations, and prediction grids have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the linear-in-parameters logic and interpretation guidance.

11.21 Common Mistakes

Mistake 1: Thinking curved means not linear

A polynomial curve can be a linear model. Linearity refers to the unknown coefficients, not the visual shape of the fitted curve in the original predictor.

Mistake 2: Transforming after the split incorrectly

For prediction workflows, transformations that learn quantities from the data, such as centering and scaling, should be learned on the training set and then applied to the test set. Otherwise, information leaks from test data into the training process.

Mistake 3: Ignoring domain restrictions

Log transformations require positive values. Square-root transformations require nonnegative values. A transformation that silently drops rows can change the analysis population.

Mistake 4: Overinterpreting individual polynomial coefficients

In polynomial models, the individual coefficients depend heavily on the chosen basis and scaling. Interpret the fitted curve and its derivative, not just isolated coefficient signs.

Mistake 5: Adding flexibility without checking test performance

More columns usually improve training fit. That does not guarantee better prediction on new observations. Use residual plots and validation.

Mistake 6: Creating rank-deficient transformed designs

A transformed design matrix can still be singular. Check for duplicate columns, redundant indicators, too many polynomial terms, and variables with too few distinct values.

Mistake 7: Treating category codes as numeric measurements

A category code is not a quantity just because it is stored as a number. Encode categories with indicator variables or another defensible contrast system.

Mistake 8: Using all category indicators plus an intercept

With an intercept, use $k - 1$ indicators for a k -level categorical predictor. Otherwise the indicator columns sum to the intercept column and the coefficients are not uniquely identified.

11.22 Operational Briefing Checklist

When briefing a transformed linear model, answer the following:

1. What is the response variable?
2. What original predictors were measured?
3. What transformed, categorical, or interaction columns were created?
4. Can the model be written as $\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}$?
5. Why are these transformations operationally or scientifically plausible?
6. Does the transformed design matrix have full column rank?
7. What residual pattern motivated the transformation?
8. Did residual behavior improve after the transformation?

9. Was model comparison based on residuals, test MSE, adjusted R^2 , AIC/BIC, nested tests, or a combination?
10. What is the reference category for each categorical predictor?
11. If interactions are included, what effect modification is being claimed?
12. How does the transformation or encoding change interpretation for the customer or commander?

11.23 Chapter Summary

The core idea is:

$$\boxed{\text{linear model} = \text{linear in unknown coefficients, not necessarily linear in original predictors.}} \quad (11.87)$$

If we can create a transformed design matrix $\mathbf{Z}$ such that

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad (11.88)$$

then, if $\mathbf{Z}$ has full column rank, ordinary least squares uses the inverse form:

$$\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}. \quad (11.89)$$

If $\mathbf{Z}^T \mathbf{Z}$ is singular, the fitted values may still be defined, but the coefficient vector is not uniquely identified. The fitted values are still a projection:

$$\hat{\mathbf{y}} = \mathbf{H}_Z \mathbf{y}. \quad (11.90)$$

The residuals are still orthogonal to the model columns:

$$\mathbf{Z}^T \mathbf{e} = \mathbf{0}. \quad (11.91)$$

Transformations let us fit more realistic structure while keeping the interpretability and inference tools of linear regression. Polynomial terms, sine/cosine terms, log terms, indicator columns, and interactions are all ways to build richer design matrices. For a k -level categorical predictor with an intercept, use $k - 1$ indicators and interpret coefficients relative to the reference category. The cost is that interpretation, rank, and overfitting must be handled carefully.

11.24 Practice and Workflow

Focused Study Checklist

Use this as a final check after proposing a transformed linear model.

1. Can you state the difference between original predictors and created features?
2. Can you write the transformed or engineered design matrix $\mathbf{Z}$?
3. Can you express the model as $\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}$?

4. Can you explain why the model is linear in the coefficients even if it is curved in the original variables?
5. Can you identify reference categories and interpret categorical contrasts?
6. Can you check whether transformed, indicator, or interaction columns create rank or identifiability problems?
7. Can you compare residual behavior before and after the transformation?
8. Can you interpret the transformed model in language that remains useful to the customer?

Chapter 12

Residual Analysis and Model Repair

12.1 The Big Picture

A fitted model is not the end of the analysis. It is the beginning of the model check.

The previous chapters built multiple linear regression from the design-matrix point of view:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}. \quad (12.1)$$

The residual vector $\mathbf{e}$ is the part of the observed response vector that the fitted model did not explain. Residual analysis asks:

Does the leftover pattern look like random noise, or does it look like structure the model failed to capture?

This chapter is about using residuals as a diagnostic tool and then deciding how to repair a model when the diagnostics look bad. The three practical repair moves emphasized in the source slides are:

1. transform the predictors and/or the response,
2. add interaction terms,
3. admit that the current linear-model frame is not adequate and move to a different modeling strategy.

The most important habit is to avoid treating a single hypothesis test, a single R^2 , or a single residual plot as the whole story. Residual analysis is evidence. Test MSE, cross-validation, domain knowledge, and interpretability are also evidence.

As in Chapters 7, 9, and 11, we use $\mathbf{y}$ for the observed response vector from the fitted data set. Chapter 10 used $\mathbf{Y}$ when the response vector was being treated as random. Here the emphasis is back on the realized fit and its residuals.

Math Foundations Connection. Residual analysis is geometric, but the supporting Math Foundations ideas are more basic: norms, orthogonality, matrix-vector multiplication, rank, column-space/residual-space geometry, and least-squares minimization. These ideas are covered in the Math Foundations textbook, Chapters 5, 7, 10, 11, and 12. The projection language used in this chapter

is a Data Analysis synthesis built from those underlying ideas. Model repair changes the design columns and therefore changes which patterns can be represented by the fitted part of the model.

12.2 Recall from Chapters 7 and 10: The Residual Vector

Chapters 7 and 10 already developed the linear-algebra machinery behind residuals. We recall only the pieces needed for diagnostics.

For a full-rank design matrix $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$, ordinary least squares gives

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \quad \hat{\mathbf{y}} = \mathbf{H} \mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (12.2)$$

The residual vector is therefore

$$\boxed{\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}.} \quad (12.3)$$

If the linear truth-plus-error model is appropriate, then

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (12.4)$$

Since $\mathbf{H}\mathbf{X} = \mathbf{X}$, the residuals can be written as

$$\boxed{\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.} \quad (12.5)$$

This identity is the key diagnostic reminder: residuals are not the true errors. They are the parts of the errors left over after projecting onto the fitted model space.

The normal equations also give

$$\mathbf{X}^T \mathbf{e} = \mathbf{0}. \quad (12.6)$$

Thus residuals are orthogonal to every column of the design matrix. If the model includes an intercept column $\mathbf{1}$, then

$$\mathbf{1}^T \mathbf{e} = \sum_{i=1}^n e_i = 0. \quad (12.7)$$

This is an algebraic fact, not a diagnostic victory. The residuals can sum to zero and still show curvature, changing spread, time dependence, or outliers. The modeling assumption we care about is closer to

$$\mathbb{E}[\epsilon_i \mid \mathbf{x}_i] = 0, \quad (12.8)$$

meaning that the unobserved errors are centered correctly at each predictor setting. Residual plots give practical evidence for or against that claim.

12.3 What Residuals Are Trying to Tell Us

The source slides summarize the residual-checking logic as follows:

Residual behavior	Suggests about the errors	Common diagnostic view
Mean-zero pattern for each x	$\mathbb{E}[\epsilon X = x] = 0$ may be plausible	residuals versus fitted values or predictors
Approximately constant spread	$\text{Var}(\epsilon X = x) = \sigma^2$ may be plausible	residuals versus fitted values; scale-location plot
Approximately normal shape	$\epsilon \sim N(0, \sigma^2)$ may be plausible	histogram or normal Q-Q plot
No sequential pattern	uncorrelated errors may be plausible	residuals versus time/order

The phrase “suggests” is important. Residual plots do not prove assumptions. They help us identify visible violations.

12.4 Residuals Under the Big Three Regression Assumptions

Residual diagnostics connect directly to the three-step assumption ladder from earlier regression chapters. This section gives the diagnostic consequences, rather than re-deriving the full residual theory.

12.4.1 Assumption 1: Mean-zero errors

If $\mathbb{E}[\epsilon | \mathbf{X}] = \mathbf{0}$, then the model is centered correctly and residuals should not show systematic mean structure. A residual plot with no visible pattern around zero supports this assumption. A curved residual pattern suggests that the mean model is missing structure, such as a transformed predictor, a polynomial term, or an interaction.

12.4.2 Assumption 2: Constant variance and uncorrelated errors

If $\text{Var}(\epsilon | \mathbf{X}) = \sigma^2 \mathbf{I}_n$, then the usual least-squares standard-error formulas are supported. Chapter 10 showed that residuals then satisfy

$$\text{Var}(\mathbf{e} | \mathbf{X}) = \sigma^2(\mathbf{I}_n - \mathbf{H}), \quad \text{Var}(e_i | \mathbf{X}) = \sigma^2(1 - h_{ii}). \quad (12.9)$$

The important practical point is that raw residuals are affected by leverage. Changing spread in residual plots suggests heteroskedasticity, while waves or runs over time/order suggest correlated errors.

12.4.3 Assumption 3: Gaussian errors

If $\epsilon | \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, then the usual finite-sample t - and F -based inference is exact under the model. Normal Q-Q plots and residual histograms are used to look for strong skewness, heavy tails, or outliers. For prediction, mild nonnormality may be less damaging than nonlinearity, heteroskedasticity, or dependence.

12.5 Residual Scaling: Raw, Standardized, and Studentized

The raw residual is

$$e_i = y_i - \hat{y}_i. \quad (12.10)$$

Raw residuals are measured in the units of the response, which makes them easy to interpret operationally.

The residual standard error is

$$RSE = \hat{\sigma} = \sqrt{\frac{RSS}{n - p - 1}}, \quad RSS = \sum_{i=1}^n e_i^2. \quad (12.11)$$

A common standardized residual is

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}. \quad (12.12)$$

This accounts for the fact that residual variance depends on leverage and is often called an internally studentized residual. Externally studentized residuals use a leave-one-out estimate of σ ; the idea is the same but the scaling is slightly more protective against the observation being tested. Large standardized or studentized residuals are potential outliers. A rough screening rule is that values beyond ± 2 deserve attention and values beyond ± 3 deserve a closer look.

Do not automate judgment. A large residual is a flag, not an automatic deletion order. Ask whether the observation is a data-entry error, an unusual but valid case, evidence of missing predictors, or evidence that the model class is too limited.

12.6 Core Diagnostic Plots

12.6.1 Residuals versus fitted values

This is the primary diagnostic plot for linear regression. The horizontal axis is $\hat{y}_i$, and the vertical axis is e_i . A good residual-versus-fitted plot looks like an unstructured cloud centered around zero.

Common patterns include:

Pattern	Likely issue	Candidate repair
Curved shape	missing nonlinear mean structure	polynomial term, log term, spline, or other transformed predictor
Fan shape	nonconstant variance	transform response, weighted least squares, robust standard errors
Separate bands by group	missing categorical effect or interaction	add group indicator or interaction
Large isolated points	outliers or data errors	inspect data, sensitivity analysis, robust method

12.6.2 Residuals versus predictors

A residual-versus-fitted plot may hide which predictor is causing the problem. Plotting residuals against individual predictors can reveal a missed nonlinear effect in one variable.

For example, if residuals plotted against x_1 show a curve, then a candidate repair is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \cdots + \epsilon. \quad (12.13)$$

If residuals plotted against x_2 show increasing spread, then a transformation or weighting strategy may be needed.

12.6.3 Normal Q-Q plot

A normal Q-Q plot compares standardized residual quantiles to normal quantiles. If the residuals are approximately normal, the points should lie roughly along a straight line.

Common departures include:

1. heavy tails: ends bend away from the line,
2. skewness: one tail departs more than the other,
3. isolated outliers: one or a few points far from the line.

12.6.4 Scale-location plot

A scale-location plot often uses

$$\sqrt{|r_i|} \quad (12.14)$$

against fitted values. It is designed to make changing variance easier to see. A rising trend suggests that the residual spread increases with the fitted value.

Funnel or inverted-funnel patterns suggest nonconstant variance. A curved smooth trend can suggest missing mean structure, such as a nonlinear predictor effect. Use the scale-location plot together with residuals-versus-fitted and residuals-versus-predictor plots before choosing a repair.

12.6.5 Residuals versus time or collection order

If the observations have a natural order, plot residuals against time, mission phase, batch number, route order, or collection order. A visible wave, trend, or clustering pattern can indicate correlated errors or missing time-varying predictors.

12.7 A Practical Diagnostic Hierarchy

Different diagnostics answer different questions. A useful order is:

1. **Test MSE or cross-validation error** for prediction performance.
2. **Residual plots** for missed structure, nonconstant variance, and outliers.

3. **Residuals versus individual predictors** to identify which variable may need a transformation or interaction.
4. **Formal tests** such as the Breusch–Pagan test as supporting evidence, not as the sole decision rule.
5. ***t*- and *F*-tests** for inference after the modeling assumptions are plausible enough for the claim being made.

Use the hierarchy to avoid overreacting to one number. A p-value can flag a problem, but a plot usually shows the shape of the problem and suggests the repair.

12.8 Diagnosing and Repairing Multicollinearity

Most diagnostics in this chapter look at residual behavior. Multicollinearity is different: it is a predictor-side problem. The fitted model may leave no obvious residual pattern, yet individual coefficients may be unstable because predictors contain overlapping information.

First-pass reconnaissance. Scatterplot matrices and predictor-predictor correlations are useful first screens. Strong pairwise association is an early warning, but low pairwise correlations do not rule out multicollinearity. One predictor can be explained by a combination of several others even when no single pair looks extreme.

Variance inflation factor. A common diagnostic is the variance inflation factor. For predictor X_j , regress X_j on the remaining predictors and compute the resulting R^2 , written $R^2_{X_j|X_{-j}}$. Then

$$\text{VIF}_j = \frac{1}{1 - R^2_{X_j|X_{-j}}}. \quad (12.15)$$

If the other predictors do not explain much of X_j , then $R^2_{X_j|X_{-j}}$ is small and VIF_j is near 1. If the other predictors explain X_j well, then $R^2_{X_j|X_{-j}}$ is near 1 and the VIF becomes large. A large VIF means the coefficient attached to X_j is harder to estimate separately from the other predictors.

Rules of thumb such as $\text{VIF} > 5$ or $\text{VIF} > 10$ are screening rules, not deletion rules. A large VIF says to look closer. It does not by itself say that the predictor must be removed. Also, do not interpret the VIF for the intercept or constant column; the intercept is not a substantive predictor.

Fit, prediction, and interpretation. Multicollinearity mainly warns about coefficient interpretation and inference. Large VIFs often correspond to inflated standard errors, unstable coefficient signs, or coefficients that change sharply when related predictors are added or removed. A model can still have useful fitted values, a strong overall test, or good validation error. If prediction is the goal and validation performance is good, keeping collinear predictors may be defensible. If interpretation or inference is the goal, the same collinearity may be a serious limitation.

Condition-number intuition. Rank, condition-number, and small-eigenvalue checks describe the design matrix as a whole rather than one predictor at a time. They can flag exact or near linear dependence, but their numerical values depend on scaling, so use them as supporting screens rather than standalone rules.

Repair and documentation. For exact dependence, remove a redundant column or use a reference category. For near-collinearity, first ask what the related predictors mean. Are they duplicates, unit conversions, totals and components, or several measurements of the same underlying construct? For interpretation, consider dropping one predictor, combining predictors into a meaningful index, or redefining the model. Ridge regression, principal components, and principal-components regression are advanced remedies, but they are future tools rather than part of this chapter's ordinary least-squares workflow.

Document the decision. A defensible note should answer: Which predictors were involved? Was the issue exact dependence or near-collinearity? Was the goal prediction or interpretation? What repair, if any, was attempted? Did the conclusion, residual behavior, or validation error change?

12.9 Problem 1: Nonlinearity

A linear model assumes that the conditional mean is linear in the columns of the design matrix:

$$\mathbb{E}[Y \mid \mathbf{x}] = \mathbf{x}^T \boldsymbol{\beta}. \quad (12.16)$$

If the true mean function is curved and the design matrix has only straight-line columns, residuals will show structure.

Diagnostic sign

A residual plot with a curve is the classic sign. For example, a straight-line fit to a parabola will often leave residuals that are negative-positive-negative across the predictor range.

Repairs

Candidate repairs include:

1. adding polynomial terms such as x^2 or x^3 ,
2. using transformations such as $\log x$, $\sqrt{x}$, or $1/x$,
3. using sine/cosine terms for cyclic predictors,
4. using splines or generalized additive models when a simple transformation is not enough,
5. moving to a different model class if the response type requires it.

Model repair as new columns

If the original model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (12.17)$$

then adding a quadratic predictor changes the design matrix to

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}. \quad (12.18)$$

The repaired model is

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}. \quad (12.19)$$

So model repair is still a projection problem; we changed the subspace.

12.10 Problem 2: Nonconstant Variance

The homoskedastic model assumes

$$\text{Var}(\epsilon_i \mid \mathbf{x}_i) = \sigma^2 \quad (12.20)$$

for every observation. If residual spread changes with fitted values or predictor values, then the constant-variance assumption is questionable.

Diagnostic sign

A common pattern is a funnel shape: residuals are tightly clustered for small fitted values and widely spread for large fitted values, or vice versa.

Why it matters

Nonconstant variance can make standard errors, confidence intervals, and hypothesis tests unreliable. It may not destroy point predictions, but it can damage uncertainty quantification.

Repairs

Candidate repairs include:

1. transform the response, especially $\log Y$ or $\sqrt{Y}$,
2. use weighted least squares when the variance pattern is understood,
3. use heteroskedasticity-robust standard errors for inference,
4. use a model family designed for the response type, such as logistic or Poisson regression.

A log response model has the form

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (12.21)$$

which often helps when variability grows multiplicatively with the mean.

A square-root response model has the form

$$\sqrt{Y_i} = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (12.22)$$

which can help for nonnegative count-like responses.

Breusch–Pagan test as supporting evidence

The Breusch–Pagan test is a formal check for evidence that error variance is not constant. Operationally, it asks whether squared residuals show systematic structure as a function of the predictors or fitted values. A small p-value is evidence against constant variance, but it does not identify the correct repair. Residual plots remain the primary diagnostic tool because they reveal the shape: funneling, curvature, group-specific spread, or a few unusual points.

Do not use the BP test in a vacuum. If the plot and test disagree, investigate why: the model may be nonlinear, the variance pattern may be subtle, or one influential observation may be driving the result.

12.11 Problem 3: Nonnormal Residuals

The Gaussian noise model assumes

$$\epsilon_i \sim N(0, \sigma^2). \quad (12.23)$$

A histogram or Q-Q plot of residuals can reveal skewness, heavy tails, or outliers.

Why it matters

Normality is most important for exact small-sample inference. If the sample is large, coefficient estimates may still behave well under weaker conditions, but heavy tails and outliers can still destabilize the fit.

Repairs

Candidate repairs include:

1. transform the response,
2. inspect and correct data-entry issues,
3. use robust regression or bootstrap uncertainty intervals,
4. use a model family more appropriate to the response distribution.

12.12 Problem 4: Leverage and Influence

An outlier is unusual in the response direction. A high-leverage point is unusual in predictor space. An influential point is one that can noticeably change the fitted model. These are related but not identical.

The leverage of observation i is the i th diagonal entry of the hat matrix:

$$h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i. \quad (12.24)$$

Here $\mathbf{x}_i^T$ is the full design row, including the intercept and any transformed, categorical, or interaction columns. High leverage means the design row is far from the center of the other design rows. The model has fewer nearby observations to stabilize the fit around that point.

Because $\text{trace}(\mathbf{H}) = p + 1$ under full column rank, the average leverage is

$$\bar{h} = \frac{p + 1}{n}. \quad (12.25)$$

Here $p + 1$ means the number of fitted coefficients, equivalently the number of columns in the current full-rank design matrix. Rules such as $h_{ii} > 2\bar{h}$ or $h_{ii} > 3\bar{h}$ are screening heuristics, not deletion rules.

Cook's distance combines residual size and leverage:

$$D_i = \frac{e_i^2}{(p + 1)\hat{\sigma}^2} \frac{h_{ii}}{(1 - h_{ii})^2}. \quad (12.26)$$

Large Cook's distance is context-dependent screening evidence that deleting observation i would noticeably change the fitted model, not a deletion rule. The most concerning observations usually have both high leverage and a large adjusted residual.

Keep, remove, or repair?

Influential is not the same as bad. Use EDA and the problem context before deciding what to do.

Situation	Reasonable response
Data-entry error, such as age 400 instead of 40	Correct if the truth is recoverable; otherwise remove or mark missing with documentation.
Valid extreme case, such as a very tall basketball player or a mansion-sized home	Usually keep; report that the model is sensitive to an operationally real case.
Influential pattern at the edge of a curved relationship	Repair the model, often with a transformation or interaction, rather than deleting the endpoints.
Observation outside the intended population	Exclude only under a pre-stated, reproducible population rule.

A defensible analysis often reports a sensitivity check: fit with and without the point, explain the difference, and justify the final choice.

12.13 Problem 5: Correlated Errors

Least squares formulas for standard errors often assume uncorrelated errors:

$$\text{Cov}(\epsilon_i, \epsilon_j \mid \mathbf{X}) = 0 \quad \text{for } i \neq j. \quad (12.27)$$

This assumption is especially vulnerable when observations are collected over time, across space, within units, or by repeated measurements on the same platform or person.

Diagnostic sign

Residuals plotted over time or collection order may show runs, waves, or clusters. Adjacent residuals may have similar signs or magnitudes.

Repairs

Candidate repairs include:

1. include time, batch, route, or group predictors,
2. include lagged predictors or lagged responses when appropriate,
3. use generalized least squares or time-series models,
4. use clustered or block-robust standard errors,
5. redesign the data collection plan when possible.

12.14 Problem 6: Missing Interactions

An additive model assumes that the effect of one predictor does not depend on another predictor. Interactions repair that limitation.

The additive two-predictor model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon. \quad (12.28)$$

The interaction model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon. \quad (12.29)$$

Equivalently,

$$Y = \beta_0 + (\beta_1 + \beta_3 x_2) x_1 + \beta_2 x_2 + \epsilon. \quad (12.30)$$

The slope in x_1 is now allowed to depend on x_2 .

Diagnostic signs

Possible signs of a missing interaction include:

1. residuals have different patterns in different groups,
2. one predictor appears helpful only at certain levels of another predictor,
3. residuals plotted by color or symbol for a second variable reveal structure,
4. domain knowledge says the variables should combine synergistically.

12.15 A Model Repair Menu

When residuals look bad, the repair should be tied to the defect.

Residual symptom	Likely interpretation	First repair candidates
Curved residual trend	wrong mean shape	transform predictor, polynomial, spline
Fan-shaped spread	heteroskedasticity	transform response, WLS, robust SEs
Heavy tails in Q-Q plot	outliers or nonnormal errors	inspect data, robust method, transformation
Runs over time/order	correlated errors	time terms, lags, GLS, clustered SEs
Different patterns by group	missing categorical effect or interaction	add indicators or interactions
Large influential point	unusual observation dominates fit	audit, sensitivity analysis, robust method
No repair makes sense	model class mismatch	change model family or data strategy

When several repairs remain defensible, later chapters develop model-selection criteria, cross-validation, and regularization tools for comparing or stabilizing candidate models; those tools supplement this diagnostic workflow rather than replace it.

12.16 Transformation Repairs

The source slides compare several model-repair candidates: a linear fit, a third-degree polynomial transformation, a square-root transformation, and a logarithmic-linear transformation. The general lesson is that there is rarely a purely mechanical answer to “which one wins.”

12.16.1 Linear fit

The simplest model is

$$Y = \beta_0 + \beta_1 x + \epsilon. \quad (12.31)$$

Use this as the baseline. It is interpretable and low-variance, but it may leave curved residual patterns.

12.16.2 Polynomial predictor transformation

A third-degree polynomial model is

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon. \quad (12.32)$$

This can capture curvature while remaining linear in the coefficients. The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 \end{bmatrix}. \quad (12.33)$$

The risk is overfitting and unstable behavior near the edges.

12.16.3 Square-root transformation

A square-root predictor model might be

$$Y = \beta_0 + \beta_1\sqrt{x} + \epsilon, \quad (12.34)$$

with $x \geq 0$. A square-root response model might be

$$\sqrt{Y} = \beta_0 + \beta_1x + \epsilon, \quad (12.35)$$

with $Y \geq 0$. These are different models and must be interpreted differently.

12.16.4 Logarithmic-linear transformation

A logarithmic predictor model is

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon, \quad (12.36)$$

with $x > 0$. This is useful for diminishing returns.

A log-response model is

$$\log(Y) = \beta_0 + \beta_1x + \epsilon, \quad (12.37)$$

with $Y > 0$. This is useful for multiplicative behavior and variance that grows with the mean.

Scale warning. Do not compare RMSE from a model fitted on Y to RMSE from a model fitted on $\log(Y)$ without putting predictions on the same scale. Model comparison must be done on a common operational scale.

12.17 Which One Wins?

The source slides emphasize that model repair includes some qualitative judgment. A defensible answer combines several checks:

1. Do residual plots look less patterned?
2. Does test MSE or cross-validated error improve on the operational prediction scale?
3. Is the transformation plausible for the domain?
4. Does the repair preserve interpretability?
5. Did the repair create new problems, such as extrapolation risk or rank deficiency?
6. Are nested-model tests, adjusted R^2 , AIC, or BIC consistent with the conclusion?

Nested-model comparison

If a reduced model is nested inside a full model, use the familiar comparison statistic:

$$F = \frac{(RSS_R - RSS_F)/(df_R - df_F)}{RSS_F/df_F}. \quad (12.38)$$

For example, the linear model

$$Y = \beta_0 + \beta_1 x + \epsilon \quad (12.39)$$

is nested inside the cubic model

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon. \quad (12.40)$$

The test asks whether the added polynomial columns explain enough additional variation to justify the added degrees of freedom.

Prediction-focused comparison

For prediction, validation should dominate training fit. The model with the smallest training RSS is often the most flexible model, not necessarily the best generalizing model. Use a train/test split or cross-validation to estimate test error.

Inference-focused comparison

For inference, residual repair should support assumptions needed for standard errors, intervals, and tests. A model with slightly worse prediction but clearer interpretation and better-behaved residuals may be the better briefing model.

12.18 Admit Defeat: When Repair Means Changing the Model Class

Sometimes bad residuals are not a minor defect. They are evidence that ordinary least squares is the wrong tool.

Examples:

Situation	Why OLS may struggle	Better direction
Binary response	predictions can fall outside $[0, 1]$	logistic regression
Count response	variance often grows with mean	Poisson or negative binomial model
Positive skewed response	additive normal errors may be implausible	log transform or Gamma GLM
Repeated measures	errors are correlated within unit	mixed model or clustered SEs
Time series	residuals may be autocorrelated	time-series regression or GLS
Strong nonlinear boundary	linear mean is structurally wrong	nonlinear model, tree, GAM, or ML method

Admitting defeat is not failure. It is a modeling conclusion: the residuals are telling us the current model family is missing something important.

12.19 Worked Example: Residual Pattern to Repair Decision

Suppose a linear fit gives the residual pattern

$$e_i \approx \text{curved function of } x_i. \quad (12.41)$$

A reasonable repair sequence is:

1. Fit the baseline model:

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i. \quad (12.42)$$

2. Plot residuals versus x and fitted values.
3. Add a transformation motivated by the pattern:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i. \quad (12.43)$$

4. Refit the model and replot residuals.
5. Compare the two models using residual behavior, test MSE, and a nested-model F -test if appropriate.
6. Brief the result in terms of the original problem, not just the equation.

A clean result would be: the repaired model reduces the visible residual curve, improves validation error on the response scale, and has a defensible interpretation.

12.20 Residual Analysis Workflow

A reusable workflow is:

1. Fit a simple, interpretable baseline model.
2. Compute fitted values and residuals.
3. Check residuals versus fitted values.
4. Check residuals versus important predictors.
5. Check residual distribution with a histogram and Q-Q plot.
6. Check residuals over time/order/group if applicable.
7. Check leverage and influence for unusual observations.
8. Propose one repair at a time.
9. Refit and repeat diagnostics.
10. Compare models on a common scale using validation or other appropriate criteria.
11. Document the decision.

The last step is critical. If an analyst tries several transformations and only reports the one that looks good, the analysis is less defensible. A good report explains what was tried and why the final choice was made.

Coding companion reference. Package-specific Python/R commands for residual plots, Q-Q plots, influence measures, transformations, interactions, and validation RMSE have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the diagnostic logic and model-repair workflow.

12.21 Common Mistakes

Mistake 1: Treating residuals as the true errors

Residuals estimate error behavior, but they are not the unobserved errors. They are constrained by the fitted model and satisfy $\mathbf{X}^T \mathbf{e} = \mathbf{0}$.

Mistake 2: Celebrating mean-zero residuals

If the model has an intercept, the residuals sum to zero automatically. The meaningful question is whether residuals have no systematic pattern across predictor values.

Mistake 3: Comparing transformed and untransformed RMSE directly

RMSE on Y and RMSE on $\log(Y)$ are not on the same scale. Compare models on the operational scale that matters.

Mistake 4: Deleting outliers without explanation

Outlier removal must be repeatable and defensible. If the point is valid and operationally relevant, deleting it may hide exactly the case the model needs to understand.

Mistake 5: Adding every possible transformation

More columns reduce training RSS but can increase test error. Add transformations because the residuals, domain knowledge, and validation evidence justify them.

Mistake 6: Ignoring data quality

Sometimes residuals look bad because the data are bad: wrong units, miscoded categories, impossible values, repeated rows, or data-entry errors. Model repair does not replace data cleaning.

Mistake 7: Taking one hypothesis test in a vacuum

A small p-value for one transformation does not automatically make it the best model. Consider residual behavior, prediction performance, assumptions, and operational meaning together.

Mistake 8: Letting the Breusch–Pagan test replace the plot

The BP test can support a heteroskedasticity concern, but it does not show the shape of the problem or tell you how to repair it. Use the plot, the test, and the modeling context together.

Mistake 9: Deleting influential points automatically

A high-leverage or high-Cook’s-distance observation is a flag for investigation. It may be a data error, a valid extreme case, or evidence that the model form is too simple.

12.22 Operational Briefing Checklist

When briefing residual analysis and model repair, answer:

1. What was the baseline model?
2. What residual pattern indicated a problem?
3. Which assumption did that pattern threaten?
4. What repair options were considered?
5. Which repair was selected, and why?
6. Did residual behavior improve after the repair?
7. Did test MSE or cross-validation error improve on the response scale?
8. Did the repair change interpretation?
9. Were outliers, high-leverage observations, or influential observations investigated?
10. Did BP or another formal test support the visual diagnostic evidence, and was it interpreted cautiously?
11. Were any observations removed? If yes, what was the defensible rule?
12. Are remaining limitations clear to the customer or commander?

12.23 Chapter Summary

The central residual-analysis loop is:

$$\boxed{\text{fit} \longrightarrow \text{diagnose residuals} \longrightarrow \text{repair model} \longrightarrow \text{re-check.}} \quad (12.44)$$

The residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}. \quad (12.45)$$

If the model is correctly specified, then

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (12.46)$$

Residual plots help us evaluate whether the leftover variation behaves like random noise or contains missed structure.

A good residual plot is not proof that the model is true. A bad residual plot is useful evidence that the model needs work. Formal tests such as Breusch–Pagan can provide supporting evidence, but they do not replace plots or judgment. Repairs include transformed predictors, transformed responses, interaction terms, different model families, robust methods, and sometimes better data. Leverage and Cook’s distance help identify observations that may drive the fit and require investigation.

The guiding principle is:

Residuals are the model’s after-action report. Read them before trusting the fit.

12.24 Practice and Workflow

Focused Study Checklist

Use this as a final check after diagnosing and repairing a fitted model.

1. Can you name the residual plot or diagnostic view you used?
2. Can you describe the observed residual pattern without overstating it?
3. Can you connect the pattern to the threatened assumption or modeling weakness?
4. Can you choose a repair: transform predictors, transform the response, add interactions, audit data, use a robust method, or change model family?
5. Can you refit and compare residual behavior, validation error, and interpretability?
6. Can you explain whether the repair changed the meaning of the coefficients or predictions?
7. Can you state the remaining limitations honestly in a briefing-style conclusion?

Chapter 13

Uncertainty for Linear Regression: Confidence and Prediction Intervals

13.1 The Big Picture

A fitted regression model gives point estimates:

$$\hat{\boldsymbol{\beta}}, \quad \hat{y}_0 = \mathbf{x}_0^T \hat{\boldsymbol{\beta}}. \quad (13.1)$$

Point estimates are useful, but they are incomplete. A briefing that only gives a fitted coefficient or a fitted value hides the fact that the estimate came from one sample.

This chapter asks a practical question:

How uncertain should we be about the coefficient, the mean response, or a future response?

There are three interval questions that students often mix together:

Question	Target	Interval type
How uncertain is a coefficient?	a true coefficient β_j	coefficient confidence interval
What is the average response at this predictor setting?	the conditional mean $\mathbb{E}[Y \mid \mathbf{x}_0]$	mean-response confidence interval
What value might a new case have?	a future observation $Y_0 \mid \mathbf{x}_0$	prediction interval

The first two intervals describe uncertainty about unknown but fixed model quantities. The third interval describes uncertainty about a new observed response, so it must include the irreducible noise in that future observation. This is why prediction intervals are wider than confidence intervals for the mean response.

Connection to earlier chapters. Chapter 10 developed the fixed-design theory behind $\hat{\boldsymbol{\beta}}$, fitted values, residuals, and residual variance. Chapter 12 emphasized that the assumptions behind those

formulas should be checked with residual diagnostics. This chapter uses those same ingredients to build interpretable uncertainty statements.

Math Foundations Connection. The interval formulas use vector notation, matrix inverses, diagonal entries of matrices, quadratic forms such as $\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$, residual norms, and least-squares projection geometry. These ideas are supported by the Math Foundations textbook, especially Chapters 2, 5, 7, 10, 11, 12, and 13.

13.2 Recall: Fixed-Design Regression Theory

We use the fixed-design multiple regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (13.2)$$

where $\mathbf{X} \in \mathbb{R}^{n \times q}$, $q = p + 1$, and the intercept column is included unless stated otherwise. In this chapter, q is the number of fitted coefficients.

Under the usual full-column-rank condition,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}. \quad (13.3)$$

If the errors have mean zero and covariance $\sigma^2\mathbf{I}_n$, then

$$\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}, \quad \text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}. \quad (13.4)$$

The unknown error variance is estimated by

$$\hat{\sigma}^2 = \frac{RSS}{n - q} = \frac{\|\mathbf{y} - \hat{\mathbf{y}}\|^2}{n - q}. \quad (13.5)$$

This is the residual variance estimate with residual degrees of freedom $n - q = n - p - 1$.

If we also assume Gaussian errors,

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n), \quad (13.6)$$

then the usual standardized regression quantities have exact t -distributions. That is the distributional fact that turns standard errors into confidence and prediction intervals.

Observed data versus random response. The theoretical formulas use $\mathbf{Y}$ because the response vector is random before we observe the data. In the actual computation, we plug in the realized data vector $\mathbf{y}$.

13.3 Confidence Intervals for Coefficients

A coefficient confidence interval describes uncertainty about one true coefficient β_j . For $j = 0, 1, \dots, p$, the estimated coefficient is $\hat{\beta}_j$. Its estimated standard error is

$$SE(\hat{\beta}_j) = \hat{\sigma} \sqrt{[(\mathbf{X}^T\mathbf{X})^{-1}]_{j+1, j+1}}, \quad (13.7)$$

where the $j + 1$ indexing accounts for the intercept β_0 being the first entry of the coefficient vector. A $100(1 - \alpha)\%$ confidence interval for β_j is

$$\boxed{\widehat{\beta}_j \pm t_{1-\alpha/2, n-q} SE(\widehat{\beta}_j)}. \quad (13.8)$$

Here $t_{1-\alpha/2, n-q}$ is the appropriate quantile of a t -distribution with $n - q$ degrees of freedom.

Interpretation. A 95% confidence interval is not a claim that the probability is 95% that the fixed coefficient lies in this particular realized interval. The coefficient is fixed in the model. The interval is random because it depends on the sample. The repeated-sampling interpretation is:

If we repeated the data-collection and interval-building process many times under the same model, about 95% of the intervals would contain the true coefficient.

Briefing language. For a coefficient interval, say what coefficient you are describing and what units it uses. For example: “After adjusting for the other predictors, the model estimates the slope for this predictor to be between these two values, under the regression assumptions.”

13.4 Confidence Intervals for the Mean Response

A coefficient interval answers a question about a slope or intercept. Often, the analyst also wants a statement about the expected response at a particular predictor setting.

Let $\mathbf{x}_0 \in \mathbb{R}^q$ be a design row for the predictor setting of interest, including the intercept entry. The fitted mean response is

$$\widehat{\mu}_0 = \mathbf{x}_0^T \widehat{\boldsymbol{\beta}}. \quad (13.9)$$

The target is

$$\mu_0 = \mathbb{E}[Y_0 \mid \mathbf{x}_0] = \mathbf{x}_0^T \boldsymbol{\beta}. \quad (13.10)$$

The variance of the fitted mean is

$$\text{Var}(\widehat{\mu}_0 \mid \mathbf{X}) = \sigma^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0. \quad (13.11)$$

Therefore the estimated standard error for the fitted mean is

$$SE(\widehat{\mu}_0) = \widehat{\sigma} \sqrt{\mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}. \quad (13.12)$$

A $100(1 - \alpha)\%$ confidence interval for the mean response at $\mathbf{x}_0$ is

$$\boxed{\widehat{\mu}_0 \pm t_{1-\alpha/2, n-q} \widehat{\sigma} \sqrt{\mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}}. \quad (13.13)$$

Existing design rows. If $\mathbf{x}_0^T$ is the i th row of the fitted design matrix, then

$$h_{ii} = \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0. \quad (13.14)$$

In that special case, the mean-response interval becomes

$$\widehat{y}_i \pm t_{1-\alpha/2, n-q} \widehat{\sigma} \sqrt{h_{ii}}. \quad (13.15)$$

The hat value h_{ii} measures how far that predictor setting is from the center of the design. Mean estimates at unusual predictor settings tend to have larger uncertainty.

13.5 Prediction Intervals for Future Observations

A prediction interval answers a different question. It is not asking for the average response at $\mathbf{x}_0$. It is asking where a new observation might fall.

Write the future response as

$$Y_0 = \mathbf{x}_0^T \boldsymbol{\beta} + \epsilon_0, \quad (13.16)$$

where ϵ_0 is the new observation's random error. The prediction error is

$$Y_0 - \hat{\mu}_0. \quad (13.17)$$

It contains two sources of variation:

1. uncertainty in the fitted mean $\hat{\mu}_0$, and
2. the new noise term ϵ_0 .

Because of the new noise term, the prediction standard error is

$$\hat{\sigma} \sqrt{1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}. \quad (13.18)$$

Thus a $100(1 - \alpha)\%$ prediction interval for a future response at $\mathbf{x}_0$ is

$$\boxed{\hat{\mu}_0 \pm t_{1-\alpha/2, n-q} \hat{\sigma} \sqrt{1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}}. \quad (13.19)$$

For a new observation at the same predictor values as the i th observed row, this becomes

$$\hat{y}_i \pm t_{1-\alpha/2, n-q} \hat{\sigma} \sqrt{1 + h_{ii}}. \quad (13.20)$$

13.6 Why Prediction Intervals Are Wider

The mean-response interval and the prediction interval both start with the same fitted value $\hat{\mu}_0$. The difference is the extra 1 inside the square root:

$$\underbrace{\hat{\sigma}^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}_{\text{uncertainty in fitted mean}} \quad \text{versus} \quad \underbrace{\hat{\sigma}^2 (1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0)}_{\text{fitted mean uncertainty plus new noise}}. \quad (13.21)$$

The prediction interval is wider because it must cover an individual future response, not just the average response.

A practical way to remember the difference is:

A confidence interval for the mean response asks, “Where is the regression surface?” A prediction interval asks, “Where might the next point land?”

13.7 Which Interval Should I Use?

Use the target of the question to choose the interval.

Analyst question	What is uncertain?	Use
How much does predictor j change the response after adjustment?	the coefficient β_j	coefficient CI
What is the expected response for this type of case?	the mean $\mathbb{E}[Y \mid \mathbf{x}_0]$	mean-response CI
What response should we expect for one future case?	a new value Y_0	prediction interval
How accurate is the model over many cases?	prediction error over cases	validation metric

The last row is included to prevent a common mistake. Intervals are uncertainty statements about coefficients, means, or future observations. Prediction accuracy over many cases is a model-performance question. It is usually evaluated with test error, cross-validation, RMSE, MAE, or a similar metric.

13.8 Assumptions Behind the Intervals

The interval formulas above are not magic. They depend on the regression assumptions.

Assumption	Why it matters for intervals
Mean-zero errors	centers the model correctly around $\mathbf{X}\beta$.
Constant variance and uncorrelated errors	supports the usual variance and standard-error formulas.
Gaussian errors	gives exact finite-sample t -based intervals under the model.
Full column rank	makes the inverse-form OLS estimator and coefficient covariance formula unique.
Fixed design, or conditioning on observed $\mathbf{X}$	clarifies that the uncertainty is in the response noise, not in the already observed design matrix.

Chapter 12's residual diagnostics matter here. If the residuals show strong nonlinearity, changing variance, dependence, heavy tails, or influential observations, then the interval formulas may still compute, but their interpretation becomes less defensible.

Operational warning. An interval is only as credible as the model behind it. If the model is misspecified, a narrow interval can give a false sense of precision.

13.9 Intervals Are Not the Same as Prediction Accuracy

The slides that motivated this chapter also discuss alternative fit measures. Those measures are useful, but they answer a different question.

Quantity	What it summarizes
RMSE or RSE	typical size of residuals in response units, with squared-error emphasis.
MAE or MAD	average absolute residual size; less sensitive to very large residuals than RMSE.
R^2	fraction of sample variation explained relative to the mean-only baseline.
Adjusted R^2	an R^2 -style summary that penalizes additional predictors.

In this book, we will usually call the average absolute residual size *MAE*, or mean absolute error. Some course slides call the same idea MAD, mean absolute deviation. Do not confuse this with the median absolute deviation, which is a different robust summary used in other statistics contexts.

Fit measures help compare models and communicate error size. They do not replace confidence intervals or prediction intervals. Later model-selection chapters use these quantities more directly when choosing among competing models.

13.10 Optional Note: Least Absolute Deviation

Ordinary least squares chooses coefficients by minimizing squared residuals. An alternative is the least absolute deviation estimator, or LAD, which minimizes absolute residuals:

$$\min_{\mathbf{b}} \frac{1}{n} \sum_{i=1}^n |y_i - \mathbf{x}_i^T \mathbf{b}|. \quad (13.22)$$

This criterion is less sensitive to extremely large residuals than squared error because $|r|$ grows more slowly than r^2 . The cost is that the absolute value is not differentiable at zero, so the calculus used for ordinary least squares does not produce the same normal equations.

For this course, LAD is a useful comparison point, not a new main method. The key lesson is that the choice of loss function changes what the fitted model is trying to optimize.

13.11 Worked Interpretation Example

Suppose we fit a regression model to predict body fat percentage from several physical measurements. The fitted model can produce three different uncertainty statements for a new predictor setting $\mathbf{x}_0$.

1. **Coefficient interval:** “The coefficient for waist circumference is estimated with this uncertainty, after adjusting for the other predictors.”
2. **Mean-response interval:** “For people with this predictor pattern, the average body fat percentage is estimated to fall in this interval.”
3. **Prediction interval:** “For one future person with this predictor pattern, the observed body fat percentage is predicted to fall in this wider interval.”

These are not interchangeable. The coefficient interval talks about the model parameter. The mean-response interval talks about the average of a subpopulation or predictor pattern. The prediction interval talks about an individual future case.

13.12 Common Mistakes

Mistake 1: Using a mean-response interval as a prediction interval

A confidence interval for $\mathbb{E}[Y \mid \mathbf{x}_0]$ is too narrow for a future individual observation. Use a prediction interval when the question is about one future case.

Mistake 2: Forgetting the extra noise in a future response

The extra 1 in

$$1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0 \quad (13.23)$$

comes from the new observation's own error term. It is not a typo.

Mistake 3: Treating the confidence level as certainty for this dataset

A 95% confidence interval is a repeated-sampling procedure. It is not a promise that the realized interval has a 95% posterior probability unless a Bayesian framework is being used.

Mistake 4: Ignoring diagnostics before trusting intervals

Intervals can be printed even when residual diagnostics look bad. That does not mean the interval interpretation is defensible.

Mistake 5: Comparing error metrics on different scales

RMSE on Y and RMSE on $\log(Y)$ are not directly comparable. Compare model performance on the operational response scale that matters.

13.13 Operational Briefing Checklist

When briefing interval results, answer the following questions:

1. What is the target: coefficient, mean response, or future observation?
2. What predictor setting $\mathbf{x}_0$ is being used, if any?
3. Is the interval a confidence interval or a prediction interval?
4. What confidence level is being used?
5. What residual degrees of freedom are used?
6. Are the model assumptions plausible enough for this interpretation?
7. Do residual diagnostics reveal nonlinearity, changing variance, dependence, or influential observations?
8. Is the interval on the same scale as the operational decision?
9. How should the decision-maker use the interval, not just the point estimate?

Coding companion reference. Package-specific Python/R commands for coefficient intervals, mean-response intervals, prediction intervals, and model-fit metrics belong in the separate Coding and Software Cookbook companion. The main chapter keeps the statistical meaning and formulas.

13.14 Chapter Summary

Regression uncertainty has three common targets:

coefficient	mean response	future observation
-------------	---------------	--------------------

(13.24)

A coefficient confidence interval describes uncertainty about β_j . A mean-response confidence interval describes uncertainty about $\mathbb{E}[Y \mid \mathbf{x}_0]$. A prediction interval describes uncertainty about a future response $Y_0 \mid \mathbf{x}_0$, so it is wider because it includes new observation noise.

The key formulas are:

$$\hat{\beta}_j \pm t_{1-\alpha/2, n-q} SE(\hat{\beta}_j), \quad (13.25)$$

$$\hat{\mu}_0 \pm t_{1-\alpha/2, n-q} \hat{\sigma} \sqrt{\mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}, \quad (13.26)$$

$$\hat{\mu}_0 \pm t_{1-\alpha/2, n-q} \hat{\sigma} \sqrt{1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}. \quad (13.27)$$

Fit metrics such as RMSE, MAE, R^2 , and adjusted R^2 summarize performance. They do not replace interval statements. A defensible analysis explains both the fitted value and the uncertainty around it.

13.15 Practice and Workflow

Focused Study Checklist

Use this as a final check after studying regression intervals.

1. Can you distinguish a coefficient confidence interval from a mean-response confidence interval?
2. Can you explain why a prediction interval is wider than a mean-response interval?
3. Can you identify whether a question is about β_j , $\mathbb{E}[Y \mid \mathbf{x}_0]$, or $Y_0 \mid \mathbf{x}_0$?
4. Can you state the residual degrees of freedom $n - p - 1$ or $n - q$?
5. Can you explain the role of $\hat{\sigma}$ in the interval formulas?
6. Can you recognize when the hat value h_{ii} appears in an interval formula?
7. Can you explain why residual diagnostics matter before reporting intervals?
8. Can you avoid using RMSE, MAE, or R^2 as a substitute for an uncertainty interval?

Chapter 14

Likelihood, GLMs, and Hypothesis Testing Beyond Least Squares

14.1 The Big Picture

Least squares gave us a way to fit a regression model by minimizing squared residuals. Chapter 13 used the Gaussian-error theory behind least squares to build coefficient confidence intervals, mean-response intervals, and prediction intervals.

This chapter asks a different question:

What if we start with a probability model for the response and fit the unknown parameters by choosing the values that make the observed data most plausible?

That idea is called *maximum likelihood*. It explains why ordinary least squares is the natural estimator under Gaussian errors, and it gives us a way to build models when least squares is not appropriate, such as logistic regression for binary responses and Poisson regression for counts.

The goal is not to turn this chapter into a complete generalized-linear-model course. The goal is to give the statistical logic needed to recognize likelihood-based models, understand their assumptions, and interpret their tests carefully.

Connection to earlier chapters. Chapters 5–10 built the linear-regression model, its assumptions, and the distributional behavior of $\hat{\beta}$. Chapter 12 emphasized diagnostics. Chapter 13 turned Gaussian fixed-design theory into interval estimates. This chapter explains the likelihood logic that sits underneath those results and extends the same modeling mindset beyond Gaussian least squares.

Math Foundations Connection. Likelihood models use functions, optimization, gradients, and sometimes second-derivative checks. Regression likelihoods also rely on matrix-vector multiplication, least-squares objectives, full-rank inverse formulas, and quadratic residual forms. These ideas are supported by the Math Foundations textbook, especially Chapters 5, 7, 10, 12, and 13.

14.2 Likelihood and Log-Likelihood

Suppose we observe data $Y_1, \dots, Y_n$ from a probability model with parameter vector $\boldsymbol{\theta}$. For each possible value of $\boldsymbol{\theta}$, the model assigns a probability density or probability mass to the observed data.

For independent observations, the likelihood is

$$L(\boldsymbol{\theta} \mid y_1, \dots, y_n) = \prod_{i=1}^n f(y_i \mid \boldsymbol{\theta}), \quad (14.1)$$

where f is the density or mass function of the response model.

The *maximum likelihood estimator*, or MLE, is

$$\hat{\boldsymbol{\theta}}_{MLE} = \arg \max_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta} \mid y_1, \dots, y_n). \quad (14.2)$$

In words, the MLE is the parameter value that makes the observed data most likely under the model.

Because products are often awkward, we usually maximize the log-likelihood

$$\ell(\boldsymbol{\theta}) = \log L(\boldsymbol{\theta} \mid y_1, \dots, y_n) = \sum_{i=1}^n \log f(y_i \mid \boldsymbol{\theta}). \quad (14.3)$$

The logarithm is increasing, so the value of $\boldsymbol{\theta}$ that maximizes the likelihood also maximizes the log-likelihood.

Important interpretation. The likelihood is a function of the parameter after the data have been observed. It is not the probability that the model is true. It is a way to compare parameter values within an assumed model family.

14.3 A First MLE Example: The Poisson Mean

Let $Y_1, \dots, Y_n$ be independent observations from a Poisson distribution with mean $\mu > 0$:

$$Y_i \sim \text{Pois}(\mu). \quad (14.4)$$

The probability mass function is

$$f(y_i \mid \mu) = \frac{e^{-\mu} \mu^{y_i}}{y_i!}. \quad (14.5)$$

The log-likelihood is

$$\ell(\mu) = \sum_{i=1}^n \log \left(\frac{e^{-\mu} \mu^{y_i}}{y_i!} \right) \quad (14.6)$$

$$= \sum_{i=1}^n [-\mu + y_i \log(\mu) - \log(y_i!)] . \quad (14.7)$$

Taking the derivative and setting it equal to zero gives

$$0 = -n + \frac{1}{\hat{\mu}_{MLE}} \sum_{i=1}^n y_i, \quad (14.8)$$

so

$$\boxed{\hat{\mu}_{MLE} = \bar{y}.} \quad (14.9)$$

When at least one observed count is positive, this is the interior maximizer. The all-zero sample is a boundary case: the likelihood is maximized at $\mu = 0$ if that boundary is allowed. For a Poisson mean, maximum likelihood gives the sample mean. This example is simple, but it illustrates the pattern: write a probability model, form the likelihood, and choose the parameter value that maximizes it.

14.4 Gaussian-Error Regression and OLS as MLE

Now return to the fixed-design linear model. Under the strongest regression assumption,

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n), \quad (14.10)$$

so

$$\mathbf{Y} \mid \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n). \quad (14.11)$$

The unknown parameters are $\boldsymbol{\beta}$ and σ^2 .

For observed response vector $\mathbf{y}$, the log-likelihood has the form

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + C, \quad (14.12)$$

where C does not depend on $\boldsymbol{\beta}$ or σ^2 .

For fixed σ^2 , maximizing (14.12) over $\boldsymbol{\beta}$ is equivalent to minimizing

$$RSS(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b}). \quad (14.13)$$

Therefore, under Gaussian errors, the maximum likelihood estimator for $\boldsymbol{\beta}$ is the ordinary least-squares estimator. If $\mathbf{X}$ has full column rank, then

$$\boxed{\hat{\boldsymbol{\beta}}_{MLE} = \hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.} \quad (14.14)$$

This is the key bridge: least squares is an algebraic projection method, but under Gaussian errors it is also a likelihood method.

14.5 Estimating σ^2 : MLE Versus Residual Variance

The MLE for σ^2 is

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \frac{RSS}{n}. \quad (14.15)$$

This is the empirical mean squared residual.

However, after fitting $q = p + 1$ coefficients, the residuals have only $n - q$ degrees of freedom. The usual unbiased residual variance estimator is

$$s^2 = \frac{RSS}{n - q} = \frac{RSS}{n - p - 1}. \quad (14.16)$$

This is the quantity used in the standard errors, t -tests, and intervals from the previous chapters.

The distinction matters:

Quantity	Formula	Main role
MLE for σ^2	RSS/n	likelihood maximization
Unbiased residual variance	$RSS/(n - q)$	standard errors and classical inference

The same issue appears in the sample-variance formula: using $1/n$ gives the MLE under a normal model, while using $1/(n - 1)$ gives the usual unbiased sample variance.

14.6 Why the Assumptions Matter

Maximum likelihood always starts with a probability model. If the model changes, the likelihood changes, and the interpretation of the estimator changes.

For ordinary least squares, the formula

$$(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (14.17)$$

enjoys several roles:

- it minimizes squared residuals;
- it is the orthogonal-projection estimator from the design matrix viewpoint;
- it is the MLE for β under Gaussian independent errors.

If the Gaussian-error assumption fails, the estimator may still be the least-squares estimator, but the exact Gaussian likelihood and exact small-sample t - and F -based conclusions no longer follow automatically.

This is why diagnostics still matter. A software package can print standard errors, confidence intervals, and p-values even when the model assumptions are questionable. The analyst is responsible for deciding whether those summaries are defensible.

14.7 Generalized Linear Models: The Three-Part Idea

Linear regression assumes a quantitative response whose conditional mean is linear in the predictors and whose errors are Gaussian. Many data-analysis problems do not fit that pattern.

A *generalized linear model*, or GLM, keeps the idea of a linear predictor but changes the response distribution and the way the mean connects to the predictors. The basic pieces are:

1. a response distribution, such as Bernoulli for binary data or Poisson for counts;
2. a linear predictor $\eta = \mathbf{x}^T \boldsymbol{\beta}$;
3. a link function connecting the conditional mean $\mu(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{x}]$ to η .

In symbols, the link function satisfies

$$g(\mu(\mathbf{x})) = \mathbf{x}^T \boldsymbol{\beta}. \quad (14.18)$$

The details of GLM theory are beyond this course. The important idea is that likelihood lets us fit models where the response is not naturally Gaussian and where a straight-line mean model would violate basic constraints.

14.8 Logistic Regression for Binary Responses

Suppose Y is binary, with $Y = 1$ for success and $Y = 0$ for failure. A linear model for the probability,

$$P(Y = 1 \mid \mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta}, \quad (14.19)$$

would be dangerous because $\mathbf{x}^T \boldsymbol{\beta}$ can be negative or greater than one.

Logistic regression avoids this problem by using the logistic function:

$$P(Y = 1 \mid \mathbf{x}) = \frac{\exp(\mathbf{x}^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}^T \boldsymbol{\beta})}. \quad (14.20)$$

This quantity is always between zero and one.

Equivalently,

$$\log \left(\frac{P(Y = 1 \mid \mathbf{x})}{1 - P(Y = 1 \mid \mathbf{x})} \right) = \mathbf{x}^T \boldsymbol{\beta}. \quad (14.21)$$

The left side is the log-odds. Thus logistic-regression coefficients are linear effects on log-odds, not direct additive effects on probability.

Classification rule. Logistic regression estimates a probability. To turn that probability into a class label, the analyst chooses a threshold, such as classifying as $Y = 1$ when $\hat{P}(Y = 1 \mid \mathbf{x}) > 0.5$. That threshold is a decision rule, not a law of the model.

Scope warning. This section introduces logistic regression as a likelihood-based model. Broader classification metrics and other classification algorithms are separate topics and are not developed here.

14.9 Poisson Regression for Count Responses

Suppose Y is a count: number of arrivals, number of defects, number of events, or number of items observed in a fixed window. A Poisson model is often a first candidate:

$$Y \mid \mathbf{x} \sim \text{Pois}(\mu(\mathbf{x})). \quad (14.22)$$

The mean $\mu(\mathbf{x})$ must be positive. A direct linear mean $\mu(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta}$ can be negative, so it is usually not a safe final model.

Poisson regression uses a log link:

$$\log \mu(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta}, \quad \text{or equivalently} \quad \mu(\mathbf{x}) = \exp(\mathbf{x}^T \boldsymbol{\beta}). \quad (14.23)$$

This guarantees a positive fitted mean.

A coefficient in a Poisson log-link model has a multiplicative interpretation. Increasing predictor x_j by one unit changes the fitted mean by a factor of $\exp(\beta_j)$, holding the other predictors fixed.

14.10 Hypothesis Testing Revisited

Earlier chapters introduced two common tests in Gaussian linear regression:

Test	Typical question	Main setting
<i>t</i> -test	Is one coefficient different from zero?	one coefficient at a time
<i>F</i> -test	Does a larger linear model improve over a smaller nested linear model?	Gaussian linear models

In this framework, ANOVA can be viewed as a special case of comparing nested Gaussian linear models, often when the predictors are categorical group indicators.

This chapter does not re-derive those tests. The key point is that they rely on the Gaussian regression setup, the residual variance estimate, and the distributional results developed earlier.

Likelihood gives a more general model-comparison idea: compare how well a smaller model and a larger model explain the observed data through their maximized log-likelihoods.

14.11 Likelihood-Ratio Tests

Suppose we have two nested likelihood-based models:

- a smaller model, with fewer free parameters;
- a larger model, which contains the smaller model as a special case.

Let ℓ_{small} and ℓ_{full} be the maximized log-likelihoods for the two models. Since the full model has at least as much flexibility,

$$\ell_{full} \geq \ell_{small}. \quad (14.24)$$

The likelihood-ratio test statistic is

$$\Lambda = 2(\ell_{full} - \ell_{small}). \quad (14.25)$$

Under regularity conditions and under the null hypothesis that the smaller model is sufficient, the large-sample reference distribution is

$$\Lambda \approx \chi_d^2, \quad (14.26)$$

where d is the difference in the number of free parameters between the full and smaller models.

Interpretation. A large likelihood-ratio statistic means that the full model improved the maximized log-likelihood substantially. That is evidence against the smaller model, provided the models are nested and the likelihood assumptions are appropriate.

Connection to the F -test. For Gaussian linear models, likelihood-ratio tests and F -tests are related because both compare nested models using residual error information. They are not the same object, and the likelihood-ratio test is more general: it applies to many maximum-likelihood models, including GLMs.

14.12 Common Mistakes

Mistake 1: Treating likelihood as the probability the model is true

Likelihood compares parameter values inside a model family after observing the data. It is not the probability that the model itself is true.

Mistake 2: Using a linear mean where the response has constraints

Probabilities must lie between zero and one. Poisson means must be positive. Link functions help enforce these constraints.

Mistake 3: Forgetting the assumptions behind classical tests

The usual t - and F -tests depend on the Gaussian-error regression framework. Diagnostics and modeling judgment still matter.

Mistake 4: Comparing non-nested models with a likelihood-ratio test

The likelihood-ratio test described here is for nested models. If two models are not nested, use a different comparison tool, such as validation error or an information criterion.

Mistake 5: Turning logistic regression into a black-box classifier too quickly

Logistic regression estimates probabilities. Classification requires a threshold and a decision context. The threshold should be chosen to match the operational problem.

14.13 Chapter Summary

Maximum likelihood fits a probabilistic model by choosing the parameter values that maximize the likelihood of the observed data. For independent data,

$$L(\boldsymbol{\theta}) = \prod_{i=1}^n f(y_i \mid \boldsymbol{\theta}), \quad \ell(\boldsymbol{\theta}) = \sum_{i=1}^n \log f(y_i \mid \boldsymbol{\theta}). \quad (14.27)$$

Under Gaussian errors,

$$\mathbf{Y} \mid \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n), \quad (14.28)$$

so the MLE for $\boldsymbol{\beta}$ is the ordinary least-squares estimator. The MLE for σ^2 is RSS/n , while the usual unbiased residual variance estimator is $RSS/(n - p - 1)$.

GLMs extend the regression idea by combining a response distribution, a linear predictor, and a link function. Logistic regression models binary responses with probabilities between zero and one. Poisson regression models count responses with positive means.

Likelihood-ratio tests compare nested likelihood-based models using the improvement in maximized log-likelihood. They are powerful and general, but they require nested models, appropriate assumptions, and careful interpretation.

14.14 Practice and Workflow

Focused Study Checklist

Use this checklist after studying likelihood and GLMs.

1. Can you explain the difference between a likelihood and a probability statement about a model?
2. Can you write the likelihood for independent observations as a product?
3. Can you explain why maximizing log-likelihood gives the same estimator as maximizing likelihood?
4. Can you show why Gaussian-error least squares is also maximum likelihood for $\boldsymbol{\beta}$?
5. Can you distinguish RSS/n from $RSS/(n - p - 1)$?
6. Can you explain why logistic regression keeps fitted probabilities between zero and one?
7. Can you explain why Poisson regression usually uses a log link?
8. Can you state when a likelihood-ratio test is appropriate?
9. Can you explain why diagnostics still matter before trusting likelihood-based inference?

Coding companion reference. Package-specific commands for fitting logistic regression, Poisson regression, GLMs, and likelihood-ratio tests belong in the separate Coding and Software Cookbook companion. This chapter keeps the statistical meaning and model structure.

Chapter 15

Model Selection and Variable Selection

15.1 The Big Picture

A regression model is not good just because it fits the training data well. Least squares chooses coefficients to make the training residuals small, and in nested least-squares models, adding predictors can only decrease or preserve the training residual sum of squares. If we choose models only by training error, we will usually choose models that are too complicated.

This chapter asks a practical question:

How should we choose among candidate models when the goal is to perform well on future data, not just on the data used to fit the model?

The answer depends on the analysis goal. If the goal is prediction, we want a model that generalizes well. If the goal is interpretation or inference, we also want a model whose coefficients are stable, meaningful, and defensible. Model selection is the process of balancing fit, complexity, interpretability, and validation evidence.

Connection to earlier chapters. Chapter 2 introduced prediction error, training error, test error, overfitting, and flexibility. Chapters 5–12 built linear-regression models and diagnostic workflows. Chapter 13 separated uncertainty intervals from prediction accuracy. Chapter 14 used likelihood to fit probabilistic models and to compare nested models with likelihood-ratio tests. This chapter asks a broader selection question: when several plausible models are available, how should we choose among them? Some tools in this chapter, such as AIC and BIC, build directly on likelihood. Others, such as adjusted R^2 , Mallows-type C_p , validation, and cross-validation, focus more directly on prediction error, model complexity, or future performance.

Math Foundations Connection. Model selection uses residual lengths, squared-error objectives, design matrices, degrees of freedom, and the geometry of least squares. These ideas are supported by the Math Foundations textbook, especially Chapters 5, 6, 7, 10, 11, and 12.

15.2 Why Training Error Is Too Optimistic

For a candidate model m , write

$$RSS_m = \|\mathbf{y} - \hat{\mathbf{y}}_m\|^2, \quad \text{MSE}_{\text{train},m} = \frac{RSS_m}{n}. \quad (15.1)$$

This training MSE measures how well the model fits the data used to estimate it. It is useful, but it is usually too optimistic as an estimate of future prediction error.

There are two reasons.

1. The model was fit to make the training residuals small.
2. A more flexible model can chase patterns that are partly real signal and partly random noise.

When one least-squares candidate model contains all predictors in another, the larger model cannot have larger training RSS. That does not mean the larger model is better for future data. A selection rule must either estimate future performance directly or penalize the model for using extra flexibility.

15.3 Selection Criteria as Fit Plus Penalty

Many model-selection criteria have the same basic structure:

$$\boxed{\text{selection score} = \text{training fit term} + \text{complexity penalty}.} \quad (15.2)$$

The fit term rewards models that explain the observed data. The penalty term discourages unnecessary complexity. Different criteria measure fit and complexity in different ways.

For linear models, let q_m be the number of fitted coefficients in model m , including the intercept. A model with more predictors has larger q_m . The selection problem is not simply to make q_m small or to make RSS small. The goal is to find a model that is accurate enough, simple enough, and defensible for the task.

15.4 Mallows-Type C_p on the MSE Scale

Mallows C_p is motivated by the gap between training MSE and expected test MSE. Under the fixed-design Gaussian linear-model setup, with mean-zero errors and constant variance, the expected optimism in training MSE is approximately

$$\frac{2}{n}\sigma^2 q_m. \quad (15.3)$$

Since σ^2 is unknown, we estimate it with a residual variance estimate s^2 , often taken from the largest model under consideration. On the MSE scale, the resulting score is

$$\boxed{C_{p,m}^{(MSE)} = \frac{RSS_m}{n} + \frac{2}{n}s^2 q_m.} \quad (15.4)$$

Smaller values are better.

The first term rewards fit. The second term penalizes the number of fitted coefficients. Adding a predictor is helpful only if the reduction in training error is large enough to justify the extra flexibility.

Why the largest model appears. The residual variance estimate s^2 should represent the noise level, not the error from a deliberately underfit small model. Using a reasonably large model helps estimate σ^2 without forcing the small candidate models to explain variation they were not allowed to model.

Cautions. Mallows-type criteria rely on regression assumptions, especially mean-zero errors and constant variance. They also depend on the chosen candidate set and the quality of s^2 . They are useful tools, not automatic truth detectors.

Convention warning. Different references scale and shift Mallows C_p in different ways. In this chapter, (15.4) is written on the training-MSE scale to match the course intuition. The important point is usually the ordering of candidate models, not the absolute numerical value.

15.5 AIC

The Akaike Information Criterion, or AIC, is a likelihood-based selection criterion. If $\ell(\hat{\boldsymbol{\theta}})$ is the maximized log-likelihood and d is the number of estimated likelihood parameters being counted in the criterion, then a common form is

$$\boxed{\text{AIC} = -2\ell(\hat{\boldsymbol{\theta}}) + 2d.} \quad (15.5)$$

Smaller AIC is better.

The term $-2\ell(\hat{\boldsymbol{\theta}})$ rewards models that give high likelihood to the observed data. The term $2d$ penalizes the number of estimated likelihood parameters; this d is a general parameter count, while q_m remains the linear-model coefficient count used for regression degrees of freedom. AIC is more general than a least-squares-only criterion because it applies to any model with a likelihood.

How to compare AIC values. Compare AIC values only among models fit to the same response data under compatible likelihoods. Software may differ by constants or scaling conventions, so do not compare AIC values across unrelated analyses or incompatible model families.

15.6 BIC

The Bayesian Information Criterion, or BIC, uses the same likelihood term as AIC but applies a stronger penalty for complexity:

$$\boxed{\text{BIC} = -2\ell(\hat{\boldsymbol{\theta}}) + d\log(n).} \quad (15.6)$$

Smaller BIC is better.

When n is reasonably large, $\log(n) > 2$, so BIC penalizes extra parameters more heavily than AIC. As a result, BIC often selects smaller models. This can be useful when the analyst wants a parsimonious model, but it can also remove predictors that help prediction.

AIC and BIC are not moral rankings of models. They are ways to compare candidate models under a defined fitting framework. They should be interpreted alongside diagnostics, validation performance, and the purpose of the analysis.

15.7 Adjusted R^2

Ordinary R^2 measures the fraction of total variation explained by the fitted model:

$$R^2 = 1 - \frac{RSS}{TSS}. \quad (15.7)$$

For models fit to the same response, ordinary R^2 cannot decrease when predictors are added. This makes it a poor standalone tool for choosing model size.

Adjusted R^2 adds a degrees-of-freedom correction:

$$R_{adj}^2 = 1 - \frac{RSS/(n - q_m)}{TSS/(n - 1)}. \quad (15.8)$$

Here q_m is the number of fitted coefficients in the candidate model, including the intercept. Larger adjusted R^2 is better.

Adjusted R^2 is easy to read from regression output, but it is only one criterion. It should not replace validation, diagnostics, or domain judgment.

15.8 Cross-Validation for Model Selection

The criteria above adjust training fit using mathematical penalties. Cross-validation takes a different approach: estimate future performance by repeatedly fitting the model on part of the data and evaluating it on held-out data.

Validation-set approach. Split the data into a training set and a validation set. Fit the model on the training set. Compute prediction error on the validation set. This is simple, but the result depends on the split.

Leave-one-out cross-validation. Leave-one-out cross-validation, or LOOCV, fits the model n times. Each time, one observation is held out and the model is trained on the remaining $n - 1$ observations. The held-out errors are averaged. LOOCV uses the data efficiently because every observation is used for validation exactly once, but a direct implementation may require n model fits.

LOOCV and analytic shortcuts. Mallows-type C_p and AIC can be viewed as analytic fit-plus-penalty shortcuts aimed at estimating future performance, or the expected optimism of training fit, under modeling assumptions. These shortcuts can be useful when repeated refitting is expensive, but they are not literally identical to LOOCV in all settings. Read them alongside diagnostics, validation evidence, and domain judgment.

k -fold cross-validation. In k -fold cross-validation, the data are split into k folds. Each fold is used once as the validation fold while the other folds are used for training. The validation errors are averaged. Common choices such as $k = 5$ or $k = 10$ often give a practical balance between computation and stability.

Workflow caution. Cross-validation should repeat the modeling workflow, not just the final fitting step. If preprocessing is learned from the data, such as scaling, imputation, or feature selection, it should be learned inside the training portion of each fold to avoid leakage.

15.9 Best Subset Selection

If there are p possible predictors, there are 2^p possible subsets of predictors. Best subset selection tries many or all of these subsets and chooses the best one according to a selection criterion.

The advantage is conceptual clarity: it asks directly which subset of predictors gives the best score. The disadvantage is computation. The number of subsets grows quickly. With 10 predictors, there are $2^{10} = 1024$ subsets. With 30 predictors, there are more than one billion.

Best subset selection is most useful when the number of candidate predictors is small and the analyst can define a meaningful candidate set in advance.

15.10 Forward, Backward, and Stepwise Selection

When best subset selection is too expensive, stepwise methods search through a smaller set of candidate models.

Forward selection. Start with a small model, often the intercept-only model. At each step, add the predictor that gives the best improvement according to the selection criterion. Stop when no addition improves the criterion enough.

Backward selection. Start with a large model. At each step, remove the predictor whose removal gives the best improvement, or the least damage, according to the selection criterion. Stop when no deletion improves the criterion enough.

Hybrid stepwise selection. A hybrid method allows both additions and deletions as the search proceeds. This can correct some earlier choices, but it is still a path-based search.

Stepwise methods are algorithms for searching model space. They are not guarantees that the selected model is the true model, the best possible model, or the most interpretable model.

15.11 The Post-Selection Inference Warning

The most dangerous mistake in model selection is to search through many models and then report the selected model as if it had been chosen before looking at the data.

If a procedure tries many predictor sets, compares many criteria, or repeatedly adds and removes predictors based on p-values, then the final model has been shaped by the data. The usual coefficient p-values in the selected model do not automatically account for that search. They can be too optimistic.

Post-selection p-values are not ordinary p-values unless the selection process is accounted for.

This does not mean stepwise methods are forbidden. It means the analyst must be honest about the selection process. If the goal is prediction, validate the selected model on held-out data or through cross-validation. If the goal is interpretation or inference, report the selection procedure and be cautious about strong claims based only on selected-model p-values.

15.12 Multiple Testing and Data-Driven Inference

The post-selection warning is one example of a broader issue: many inferential tools are calibrated for a planned analysis, not for unlimited searching. A single p-value has a precise meaning only after the model, hypothesis, test statistic, and assumptions have been specified. If the analyst repeatedly changes the model, tries many transformations, tests many subgroups, or searches until something looks significant, the reported p-values no longer describe a single planned test.

This does not make exploratory analysis wrong. Exploration is often how good analyses begin. The problem is pretending afterward that an exploratory result was confirmatory. A defensible analysis separates what was planned in advance from what was discovered by searching.

One planned test versus many chances to find something

For one planned test at level α , the Type I error rate is controlled at α , assuming the model and test assumptions hold. For example, a test run at $\alpha = 0.05$ is designed so that, under the null hypothesis, it rejects about 5% of the time in repeated use.

That calibration changes when the analyst has many chances to find something. Examples include testing many individual coefficients, deleting predictors one at a time based on p-values, trying several transformations of the same variable, searching across many candidate models, checking many subgroups, trying several response definitions, or using a variable-selection method, such as the LASSO introduced in the next chapter, and then refitting ordinary least squares as if the selected variables had been chosen in advance. Each additional search creates another opportunity for a chance pattern in the data to look meaningful.

The key question is not simply, “What is the p-value in the final model?” The key question is, “How many ways did we give the data a chance to produce this result?”

Family-wise error rate

A *family* is the collection of tests or comparisons that should be considered together for a particular claim. The *family-wise error rate*, or FWER, is

$$\text{FWER} = P(\text{at least one false positive in the family}). \quad (15.9)$$

This is different from the Type I error rate of one individual test.

If all m null hypotheses in a family are true, the tests are independent under the null, and each test is run at level α , then the probability of at least one false positive is

$$1 - (1 - \alpha)^m. \quad (15.10)$$

This formula is useful intuition under those assumptions; it is not a universal exact formula for every multiple-testing situation. Still, it shows the scale of the problem. If $m = 20$ independent true null hypotheses are each tested at $\alpha = 0.05$, then

$$1 - (1 - 0.05)^{20} \approx 0.64. \quad (15.11)$$

Even when every null hypothesis is true, there is about a 64% chance of at least one false positive somewhere in the family.

Bonferroni correction

The simplest correction is the Bonferroni correction. If the analyst wants to control the family-wise error rate at level α across m planned comparisons, then test each comparison at level

$$\frac{\alpha}{m}. \quad (15.12)$$

Equivalently, one may multiply each p-value by m and compare the adjusted p-values with α , with adjusted p-values truncated at 1 when necessary.

Bonferroni is simple, transparent, and conservative. Its strongest simple justification does not require the tests to be independent; it follows from the fact that the probability of at least one error is no larger than the sum of the individual error probabilities. The tradeoff is power. When m is large, α/m can be very small, making it harder to detect real effects. Bonferroni is most useful when the family of planned comparisons is clearly defined; it does not automatically validate a long, undocumented modeling search.

False discovery rate

Family-wise error rate asks a strict question: did we make any false positive conclusion in the family? The *false discovery rate*, or FDR, asks a different question: among the discoveries we report, what fraction are false on average? In words, FWER focuses on whether at least one false positive occurred, while FDR focuses on the expected share of reported discoveries that are false.

FDR control can be more useful when the analysis involves many potential signals and the goal is to identify a manageable set for follow-up. Common FDR procedures order the p-values and use a gradually changing cutoff rather than the single α/m cutoff used by Bonferroni. The details are beyond this chapter, but the interpretation matters: FDR is still a multiplicity-aware procedure, not permission to search without documentation.

Resampling intuition

Permutation and resampling approaches provide another way to ask what counts as surprising. The basic idea is to approximate the behavior of a statistic under a null setup by repeatedly reshuffling or resampling the data in a way that preserves the structure assumed under the null hypothesis. The observed statistic is then compared with the distribution generated under those null reshufflings or resamples.

The important word is *appropriate*. A resampling scheme must match the structure of the question. Time ordering, clustered observations, paired measurements, repeated observations, and train/test

separation can all affect what reshuffling is legitimate. If the original analysis searched across many predictors or models, a resampling check should ideally mimic the same search rather than calibrating only one fixed test.

Reporting data-driven inference

A clear report should distinguish confirmatory analysis from exploratory analysis. Confirmatory claims are tied to hypotheses, models, and comparisons that were specified before looking for the result. Exploratory claims may be useful and important, but they should be labeled as discoveries that need follow-up, validation, or replication.

When reporting an analysis that involved many tests or searches, state what family of tests, comparisons, models, transformations, thresholds, or subgroups was considered; whether the analysis was planned, exploratory, or a mixture of both; whether variable selection, model selection, tuning, or repeated testing occurred before the reported inference; whether any multiplicity correction, validation strategy, or resampling check was used; and which conclusions are confirmatory versus exploratory.

Prediction validation is valuable, but it answers a different question. A held-out test set or cross-validation can estimate future predictive performance. It does not automatically repair coefficient-level p-values or confidence intervals after data-driven variable selection. Similarly, Bonferroni, FDR, and resampling can improve calibration for defined families of tests, but they do not make every modeling search valid. The defensible habit is to analyze the data as a coherent whole, document the search process, and match the strength of the conclusion to the way the result was found.

15.13 A Defensible Model-Selection Workflow

Use the following workflow to keep model selection defensible.

1. **State the goal.** Is the model primarily for prediction, interpretation, or both?
2. **Define the candidate predictor set.** Use the analytic question, data quality checks, and subject-matter knowledge.
3. **Fit a baseline model.** Start with a simple, defensible model before expanding.
4. **Choose comparison tools.** Decide whether to use adjusted R^2 , C_p , AIC, BIC, cross-validation, multiplicity-aware inference, or a combination.
5. **Search carefully.** Use best subset or stepwise methods only as tools, not as substitutes for judgment; avoid changing the model until the p-values look good.
6. **Check diagnostics.** A selected model can still have nonlinearity, heteroskedasticity, influential observations, or collinearity problems.
7. **Validate prediction performance.** When prediction matters, estimate performance on data not used to choose the model.
8. **Document the process.** Record what models, transformations, tests, and subgroups were considered; what criterion was used; what was selected; and which claims remain exploratory.

The final model should be easy to explain: why those predictors, why that complexity, why that criterion, and why the conclusion is defensible. If the analysis involved many tests or searches, the report should also state whether any multiplicity correction, resampling check, or validation strategy was used.

The next chapter develops another family of tools for controlling complexity and instability, including regularization and dimension reduction. Those methods complement model selection; they do not remove the need to define the goal, validate prediction performance, and document the modeling process.

15.14 Common Mistakes

Mistake 1: Choosing the model with the lowest training RMSE

Training RMSE rewards flexibility. It usually decreases as predictors are added. Use a penalty or validation strategy when comparing models of different sizes.

Mistake 2: Using ordinary R^2 alone

Ordinary R^2 does not penalize extra predictors. It can make larger models look better even when they do not generalize better.

Mistake 3: Treating AIC or BIC as universal truth

AIC and BIC compare candidate models under assumptions and conventions. They do not replace diagnostics, validation, or subject-matter reasoning.

Mistake 4: Reporting selected-model p-values without a selection caveat

If the model was selected after searching, the p-values in the final model can be optimistic. Say how the model was selected.

Mistake 5: Changing the model until the p-values look good

This is not defensible inference. It is data-driven searching disguised as hypothesis testing.

Mistake 6: Ignoring diagnostics after selection

Selection criteria do not check every modeling assumption. Residual diagnostics, leverage, influence, and practical interpretation still matter.

15.15 Chapter Summary

Model selection is about choosing a model that supports the analysis goal. Training error alone is not enough because it is usually too optimistic.

Criteria such as Mallows-type C_p , AIC, BIC, and adjusted R^2 combine fit with a complexity penalty. Cross-validation estimates prediction performance more directly by evaluating models on held-out data. LOOCV uses the data efficiently but may require many refits, while analytic criteria such as C_p and AIC offer assumption-based shortcuts for correcting training-fit optimism. Best subset and stepwise methods search through candidate predictor sets, but they do not remove the need for judgment.

The most important caution is post-selection inference, which is one form of multiple testing and data-driven inference. If the data were used to select the model, choose transformations, screen subgroups, tune a procedure, or search across many tests, ordinary p-values from the final analysis should be interpreted carefully. A defensible analysis documents the search process, distinguishes confirmatory claims from exploratory ones, checks diagnostics, validates when prediction matters, and explains why the selected model is appropriate for the task.

15.16 Practice and Workflow

Focused Study Checklist

Use this checklist after studying model selection.

1. Can you explain why training error is usually too optimistic?
2. Can you distinguish ordinary R^2 from adjusted R^2 ?
3. Can you explain the fit-plus-penalty structure behind C_p , AIC, and BIC?
4. Can you state whether lower or higher is better for each criterion?
5. Can you explain why cross-validation estimates future performance more directly?
6. Can you explain why LOOCV is data-efficient but can be computationally expensive, and how C_p and AIC serve as assumption-based shortcuts?
7. Can you distinguish best subset selection from forward and backward selection?
8. Can you explain why selected-model p-values need caution?
9. Can you define family-wise error rate and explain why many tests increase the chance of at least one false positive?
10. Can you distinguish Bonferroni-style family-wise error control from false discovery rate control?
11. Can you describe how you would document a model-selection workflow?
12. Can you decide when prediction validation matters more than coefficient inference?

Coding companion reference. Package-specific commands for AIC and BIC extraction, cross-validation, subset selection, stepwise selection, and model comparison tables belong in the separate Coding and Software Cookbook companion. This chapter keeps the statistical meaning and workflow logic.

Chapter 16

Regularization and Dimension Reduction: Ridge, LASSO, PCA, and PCR

16.1 The Big Picture

Ordinary least squares is a powerful starting point, but it is not always stable. When predictors are numerous, highly correlated, or nearly redundant, the least-squares coefficients can become fragile. Small changes in the data can produce large changes in individual coefficient estimates, even when the fitted values remain reasonable.

Chapter 15 studied model selection: choose among candidate models, often by choosing a subset of predictors and then fitting an ordinary linear model to that chosen set. This chapter studies a different family of ideas:

Instead of only choosing columns, change the fitting problem or replace the predictor columns with more stable directions.

The main tools are *regularization* and *dimension reduction*. Regularization keeps the original predictors but penalizes coefficient size. Dimension reduction replaces the original predictors with a smaller number of constructed variables.

The goal is not to find a magic replacement for ordinary least squares. The goal is to understand the tradeoff: more stability and better prediction may come at the cost of bias, less direct interpretation, or more complicated inference.

Connection to earlier chapters. Chapters 7 and 8 explained rank, column space, and identifiability. Chapter 12 diagnosed leverage, influence, and model repair. Chapter 15 chose among candidate models. This chapter connects those ideas to ridge regression, the LASSO, elastic net, principal component analysis, and principal component regression.

Math Foundations Connection. Regularization and dimension reduction use norms, linear combinations, bases, rank, subspaces, least-squares objectives, and quadratic forms. These ideas are supported by the Math Foundations textbook, especially Chapters 3, 5, 6, 7, 10, 11, 12, and 13.

16.2 OLS Fragility and Multicollinearity Recap

Suppose a design matrix $\mathbf{X}$ contains an intercept and p predictor columns. If one column is an exact linear combination of other columns, then $\mathbf{X}$ does not have full column rank. The inverse form

$$(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (16.1)$$

does not exist, and the coefficient vector is not uniquely identified.

Near dependence is different. If one predictor column is almost a linear combination of the others, then the inverse may exist, but it can be numerically unstable. The coefficient vector is unique, but fragile. The variance formula from Chapter 10 helps explain why:

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}. \quad (16.2)$$

When $\mathbf{X}^T \mathbf{X}$ is close to singular, entries of the inverse can become large. Large entries mean unstable coefficients and inflated standard errors.

This does not always destroy prediction. Highly correlated predictors may jointly contain useful information, so fitted values can remain accurate even when individual coefficients are hard to interpret. Multicollinearity is therefore mainly an interpretation and inference warning. It becomes a prediction problem when it hurts validation performance or makes the fitted model unstable on future data.

16.3 Shrinkage as a Bias-Variance Tradeoff

Least squares chooses coefficients by minimizing training error:

$$\min_{\mathbf{b}} \frac{1}{n} (\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b}). \quad (16.3)$$

Regularized methods add a penalty:

regularized objective = training error + penalty for coefficient size.

(16.4)

The penalty discourages large coefficients. This usually introduces bias, because the regularized estimates are intentionally pulled away from the unpenalized least-squares estimates. In exchange, the estimates can have lower variance and better prediction stability.

The tuning parameter, usually written λ , controls the strength of the penalty. When $\lambda = 0$, the penalized objective becomes the ordinary least-squares objective, assuming the unpenalized least-squares solution is well defined. As λ increases, the penalty becomes stronger.

16.4 Ridge Regression

Ridge regression adds a squared two-norm penalty to the least-squares objective. With intercept b_0 and predictor coefficients $b_1, \dots, b_p$, a common form is

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\mathbf{b}} \left\{ \frac{1}{n} (\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b}) + \lambda \sum_{j=1}^p b_j^2 \right\}.$$

(16.5)

The sum starts at $j = 1$, not $j = 0$, because the intercept is usually not penalized.

Ridge regression shrinks coefficients toward zero. It usually does not set coefficients exactly equal to zero. Instead, it stabilizes them. This is useful when predictors are correlated, because ridge is less sensitive to directions in predictor space that are nearly redundant.

Math note: the ridge closed form. For standardized predictor columns, with the intercept handled separately and left unpenalized, a common closed-form ridge estimator is

$$\hat{\beta}_{\text{ridge}, -0} = (\mathbf{X}_s^T \mathbf{X}_s + n\lambda \mathbf{I}_p)^{-1} \mathbf{X}_s^T \mathbf{y}_c, \quad (16.6)$$

where $\mathbf{X}_s$ contains the standardized non-intercept predictor columns and $\mathbf{y}_c$ is the centered response. The factor $n\lambda$ appears because this chapter writes the objective in (16.5) with a factor of $1/n$ in front of the training error. References that omit that $1/n$ convention often absorb this factor into the tuning parameter. The main idea is unchanged.

Ordinary least squares relies on $(\mathbf{X}_s^T \mathbf{X}_s)^{-1}$, which can be unstable when $\mathbf{X}_s^T \mathbf{X}_s$ is nearly singular. Ridge adds $n\lambda \mathbf{I}_p$ under this scaling, stabilizing the inverse by moving small eigenvalues away from zero. This introduces bias, but it can reduce variance and improve coefficient stability or prediction performance.

Math note: truth plus error for ridge. The closed form also lets us see the bias–variance tradeoff algebraically. In this note, treat λ as fixed and use the same standardized, non-intercept setup as above. At the model level, write the centered random response as

$$\mathbf{Y}_c = \mathbf{X}_s \beta_{-0} + \epsilon, \quad E[\epsilon \mid \mathbf{X}_s] = \mathbf{0}, \quad \text{Var}(\epsilon \mid \mathbf{X}_s) = \sigma^2 \mathbf{I}_n.$$

For the observed data, the estimator uses the centered response vector $\mathbf{y}_c$. Because the columns of $\mathbf{X}_s$ are centered and the intercept is handled separately, the ridge cross-product uses

$$\mathbf{X}_s^T \mathbf{y}_c = \mathbf{X}_s^T (\mathbf{X}_s \beta_{-0} + \epsilon).$$

Let

$$\mathbf{B}_\lambda = (\mathbf{X}_s^T \mathbf{X}_s + n\lambda \mathbf{I}_p)^{-1}.$$

Then the ridge estimator for the non-intercept coefficients can be written

$$\begin{aligned} \hat{\beta}_{\lambda, -0} &= \mathbf{B}_\lambda \mathbf{X}_s^T \mathbf{y}_c \\ &= \mathbf{B}_\lambda \mathbf{X}_s^T (\mathbf{X}_s \beta_{-0} + \epsilon) \\ &= \mathbf{B}_\lambda \mathbf{X}_s^T \mathbf{X}_s \beta_{-0} + \mathbf{B}_\lambda \mathbf{X}_s^T \epsilon. \end{aligned}$$

Define

$$\mathbf{M}_\lambda = \mathbf{B}_\lambda \mathbf{X}_s^T \mathbf{X}_s.$$

For ordinary least squares, the corresponding product cancels to the identity matrix when $\mathbf{X}_s^T \mathbf{X}_s$ is invertible. For ridge,

$$(\mathbf{X}_s^T \mathbf{X}_s + n\lambda \mathbf{I}_p)^{-1} \mathbf{X}_s^T \mathbf{X}_s \neq \mathbf{I}_p$$

unless $\lambda = 0$. This is the no-cancellation step: ridge changes the target of the estimator.

Taking conditional expectation gives

$$E[\hat{\beta}_{\lambda,-0} | \mathbf{X}_s] = \mathbf{M}_\lambda \beta_{-0},$$

so the conditional bias is

$$\text{Bias}(\hat{\beta}_{\lambda,-0} | \mathbf{X}_s) = (\mathbf{M}_\lambda - \mathbf{I}_p) \beta_{-0} = -n\lambda(\mathbf{X}_s^T \mathbf{X}_s + n\lambda \mathbf{I}_p)^{-1} \beta_{-0}.$$

Thus ridge is generally biased when $\lambda > 0$. The vector estimator is unbiased only in special cases, such as $\beta_{-0} = \mathbf{0}$; in the eigenvector basis, a direction with true coefficient component zero has no shrinkage bias in that direction.

The conditional variance comes from the error term:

$$\text{Var}(\hat{\beta}_{\lambda,-0} | \mathbf{X}_s) = \sigma^2 (\mathbf{X}_s^T \mathbf{X}_s + n\lambda \mathbf{I}_p)^{-1} \mathbf{X}_s^T \mathbf{X}_s (\mathbf{X}_s^T \mathbf{X}_s + n\lambda \mathbf{I}_p)^{-1}.$$

This is the algebraic version of the tradeoff. Ridge introduces bias, but it changes the variance behavior in fragile directions.

To connect this to the multicollinearity discussion, suppose

$$\mathbf{X}_s^T \mathbf{X}_s = \mathbf{V} \mathbf{D} \mathbf{V}^T,$$

where $\mathbf{D}$ has eigenvalues $d_1, \dots, d_p$. Along eigen-direction j , the signal part is multiplied by

$$\frac{d_j}{d_j + n\lambda},$$

and the variance contribution is

$$\sigma^2 \frac{d_j}{(d_j + n\lambda)^2}$$

instead of the OLS value σ^2/d_j . When d_j is very small, OLS can be extremely unstable in that direction. Ridge replaces that unstable inverse behavior with controlled shrinkage. It does not create information in a low-variance direction; it deliberately damps that direction and accepts bias in exchange for reduced variance and greater stability.

This calculation treats λ as fixed. When λ is chosen by validation or cross-validation, the derivation is still useful for intuition, but ordinary coefficient-level p-values and confidence intervals still do not automatically follow.

Why standardization matters. The ridge penalty uses coefficient size. If one predictor is measured in feet and another in miles, their coefficients have different units and scales. Penalizing their raw coefficients equally may be misleading. For this reason, predictors are usually standardized before ridge regression when units differ. The intercept is then handled separately.

What ridge does and does not do. Ridge can improve prediction and stability, especially when ordinary least squares is fragile. It does not automatically make coefficients easy to interpret, and it does not perform variable selection in the usual sense because it typically keeps all predictors in the model.

16.5 The One-Norm and the LASSO

The LASSO uses the one-norm penalty. For a vector of predictor coefficients $(b_1, \dots, b_p)$, the one-norm is

$$\|\mathbf{b}_{-0}\|_1 = \sum_{j=1}^p |b_j|, \quad (16.7)$$

where $\mathbf{b}_{-0}$ means the coefficient vector without the intercept.

The LASSO objective is

$$\hat{\boldsymbol{\beta}}_{\text{LASSO}} = \arg \min_{\mathbf{b}} \left\{ \frac{1}{n} (\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b}) + \lambda \sum_{j=1}^p |b_j| \right\}. \quad (16.8)$$

Again, the intercept is usually not penalized.

Like ridge regression, the LASSO shrinks coefficients. Unlike ridge regression, the LASSO can set some coefficients exactly equal to zero. This makes it useful for variable selection when many candidate predictors are available but only some are expected to matter.

The important tradeoff is this: LASSO-selected coefficients are biased by the penalty. A coefficient equal to zero is a modeling result from a penalized fit, not a proof that the corresponding variable has no scientific or operational relevance.

16.6 Why the LASSO Can Be Sparse

The LASSO's sparsity comes from the shape of the one-norm penalty. In two dimensions, the constraint

$$|b_1| + |b_2| \leq \gamma \quad (16.9)$$

forms a diamond. The corners of this diamond lie on the coordinate axes. When the least-squares contours touch the boundary of the feasible region, they often touch at a corner. At a corner, one coefficient is zero.

Ridge regression has a rounded constraint shape instead. Rounded boundaries tend to shrink coefficients without forcing them exactly to zero. This geometric difference explains why ridge primarily stabilizes while the LASSO can both stabilize and select.

This geometric picture is only intuition. The textbook point is that the penalty shape affects the kind of model the method prefers.

16.7 Elastic Net

Elastic net combines ridge and LASSO penalties. A common schematic form is

$$\frac{1}{n} (\mathbf{y} - \mathbf{X}\mathbf{b})^T (\mathbf{y} - \mathbf{X}\mathbf{b}) + \lambda \left[\alpha \sum_{j=1}^p |b_j| + (1 - \alpha) \sum_{j=1}^p b_j^2 \right], \quad (16.10)$$

where $0 \leq \alpha \leq 1$. When $\alpha = 1$, the penalty is LASSO-like. When $\alpha = 0$, the penalty is ridge-like.

Elastic net is useful when predictors are correlated and the analyst wants some sparsity without relying on the LASSO alone. It has more tuning choices than ridge or LASSO, so validation becomes even more important. Different conventions may scale the penalty slightly differently, but the conceptual role is the same: blend shrinkage and sparsity.

16.8 PCA as a Change of Basis

Regularization keeps the original predictor columns and changes the fitting objective. Principal component analysis, or PCA, takes a different approach: replace the original predictor columns with new directions.

Let $\mathbf{W}$ be the predictor matrix after removing the intercept and centering or standardizing the columns as appropriate. PCA finds directions $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ in predictor space. The first direction captures the largest possible predictor variance. The second captures the largest remaining variance among directions orthogonal to the first. The process continues until all directions are accounted for.

The principal component scores are

$$\mathbf{z}_k = \mathbf{W}\mathbf{v}_k, \quad k = 1, \dots, p. \quad (16.11)$$

Each score vector $\mathbf{z}_k$ is a linear combination of the original predictors. The vectors $\mathbf{v}_k$ are the principal component directions. The associated eigenvalues describe how much predictor variance is captured along those directions.

Math note: eigendecomposition behind PCA. For a centered or standardized predictor matrix $\mathbf{W}$, a covariance or Gram-type matrix such as

$$\mathbf{S} = \frac{1}{n-1} \mathbf{W}^T \mathbf{W} \quad (16.12)$$

is symmetric positive semidefinite. Therefore it has orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ and nonnegative eigenvalues $\lambda_1, \dots, \lambda_p$ satisfying

$$\mathbf{S}\mathbf{v}_k = \lambda_k \mathbf{v}_k. \quad (16.13)$$

Here $\mathbf{v}_k$ is a direction in predictor space, and λ_k is the predictor variance along that direction. In this PCA context, λ_k denotes an eigenvalue, not the ridge or LASSO penalty parameter λ . Large eigenvalues point toward high-variance directions. Small eigenvalues indicate low-variance or nearly redundant directions. Zero eigenvalues indicate exact linear dependence among predictor columns. We use this result as an interpretive tool; the spectral theorem itself is not proved here.

Geometric intuition. Imagine the cloud of predictor values as an ellipse or hyperellipsoid. PCA rotates the coordinate system to line up with the main axes of that cloud. Long axes correspond to large variance. Short axes correspond to small variance.

Scaling warning. PCA is sensitive to units. If one predictor is measured on a much larger scale than another, it can dominate the principal components. When predictor units differ, PCA is usually performed on standardized predictors.

16.9 PCA, Degeneracy, and Multicollinearity

PCA also explains multicollinearity. A zero-variance direction means there is an exact linear combination of predictors with no variation. That is exact degeneracy. A very small variance direction means there is a nearly redundant linear combination of predictors. That is near-collinearity.

In matrix language, these directions are connected to small eigenvalues of the predictor covariance matrix or of a related Gram matrix. A small eigenvalue does not automatically tell the analyst what to delete, but it warns that some combination of predictors is weakly determined.

This is why PCA can be a useful diagnostic companion to VIFs and pairwise correlations. Pairwise correlations look at pairs of predictors. VIFs look at how each predictor is explained by the rest. PCA looks for low-variance directions across the whole predictor set.

Caution: dropping collinear predictors and omitted-variable bias. One tempting response to multicollinearity is to drop one of the correlated predictors. This can reduce coefficient instability, but it can also change the meaning of the remaining coefficients.

Consider the two-predictor setup

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon. \quad (16.14)$$

If we instead fit a reduced model that omits x_2 ,

$$y = \alpha_0 + \alpha_1 x_1 + u, \quad (16.15)$$

then the slope on x_1 in the reduced model generally combines two pieces: the direct x_1 association from the fuller linear model and the omitted x_2 association carried through the relationship between x_1 and x_2 . In the usual two-variable linear projection calculation,

$$\alpha_1 = \beta_1 + \beta_2 \delta_1, \quad (16.16)$$

where δ_1 is the slope from regressing x_2 on x_1 . In words,

reduced-model slope = direct x_1 association + omitted x_2 association $\times$ association
between x_1 and x_2 .

This formula describes how coefficients behave in a linear model; it does not by itself prove that either predictor has a causal effect. The practical caution is that dropping a collinear predictor may trade lower variance for additional bias or a different coefficient interpretation. For prediction, compare validation performance. For interpretation, ask whether the dropped variable is a duplicate measurement, a mechanical transformation, a proxy, or a theoretically important variable.

16.10 Principal Component Regression

Principal component regression, or PCR, uses principal components as predictors in a regression model. The workflow is:

1. Center or standardize the predictor matrix, as appropriate.
2. Compute principal components from the predictors.

3. Keep the first K principal components.
4. Regress the response on those K component scores.
5. Choose K using validation or cross-validation.

PCR can help when predictors are strongly correlated. Instead of trying to estimate separate coefficients for several nearly redundant predictors, PCR fits the response using a smaller number of orthogonal component directions.

There is a cost. Principal components are combinations of predictors, so they are usually less directly interpretable than the original variables. Also, PCA uses only the predictors, not the response. A low-variance direction can sometimes matter for predicting the response, so dropping components purely because their eigenvalues are small is not always safe.

16.11 Choosing Tuning Parameters and Components

Ridge, LASSO, elastic net, and PCR all require tuning choices:

Method	Main tuning choice
Ridge	penalty strength λ
LASSO	penalty strength λ
Elastic net	penalty strength λ and mixing parameter α
PCR	number of principal components K

The usual way to choose these values is validation or cross-validation. Chapter 15 already explained why training error alone is too optimistic. The same warning applies here. Increasing flexibility or decreasing the penalty may improve training fit without improving future performance.

Heuristics for component choice. For PCA and PCR, a scree plot displays eigenvalues in decreasing order; the “elbow” heuristic looks for the point where the eigenvalues begin to flatten. Another rule of thumb, the Kaiser criterion, keeps components with eigenvalues greater than 1 when the predictors have been standardized. These are screening heuristics, not proofs and not automatic selection rules. For prediction-oriented PCR, validation or cross-validation remains the primary guide for choosing K . For interpretation, the analyst should also examine whether the retained components are meaningful for the problem.

The correct tuning choice depends on the goal. If prediction is the goal, validation error is central. If interpretation is the goal, sparsity, stability, and clarity matter too.

16.12 Comparing the Methods

OLS. Fits unpenalized least squares. It is simple and interpretable when the design is stable, but can be fragile under collinearity or high dimension.

Ridge. Penalizes squared coefficient size. It stabilizes correlated predictors, keeps all predictors, and introduces bias.

LASSO. Penalizes absolute coefficient size. It can select variables by setting coefficients to zero, but its coefficients are biased and ordinary OLS inference does not directly apply.

Elastic net. Blends ridge and LASSO penalties. It can give sparse models with more stability among correlated predictors, at the cost of another tuning choice.

PCR. Regresses on leading principal components. It reduces dimension and collinearity, but components are less directly interpretable than original predictors.

No method is best for every analysis. A defensible choice depends on the purpose, data quality, validation performance, and the need for interpretation.

16.13 Inference After Regularization

Ridge and LASSO estimates are biased by design. That is not a flaw; it is the point. The penalty trades some bias for reduced variance or sparsity. But this also means ordinary OLS standard errors, confidence intervals, and coefficient t -tests do not carry over automatically.

The LASSO is especially tempting because it performs variable selection. A common workflow is:

1. use LASSO to choose a smaller set of predictors;
2. refit an ordinary linear model on the selected predictors;
3. report the refit model.

This can be useful, but it does not erase the selection step. The selected variables were chosen using the data, so the post-selection inference warnings from Chapter 15 still apply.

For prediction-focused work, validation performance may be more important than classical coefficient inference. For interpretation-focused work, regularization can help stabilize a model, but the analyst must explain how the method changes coefficient meaning and uncertainty.

16.14 Common Mistakes

Mistake 1: Penalizing unstandardized predictors without thinking about units

Ridge, LASSO, elastic net, and PCA are sensitive to predictor scale. If predictors have different units, standardize or justify why not.

Mistake 2: Treating LASSO zeros as proof of irrelevance

A zero LASSO coefficient means the penalized fitting procedure removed that predictor under a chosen tuning parameter. It is not scientific proof that the variable has no relationship to the response.

Mistake 3: Reporting ordinary p-values after data-driven selection

If a model was selected using LASSO, stepwise selection, or many candidate comparisons, ordinary selected-model p-values need caution.

Mistake 4: Treating principal components as automatically interpretable

A principal component is a linear combination of predictors. It may be useful for prediction but hard to brief as a simple operational variable.

Mistake 5: Using all principal components and expecting dimension reduction

If all components are kept, PCR has not reduced dimension. The point of PCR is to keep a smaller set of components when that improves stability or prediction.

Mistake 6: Choosing tuning parameters by training error alone

Training error usually improves when the penalty weakens or more components are kept. Use validation or cross-validation.

16.15 Chapter Summary

Ordinary least squares can be unstable when predictors are numerous, redundant, or highly correlated. Regularization and dimension reduction are two ways to address that problem.

Ridge regression adds a squared two-norm penalty. It shrinks coefficients and can stabilize correlated predictors, but it usually keeps all predictors. The ridge truth-plus-error representation makes the tradeoff explicit: for fixed $\lambda > 0$, ridge is biased, but its variance is damped in unstable predictor directions. The LASSO adds a one-norm penalty. It shrinks coefficients and can set some exactly to zero, making it useful for variable selection. Elastic net blends ridge and LASSO penalties.

PCA replaces the original predictor coordinates with orthogonal directions of predictor variation. PCR regresses the response on a selected number of principal components. This can reduce dimension and improve stability, but it makes interpretation less direct.

All of these methods require judgment. Tuning parameters and number of components should be chosen with validation in mind. Regularized estimates are biased by design, and ordinary OLS inference does not automatically apply after shrinkage or data-driven selection. The next chapter takes the same prediction-and-validation mindset one step farther by treating neural networks as flexible regression models that build prediction functions from repeated weighted sums and nonlinear transformations.

16.16 Practice and Workflow

Focused Study Checklist

Use this checklist after studying regularization and dimension reduction.

1. Can you explain why exact linear dependence causes non-identifiability?
2. Can you explain why near dependence makes least-squares coefficients fragile?
3. Can you distinguish model selection from regularization?
4. Can you write the ridge objective and explain the role of λ ?
5. Can you derive or explain the ridge truth-plus-error representation and identify where bias enters?
6. Can you write the LASSO objective and explain why it can set coefficients to zero?
7. Can you explain why the intercept is usually not penalized?
8. Can you explain why standardization matters before ridge, LASSO, elastic net, or PCA?
9. Can you describe PCA as a change of basis for the predictor matrix?
10. Can you explain what PCR does and why its components may be less interpretable?
11. Can you state why ordinary OLS inference does not automatically apply after regularization or data-driven selection?

Coding companion reference. Package-specific commands for ridge, LASSO, elastic net, PCA, PCR, standardization, cross-validated tuning, and validation comparisons belong in the separate Coding and Software Cookbook companion. This chapter keeps the statistical meaning, formulas, and interpretation logic.

Chapter 17

Neural Networks as Flexible Regression Models

17.1 The Big Picture

Linear models have been the backbone of this course because they are understandable, mathematically tractable, and often surprisingly effective. They also have limits. A linear model can include transformed columns, interactions, categorical indicators, regularization, and principal components, but it still asks the analyst to decide which structure to put into the design matrix.

Neural networks are a different way to build flexible prediction models. They combine many weighted sums with nonlinear transformations. In that sense, they are not unrelated to linear regression. They extend the same basic ingredients:

take inputs, form weighted combinations, transform them, and fit the resulting prediction function to data.

The tradeoff is important. Neural networks can model nonlinear relationships that are hard to see or specify by hand. They can be useful prediction benchmarks. But they are usually less interpretable than linear models, harder to audit, and less connected to the classical inference tools used earlier in the textbook.

This chapter treats neural networks only as regression tools for quantitative responses. Classification neural networks, convolutional networks, recurrent networks, and deep-learning architecture design are outside the scope of this chapter. The classification ideas from Chapter 14 have neural-network analogues, but this chapter stays with quantitative responses so that the connection to squared-error regression, validation, and baseline comparison remains clear.

Connection to earlier chapters. Chapter 2 introduced prediction error, flexibility, and overfitting. Chapters 5–14 built linear regression, inference, likelihood, and generalized linear-model intuition. Chapters 15 and 16 studied model selection, regularization, and dimension reduction. Neural networks keep the prediction and validation mindset but give up much of the simple coefficient interpretation from linear regression.

Math Foundations Connection. Neural networks use matrix-vector multiplication, repeated linear transformations, functions, gradients, and loss minimization. These ideas are supported by the Math Foundations textbook, especially Chapters 7, 12, and 13.

17.2 From Linear Models to Layered Models

A linear regression model predicts with a weighted sum. For a predictor vector $\mathbf{x}$, a fitted linear model has the form

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p. \quad (17.1)$$

Equivalently, after including an intercept in the design row, we can write the prediction as an inner product.

A neural network still uses weighted sums, but it does so repeatedly. A first set of weighted sums produces intermediate values. Those values are transformed by a nonlinear function. A later set of weighted sums combines those transformed values into a prediction.

A simple schematic is

$$\begin{aligned} \mathbf{x} &\longrightarrow \text{weighted sums} \longrightarrow \text{nonlinear transformation} \\ &\longrightarrow \text{more weighted sums} \longrightarrow \hat{y}. \end{aligned} \quad (17.2)$$

The nonlinear transformation is the key difference. Without it, stacking linear transformations would not create a genuinely new model class. The product of several linear transformations is still one linear transformation. Nonlinear activation functions prevent the layers from collapsing into one linear model.

17.3 Activation Functions

An *activation function* is a nonlinear function applied to intermediate weighted sums in a neural network. Many sources write the activation function as $\sigma(\cdot)$. To avoid confusion with the error standard deviation σ , this chapter writes the activation as $g(\cdot)$.

A common example is the rectified linear unit, or ReLU:

$$g(z) = \max(z, 0). \quad (17.3)$$

ReLU leaves positive inputs alone and turns negative inputs into zero. It is simple, computationally convenient, and nonlinear.

Activation functions are usually applied element by element. If $\mathbf{u}$ is a vector of intermediate values, then $g(\mathbf{u})$ means apply g to each entry of $\mathbf{u}$.

The activation function gives the network flexibility. Each layer forms weighted combinations; the activation bends those combinations; later layers combine the bent pieces. This repeated process can approximate nonlinear patterns that would be difficult to represent with a small number of hand-built interaction or polynomial terms.

17.4 Network Terminology

Only a small amount of vocabulary is needed for this chapter.

Term	Meaning in this chapter
Input layer	The predictor values supplied to the model.
Hidden layer	An intermediate layer between inputs and the output.
Output layer	The final layer that returns the prediction $\hat{y}$.
Weights	The numerical parameters learned during training.
Activation function	A nonlinear function applied between weighted sums.
Architecture	The number of layers, number of units, and pattern of connections.
Shallow network	A network with one hidden layer.
Deep network	A network with more than one hidden layer.

The word *layer* is sometimes used for the collection of weighted sums and sometimes for the transformed outputs of those sums. For this course, the exact convention is less important than the main idea: neural networks build prediction functions by composing linear transformations and nonlinear activations.

17.5 A Neural-Network Regression Model

Let $f_A(\mathbf{x})$ denote the prediction produced by a neural network with weights collected in A . The subscript A is a reminder that the prediction function depends on many fitted weights, not on a single coefficient vector.

For a quantitative response, a neural-network regression model has the form

$$\hat{y} = f_A(\mathbf{x}). \quad (17.4)$$

A one-hidden-layer version can be written schematically as

$$f_A(\mathbf{x}) = a_0 + \sum_{m=1}^M a_m g(c_m + \mathbf{w}_m^T \mathbf{x}), \quad (17.5)$$

where M is the number of hidden units, $\mathbf{w}_m$ is a vector of input weights for hidden unit m , c_m is a hidden-unit intercept, g is the activation function, and a_m are output weights.

Equation (17.5) is not the only possible neural-network form. It is useful because it shows the pattern clearly:

1. form weighted sums $c_m + \mathbf{w}_m^T \mathbf{x}$;
2. apply a nonlinear activation g ;
3. combine the activated values into a prediction.

With more hidden layers, the same idea is repeated. Later layers take transformed outputs from earlier layers as their inputs.

17.6 Fitting Neural Networks by Minimizing Loss

For regression, neural networks are often fit by minimizing mean squared error:

$$\min_A \frac{1}{n} \sum_{i=1}^n (y_i - f_A(\mathbf{x}_i))^2. \quad (17.6)$$

This should look familiar. The loss is the same kind of squared-error loss used throughout the regression chapters. What changes is the prediction function. Instead of $\mathbf{x}_i^T \boldsymbol{\beta}$, the model uses the more flexible function $f_A(\mathbf{x}_i)$.

The optimization problem is harder than ordinary least squares. For linear regression, the least-squares objective is a quadratic function of the coefficients, and we can derive normal equations. For neural networks, the parameters appear inside repeated nonlinear transformations. The objective can have many local features and often requires iterative numerical optimization.

That computational complexity is one reason neural networks need careful workflow discipline. A model can be flexible and still be poorly trained, over-tuned, or evaluated incorrectly.

17.7 What Still Carries Over

Neural networks do not erase the earlier course. Many habits from linear modeling remain essential.

Train/test thinking. A neural network should be judged on data not used to fit its weights. Training error alone is not enough.

Validation and cross-validation. Flexible models need tuning. Validation helps decide whether extra flexibility improves future performance or only memorizes training data.

Prediction metrics. RMSE, MAE, and similar metrics are still useful for quantitative responses.

Scaling. Predictors should usually be scaled before fitting a neural network. Scale differences can make training slower or unstable.

Bias-variance thinking. More flexibility can reduce bias but increase variance and overfitting risk.

Regularization intuition. Penalization, early stopping, and architecture choices all control flexibility, much like the regularization ideas from Chapter 16.

Baselines. A neural network should be compared against simpler models. A black-box model must earn its complexity.

The main workflow is the same: define the question, split or validate properly, fit defensible candidate models, compare performance, and document the tradeoffs.

17.8 What Does Not Carry Over Cleanly

Several familiar linear-model tools do not carry over in the same way.

Coefficient interpretation. A linear regression coefficient has a direct adjusted-effect interpretation under the model. Neural-network weights do not usually have that kind of simple interpretation.

Classical coefficient tests. Ordinary coefficient t -tests and regression F -tests do not apply directly to a neural network's weights.

Linear-model confidence intervals. The coefficient, mean-response, and prediction interval formulas from Chapter 13 depend on linear-model structure and Gaussian fixed-design theory. They are not automatically available for neural networks.

VIF and linear collinearity diagnostics. VIFs diagnose instability in linear coefficient estimates. They are not neural-network inference tools, though correlated predictors can still affect training and interpretation.

Residual assumption ladder. Residual plots can still be useful descriptively, but they no longer validate the same linear-model assumption ladder from Chapters 5 and 12.

This does not make neural networks invalid. It means the analyst must be clear about the goal. If the goal is prediction, validation performance may be the central evidence. If the goal is explanation or inference, a simpler model may be more defensible.

17.9 Neural Networks as Benchmarks

A useful way to treat neural networks in this course is as prediction benchmarks. The workflow is simple:

1. Build a defensible linear, regularized, or transformed model.
2. Fit a simple neural-network regression model.
3. Compare validation performance using the same response scale and prediction metric.
4. Decide whether the performance gain is worth the loss of interpretability.

If the neural network performs only slightly better than an interpretable model, the interpretable model may be preferable for briefing, diagnosis, and decision support. If the neural network performs much better, that is evidence that the simpler model may be missing nonlinear structure. The next step is not automatically to choose the black-box model. The next step is to ask what structure the black-box model may have found and whether that structure can be made defensible.

This benchmark mindset is especially useful in operations analysis. Sponsors may be attracted to flexible machine-learning models, but a model that is harder to explain must deliver enough performance improvement to justify the cost in interpretability.

17.10 Practical Cautions

Neural networks are powerful but easy to misuse.

Data size

Flexible models usually need more data than simple models. With small data sets, a neural network can overfit badly and perform worse than a carefully built linear model.

Tuning choices

Architecture, activation functions, regularization settings, learning rates, and stopping rules affect performance. These choices should not be tuned on the final test set.

Scaling and preprocessing

Scaling and other learned preprocessing steps should be estimated from the training data, then applied consistently to validation or test data. Letting validation or test data influence preprocessing creates leakage.

Reproducibility

Neural-network training can involve random initialization and stochastic optimization. Record the train/test split, preprocessing, tuning choices, random seeds, and evaluation metric.

Computational cost

Neural networks can take more time and computation to tune than simpler regression models. That cost should be justified by enough validation improvement to matter for the task.

Interpretability

A neural network may predict well without explaining why. That can be acceptable for some prediction tasks and unacceptable for some inference or policy tasks.

17.11 Common Mistakes

Mistake 1: Treating neural networks as automatically better

A neural network is more flexible than a linear model, but flexibility is not the same as better performance on future data. Compare validation error.

Mistake 2: Comparing models only on training error

Training error usually rewards flexibility. Use validation or test performance.

Mistake 3: Skipping the simple baseline

A neural network is easier to justify when it beats a strong simple baseline. Fit the baseline first.

Mistake 4: Claiming coefficient-style inference

Neural-network weights are not regression coefficients with simple adjusted-effect interpretations. Do not report them as if they were ordinary linear-model coefficients.

Mistake 5: Over-tuning on the test set

If the test set is used repeatedly to choose architecture or tuning settings, it becomes part of the training process. Keep a final test set or use a disciplined validation procedure.

17.12 Chapter Summary

Neural networks extend linear models by composing weighted sums with nonlinear activation functions. Without the nonlinear activations, stacked linear transformations would collapse into one linear transformation.

For regression, neural networks can be fit by minimizing mean squared error. The loss is familiar, but the prediction function is much more flexible and the optimization problem is more complex.

Train/test thinking, validation, prediction metrics, scaling, regularization intuition, and baseline comparisons still matter. Simple coefficient interpretation, ordinary coefficient tests, and the linear-model interval formulas do not carry over directly.

A neural network is best treated as a flexible prediction tool and benchmark. It may be valuable when it substantially improves prediction, but it is not a substitute for defensible modeling judgment.

17.13 Practice and Workflow

Focused Study Checklist

Use this checklist after studying neural-network regression.

1. Can you explain how a neural network extends a linear model?
2. Can you explain why nonlinear activation functions are necessary?
3. Can you write the MSE fitting objective for neural-network regression?
4. Can you distinguish weights in a network from coefficients in a linear regression?
5. Can you name tools from the course that still matter for neural networks?
6. Can you name classical linear-model inference tools that do not carry over directly?
7. Can you explain why a neural network should be compared against a simpler baseline?

8. Can you explain when a black-box prediction gain might or might not be worth the loss of interpretability?

Coding companion reference. Package-specific commands for fitting neural-network regressors, scaling predictors, setting random seeds, and comparing validation performance belong in the separate Coding and Software Cookbook companion. This chapter keeps the statistical meaning, modeling logic, and interpretation cautions.

Appendix A

Regression Assumptions, Diagnostics, and What They Support

A.1 Why This Appendix Exists

The regression chapters use one recurring assumption ladder. Chapter 5 introduces it in simple linear regression, Chapter 6 uses it to justify classical tests, Chapter 10 writes it in fixed-design matrix form, Chapter 12 turns it into residual diagnostics, Chapter 13 uses it for intervals, and Chapter 14 connects the Gaussian version to likelihood.

This appendix gathers those ideas in one place. It is not a new theory chapter. It is a review map for answering three practical questions:

1. What are the major regression assumptions?
2. What does each assumption support?
3. What should an analyst say when a diagnostic check looks weak?

The key habit is to separate *fitting* from *trusting inferential claims*. Least squares will produce fitted values and residuals whenever the computation is well defined. The assumptions determine how far we can go when interpreting coefficients, standard errors, tests, intervals, and model repairs.

A.2 The Regression Model and the Assumption Ladder

For the fixed-design regression model, write

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{A.1}$$

where

$$\mathbf{Y} \in \mathbb{R}^n, \quad \mathbf{X} \in \mathbb{R}^{n \times q}, \quad \boldsymbol{\beta} \in \mathbb{R}^q, \quad \boldsymbol{\epsilon} \in \mathbb{R}^n. \tag{A.2}$$

Here $q = p + 1$ when the model includes an intercept and p ordinary predictors. The design matrix $\mathbf{X}$ is treated as fixed, or the analysis is read conditionally on the observed predictor values.

The three-step assumption ladder is:

Level	Mathematical statement	Plain-language meaning
Mean-zero errors	$\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}$	The model is centered correctly. After accounting for the predictors in $\mathbf{X}$, the remaining error does not systematically lean positive or negative.
Constant variance and uncorrelated errors	$\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$	Errors have a common variance and are uncorrelated across observations after conditioning on the design.
Gaussian errors	$\epsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$	The conditional errors are normally distributed with mean zero, common variance, and no correlation.

These assumptions are increasingly strong. Mean-zero errors are about the location of the regression surface. Constant variance and uncorrelated errors are about the error covariance structure. Gaussian errors add a distributional shape assumption.

A structural condition, not one of the Big Three. Full column rank is also important, but it is not an error assumption. It is a design-matrix condition. The inverse-form estimator

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (\text{A.3})$$

requires $\mathbf{X}$ to have full column rank. If the columns are exactly dependent, the ordinary coefficient vector is not uniquely identified. If the columns are nearly dependent, the coefficients may be unstable even though fitted values can still be useful.

A.3 What Each Assumption Supports

The assumptions support different claims. Do not use a stronger conclusion than the assumptions justify.

Assumption or condition	Supports	Does not automatically support
Full column rank of $\mathbf{X}$	Unique inverse-form OLS coefficient estimate; identifiable coefficients in the chosen design	Correct mean model; valid standard errors; normality; causal interpretation
Mean-zero errors	$\mathbf{X}\boldsymbol{\beta}$ is the conditional mean; OLS is centered on the true coefficient vector under the fixed-design model	Constant variance; independent errors; exact t - or F -based inference
Constant variance and uncorrelated errors	Classical coefficient covariance formulas; usual standard errors; residual variance logic	Correct mean model if mean-zero fails; Gaussian finite-sample inference
Gaussian errors	Exact finite-sample t -tests, F -tests, and standard intervals under the linear model	Immunity to misspecification, outliers, dependence, or bad design
Reasonable residual behavior	Practical evidence that the model assumptions may be defensible	Proof that the assumptions are true

A common mistake is to treat a fitted regression output table as self-validating. The table can compute coefficients, standard errors, and p -values. It cannot prove that the model assumptions are credible.

A.4 Errors Are Not Residuals

The true error vector $\boldsymbol{\epsilon}$ belongs to the data-generating model. It is not observed. The residual vector $\mathbf{e}$ is computed after fitting a model:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}. \quad (\text{A.4})$$

When ordinary least squares is fit with hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T, \quad (\text{A.5})$$

the residuals satisfy

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}. \quad (\text{A.6})$$

If the truth-plus-error model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ is appropriate and $\mathbf{H}\mathbf{X} = \mathbf{X}$, then

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (\text{A.7})$$

Thus residuals are not the errors themselves. They are the portion of the errors left after projection away from the fitted model space.

This distinction matters for diagnostics. Residual plots are useful because residuals are the observable shadows of errors. But residual plots do not prove assumptions. They reveal visible patterns that would be surprising if the assumptions were a good description of the data.

A.5 Diagnostics-to-Assumptions Map

Residual diagnostics turn the assumption ladder into practical checks.

Diagnostic evidence	Main assumption being checked	Common diagnostic view
Residuals show no systematic curve or trend around zero	Mean-zero errors; mean model is not visibly missing structure	Residuals versus fitted values; residuals versus predictors
Residual spread is roughly stable across fitted values and predictors	Constant variance	Residuals versus fitted values; scale-location plot
Residuals show no runs, waves, or time-ordered pattern	Uncorrelated errors	Residuals versus time, collection order, or spatial/operational grouping
Residual shape is roughly symmetric and Q-Q plot has no severe tail departures	Gaussian error plausibility	Histogram; normal Q-Q plot
A few observations have high leverage or large Cook's distance	Influence or design sensitivity, not exactly one of the Big Three	Leverage plot; studentized residuals; Cook's distance
Predictors are nearly redundant	Design instability, not an error assumption	Scatterplot matrix; VIF; condition-number or rank check

The diagnostic hierarchy in Chapter 12 is useful: first check whether the mean structure is missing something important, then check variance and dependence, then check normality and influence. A normal Q-Q plot is not very comforting if the residuals versus fitted plot shows obvious curvature.

A.6 What Breaks When an Assumption Fails?

Assumption failures do not all have the same consequences. The right response depends on the goal of the analysis.

Problem	Main risk	Smallest responsible response
Mean structure is wrong	Coefficients and fitted means describe the wrong conditional mean relationship	Add missing predictors, transform predictors, add interactions, or change model class
Nonconstant variance	Usual standard errors, tests, and intervals may be misleading; prediction uncertainty may vary across cases	Transform response if appropriate, use a variance-aware method if covered, or report inference with caution
Correlated errors	Standard errors and tests may be too optimistic because observations contain less independent information than assumed	Model dependence, use time/order-aware validation, or report the limitation clearly
Strong nonnormality	Exact small-sample t - and F -based claims weaken	Check whether sample size makes approximation reasonable; transform, use robust alternatives if covered, or caveat inference
High leverage or influential observations	A small number of cases may dominate coefficients or fitted pattern	Investigate data quality and context; compare keep/remove/repair decisions; document the choice
Exact rank deficiency	Coefficients are not uniquely identified	Remove or recode redundant columns; fix dummy-variable trap or duplicate predictors
Near collinearity	Coefficients and standard errors may be unstable even when predictions remain acceptable	Clarify prediction versus interpretation goal; consider combining variables, dropping redundancy, ridge, LASSO, or PCR if appropriate

The goal is not to delete every uncomfortable point or chase every diagnostic plot until it looks perfect. The goal is to decide whether the model is defensible for the question being asked.

A.7 Which Claims Are Still Defensible?

When assumptions are imperfect, avoid all-or-nothing thinking. A model may be weak for one purpose and still useful for another.

Claim	Needs most	If assumptions look weak
Training fit is small	Computation of fitted values and residuals	Still descriptive, but may not generalize
Prediction on new data is good	Validation or cross-validation evidence	Prefer test/CV error over assumption-heavy claims
A coefficient has a clear interpretation	Correct mean model, identifiable design, and meaningful units	Interpret cautiously if collinearity, omitted structure, or transformations complicate meaning
A coefficient test is reliable	Mean-zero errors, usable standard errors, degrees of freedom, and distributional approximation	Caveat or avoid strong significance language if diagnostics are poor
A confidence interval is defensible	Same assumptions supporting standard errors and t -based reference distribution	Report interval with caveat, repair model, or use an alternative interval method if covered
A prediction interval is defensible	Correct mean model, variance model, and future case comparable to training design	Be cautious if variance changes, extrapolation occurs, or future conditions differ
A causal claim is warranted	Research design or causal assumptions beyond ordinary regression output	Do not infer causality from regression association alone

For operational work, this table often matters more than the formal formulas. The analyst must decide what claim the evidence can support.

A.8 A Short Project-Report Template

A regression report should say what was checked and how that affected interpretation. The following language can be adapted without overstating the evidence.

We fit a linear regression model with *[response]* as the response and *[predictors]* as predictors. We checked the residual behavior against the regression assumption ladder. The residuals showed *[no major pattern, curvature, changing spread, possible nonnormality, possible dependence, or influential cases]*. Based on these diagnostics, the model is *[reasonably defensible for inference, useful mainly for prediction, descriptive only, or in need of repair]*. We interpret coefficients, p -values, and intervals with *[standard confidence or caution]* because *[reason]*.

For a briefing, shorten the message:

The model fits the main trend, but the residuals show *[issue]*. That means *[which claim is weakened]*. The model is still useful for *[purpose]*, but we should not overstate *[p-values, intervals, coefficient interpretation, or prediction outside the observed range]*.

A.9 Chapter Map

The same assumption ladder appears in different forms across the regression chapters.

Chapter	Role of the assumption ladder
Chapter 5	Introduces the Big Three assumptions in simple linear regression and explains what each level supports.
Chapter 6	Uses the assumptions to justify F -tests, slope t -tests, p -values, and the interpretation of software output.
Chapter 10	Restates the assumptions in fixed-design matrix form and derives how they affect $\hat{\beta}$, residuals, and residual variance.
Chapter 12	Turns the assumptions into practical diagnostic checks and model-repair decisions.
Chapter 13	Uses the assumptions to explain coefficient intervals, mean-response intervals, and prediction intervals.
Chapter 14	Shows that the Gaussian-error version of the linear model makes ordinary least squares a maximum likelihood estimator.

A.10 Appendix Summary

Carry forward these habits:

1. State the regression assumption ladder before making strong inference claims.
2. Remember that full rank is a design condition; it is not an error assumption.
3. Keep true errors ϵ separate from residuals $\mathbf{e}$.
4. Use residual plots as evidence, not proof.
5. Match the diagnostic to the assumption: pattern for mean structure, spread for variance, order for dependence, Q-Q for normality.
6. Decide what claim remains defensible: description, prediction, coefficient interpretation, testing, intervals, or causal language.
7. Report limitations plainly. A useful model is not the same thing as a perfect model.

Appendix B

Math Foundations Source Map

The chapter-level *Math Foundations Connection* paragraphs remain in the chapters where they help explain why a mathematical idea matters. This appendix is a consolidated lookup map. It is meant for review, cross-checking, and finding the supporting Math Foundations material quickly; it is not required sequential reading.

The entries below are organized by mathematical theme rather than by every Data Analysis chapter. Repeated chapter-level source-map rows have been merged into broader references. When a topic appears in many places, the listed chapters are the main places where the idea is used.

B.1 How to Use This Appendix

Use this appendix when you meet a mathematical object in the Data Analysis text and want to review the supporting Math Foundations idea. Each entry gives four pieces of information:

- **Used in Data Analysis for:** the statistical or modeling task.
- **Math Foundations support:** the mathematical idea being used.
- **Primary Data Analysis chapters:** the main chapters where the idea appears.
- **Math Foundations reference:** the chapter or chapters to review.

B.2 Logic, Sets, Conditioning, and Assumption Language

Used in Data Analysis for

Null and alternative hypotheses, modeling assumptions, event language, categorical variable domains, filtering rows, conditioning on predictor values, and interpreting statements such as “if the assumptions hold, then the test statistic has this reference distribution.”

Math Foundations support

Logical statements, implications, negation, quantified statements, set membership, subsets, set operations, and partitions of cases or groups.

Primary Data Analysis chapters

Chapters 1, 3, 5, 6, 10, 12, 13, 14; Appendix A.

Math Foundations reference

Chapter 1: Logic and Deductive Reasoning; Chapter 4: Sets, Set Operations, and Quantifiers.

B.3 Data Vectors, Matrix Notation, and Design Matrices

Used in Data Analysis for

Treating variables as vectors, collecting observations into response vectors, writing fitted values as matrix-vector products, building design matrices for simple regression, multiple regression, categorical predictors, transformations, interactions, and engineered columns.

Math Foundations support

Vectors in $\mathbb{R}^n$, vector entries and subvectors, special vectors such as $\mathbf{1}$, matrix dimensions, transposes, matrix-vector multiplication, and row/column views of matrix multiplication.

Primary Data Analysis chapters

Chapters 1–5, 7–11, 13–17.

Math Foundations reference

Chapter 2: Vectors in $\mathbb{R}^n$; Chapter 3: Vector Operations and Linear Combinations; Chapter 7: Matrices and Matrix-Vector Multiplication.

B.4 Dot Products, Norms, Distance, and Error Measures

Used in Data Analysis for

Inner-product model notation $\beta^T \mathbf{x}$, squared error, residual length, RSS, MSE, RMSE, correlation as linear alignment, distance from fitted values, and penalty terms such as $\|b\|_1$ and $\|b\|_2^2$.

Math Foundations support

Dot products, weighted sums, Euclidean norm, squared norm, distance between vectors, absolute value, Euclidean size ideas, angles, and orthogonality.

Primary Data Analysis chapters

Chapters 1, 2, 4–7, 10, 12, 13, 15, 16, 17.

Math Foundations reference

Chapter 5: Dot Products, Norms, Distance, Angles, and Orthogonality; Chapter 3: Vector Operations and Linear Combinations.

B.5 Least Squares, Orthogonality, Projection, and Residual Geometry

Used in Data Analysis for

Least-squares fitting, normal equations, hat matrix interpretation, fitted values as projections, residual-maker matrix, residual orthogonality, sum-of-squares decompositions, and the geometry behind residual diagnostics.

Math Foundations support

Orthogonal projection, column-space ideas, dot-product orthogonality, least-squares minimization, normal equations, gradients of squared-error objectives, and residuals as the part of the response vector not explained by the chosen column space.

Primary Data Analysis chapters

Chapters 5–10, 12, 13, 15, 17; Appendix A.

Math Foundations reference

Chapter 5: Dot Products, Norms, Distance, Angles, and Orthogonality; Chapter 9: Orthogonal Bases, Gram-Schmidt, and QR; Chapter 11: Subspaces: Column Space, Row Space, and Null Space; Chapter 12: Multivariable Functions, Gradients, and Least Squares.

B.6 Rank, Identifiability, Inverse Formulas, and Collinearity

Used in Data Analysis for

Full column rank, unique least-squares coefficients, non-identifiability, dummy-variable trap, multicollinearity, unstable coefficient interpretation, and when the normal-equation inverse formula is valid.

Math Foundations support

Linear independence, span, bases, dimension, rank, inverses, pseudoinverses, column space, row space, null space, and full column rank criteria.

Primary Data Analysis chapters

Chapters 7–12, 14, 16; Appendix A.

Math Foundations reference

Chapter 6: Linear Independence, Span, Bases, and Dimension; Chapter 10: Matrix Inverses, Rank, and Pseudoinverses; Chapter 11: Subspaces: Column Space, Row Space, and Null Space.

B.7 Variance, Standard Errors, Intervals, and Quadratic Forms

Used in Data Analysis for

Estimator variance, standard errors, residual variance, confidence intervals, prediction intervals, leverage-adjusted residuals, covariance matrices, quadratic penalty terms, and the difference between RSS/n and $RSS/(n - p - 1)$.

Math Foundations support

Matrix products, symmetric matrices, positive-semidefinite quadratic forms, squared lengths, Hessian/curvature intuition, and the matrix structure behind variance expressions such as $\sigma^2(X^T X)^{-1}$.

Primary Data Analysis chapters

Chapters 6, 8, 10, 12–16.

Math Foundations reference

Chapter 7: Matrices and Matrix-Vector Multiplication; Chapter 10: Matrix Inverses, Rank, and

Pseudoinverses; Chapter 12: Multivariable Functions, Gradients, and Least Squares; Chapter 13: Hessians, Quadratic Forms, and Second-Order Conditions.

B.8 Optimization and Penalized Objectives

Used in Data Analysis for

Deriving least-squares estimators, understanding maximum likelihood as an optimization problem, comparing likelihood-based models, minimizing prediction error, using AIC/BIC-type criteria, and fitting ridge, LASSO, and elastic-net objectives.

Math Foundations support

Functions of several variables, gradients, stationary points, least-squares objectives, normal equations, Hessian and curvature intuition, and objective functions with added penalty terms.

Primary Data Analysis chapters

Chapters 5, 7, 10, 12, 14–17.

Math Foundations reference

Chapter 12: Multivariable Functions, Gradients, and Least Squares; Chapter 13: Hessians, Quadratic Forms, and Second-Order Conditions; Chapter 5: Dot Products, Norms, Distance, Angles, and Orthogonality.

B.9 Eigenvalues, Bases, PCA, and Dimension Reduction

Used in Data Analysis for

Principal components, principal component regression, dimension reduction, low-variance predictor directions, collinearity diagnostics by nearly degenerate directions, and replacing original predictors with linear combinations.

Math Foundations support

Linear combinations, bases, dimension, coordinate representations, orthogonal directions, rank and subspace ideas. The Data Analysis text uses eigenvalue language for PCA, but Math Foundations primarily supports the surrounding basis, rank, and subspace intuition rather than a full eigentheory course.

Primary Data Analysis chapters

Chapter 16, with background from Chapters 7–12.

Math Foundations reference

Chapter 3: Vector Operations and Linear Combinations; Chapter 6: Linear Independence, Span, Bases, and Dimension; Chapter 9: Orthogonal Bases, Gram-Schmidt, and QR; Chapter 10: Matrix Inverses, Rank, and Pseudoinverses; Chapter 11: Subspaces: Column Space, Row Space, and Null Space.

B.10 Neural-Network Matrix and Function Composition

Used in Data Analysis for

Viewing neural networks as repeated weighted sums followed by nonlinear activation functions, understanding layer matrices, comparing neural-network predictions to linear-model baselines, and fitting squared-error losses.

Math Foundations support

Matrix-vector multiplication, composition of linear transformations, functions of several variables, componentwise nonlinear transformations, MSE as a squared-error objective, and gradient/optimization intuition.

Primary Data Analysis chapters

Chapters 1 and 17, with support from Chapters 2, 15, and 16.

Math Foundations reference

Chapter 7: Matrices and Matrix-Vector Multiplication; Chapter 12: Multivariable Functions, Gradients, and Least Squares; Chapter 13: Hessians, Quadratic Forms, and Second-Order Conditions.

B.11 Quick Lookup Summary

- For **vectors, rows, columns, and data tables**, review Math Foundations Chapters 2, 3, and 7.
- For **error length, RSS, RMSE, angles, and orthogonality**, review Chapter 5.
- For **design-matrix rank, identifiability, and collinearity**, review Chapters 6, 10, and 11.
- For **least squares, gradients, and optimization**, review Chapter 12.
- For **quadratic penalties, variance formulas, and curvature**, review Chapter 13.
- For **PCA/PCR intuition**, review bases, orthogonal directions, rank, and subspaces in Chapters 6, 9, 10, and 11.

B.12 Appendix Summary

The Data Analysis textbook uses Math Foundations in a recurring way: data become vectors and matrices; fitted values are linear combinations; least squares is a projection problem; residuals are measured with norms; inference uses variance and quadratic-form structure; collinearity is a rank and subspace problem; regularization adds penalty terms to an objective; PCA replaces predictors with new basis directions; and neural networks repeat matrix operations with nonlinear functions. The chapter-level connection paragraphs explain these ideas where they first matter. This appendix provides a shorter lookup map for review.